CLEANROOM AND CHARACTERIZATION REPORT

PROGRAMME

Indo-Taiwan Advanced Semiconductor Manufacturing Program

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March 19, 2026

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Overview of the Semiconductor Fabrication Process Flow

On Day 1 the cleanroom session began with an introduction to the complete semiconductor fabrication flow carried out over the training period. Each step depends on the quality of the previous one, so understanding the full sequence is essential before any hands-on work. The flow is illustrated in Figure 1 and summarised in Table 1.

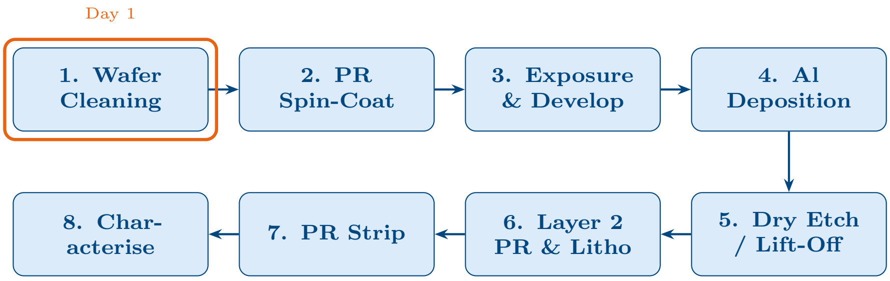


Figure 1: Complete 8-step fabrication process flow for the cleanroom training programme. The orange border marks the Day 1 activity (wafer cleaning).

Table 1: Summary of each process step.

Step	Process	Purpose
1	Wafer Cleaning	Remove organic, ionic, and metallic contaminants.
2	PR Spin-Coat	Apply light-sensitive photoresist polymer.
3	Exposure & Develop	UV pattern transfer through a mask; develop to reveal pattern.
4	Al Deposition	Deposit aluminium by sputtering or evaporation.
5	Dry Etch / Lift-Off	Pattern Al layer using RIE or lift-off.
6	Layer-2 PR	Second resist for multi-level patterning.
7	PR Strip	Remove residual resist (solvent or plasma ash).
8	Characterisation	Optical microscopy, profilometry, or SEM inspection.

Day 1 focus: Wafer orientation/identification and wet chemical cleaning using H_2SO_4/DI water with N_2 agitation.

Wafer Orientation and Identification

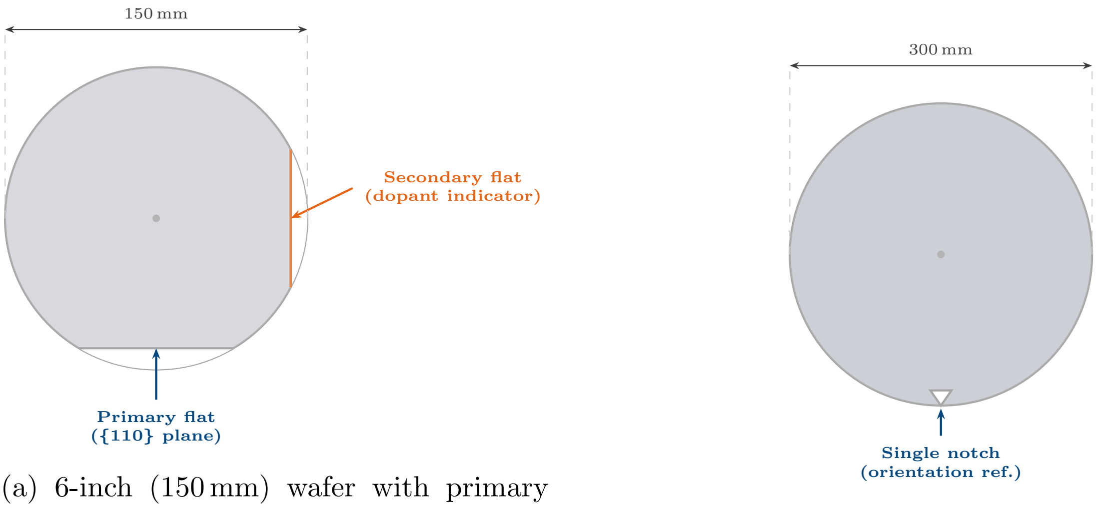
Introduction to Silicon Wafers

A silicon wafer is a thin, polished disc of crystalline silicon used as the substrate for integrated circuit fabrication. Wafers are characterised by diameter, crystal orientation, dopant type, and surface coating. During Day 1, trainees were introduced to these physical properties and learned to identify key features by visual inspection alone.

Wafer Sizes: 6-inch and 12-inch

Table 2: Comparison of 6-inch and 12-inch silicon wafers.

Parameter	**6-inch (150 mm)**	**12-inch (300 mm)**
Diameter	150 mm	300 mm
Thickness	~675 µm	~775 µm
Orientation marker	Primary + secondary flats	Single notch
Industry application	R&D, MEMS, power devices	Advanced CMOS



(a) 6-inch (150 mm) wafer with primary (grey) and secondary (orange) orientation flats.

(b) 12-inch (300 mm) wafer: a single small notch replaces the flat system.

Figure 2: 6-inch and 12-inch silicon wafers. The 6-inch wafer uses SEMI-standard flats for orientation and dopant-type identification; the 12-inch wafer uses a single notch whose dopant/orientation data are stored in the wafer barcode.

Wafer Orientation Identification

How to Identify Wafer Orientation

- **6-inch:** Primary flat (~57.5 mm) marks the {110} crystal plane. Secondary flat angular position encodes dopant type and orientation (SEMI standard).
- **12-inch:** Single notch (~1 mm deep); dopant/type from barcode.
- **(100) wafers:** Cleaving along the flat yields smooth {110} planes at 90° to each other.

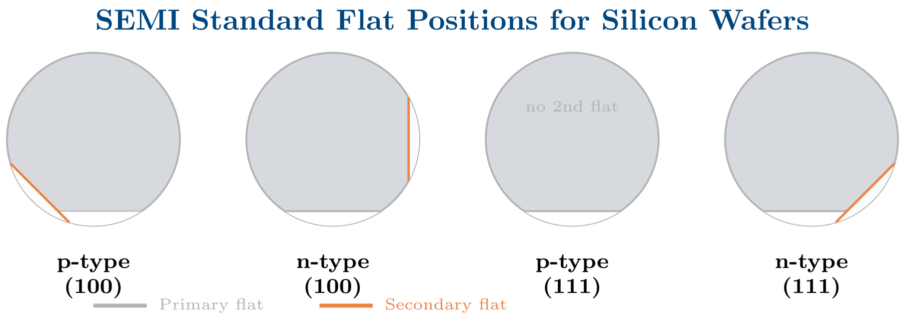


Figure 3: SEMI standard flat positions for common silicon wafer types. The primary flat (grey line) identifies the {110} plane; the secondary flat (orange) encodes dopant type and crystal orientation. P-type {111} wafers have no secondary flat.

Identifying Surface Coating: SiO_2 vs. Si_3N_4

Different surface coatings are distinguishable visually through colour and reflectivity arising from thin-film optical interference.

Table 3: Visual identification of surface coatings on silicon wafers.

Coating	Colour / Appearance	Refractive Index
Bare silicon	Mirror-like, grey-silver	—
SiO_2 (thermal oxide)	Blue, purple, or gold (thickness-dependent)	$n \approx 1.46$
Si_3N_4 (LPCVD)	Amber / yellow, dull matte	$n \approx 2.0$

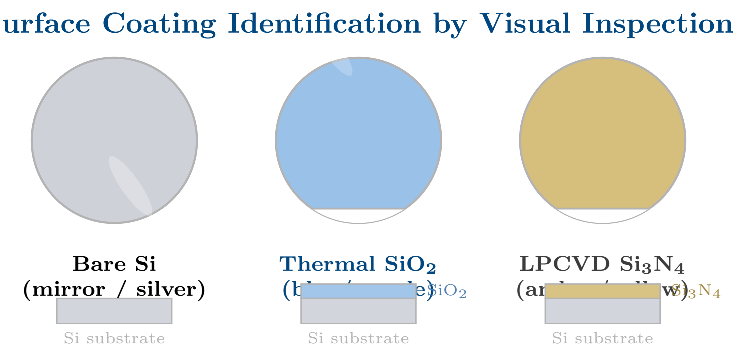


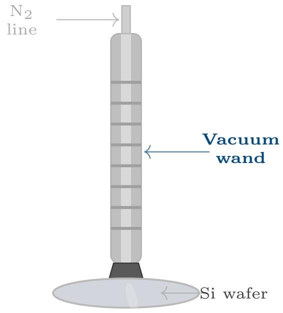
Figure 4: Visual appearance of silicon wafers with different surface coatings (top view) and corresponding cross-section insets. SiO_2 films display characteristic blue/purple colours from thin-film interference; Si_3N_4 appears amber and less reflective due to its higher refractive index ($n \approx 2.0$).

Wafer Handling Equipment

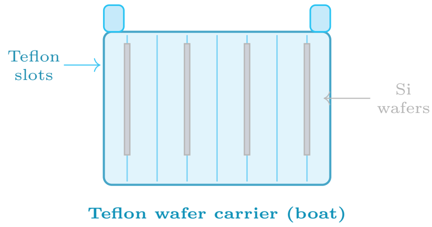
Correct handling prevents particle contamination, scratches, and electrostatic damage:

- **Vacuum wand:** soft tip; picks up wafer at edge without touching the active surface.

- **Teflon wafer carriers (boats):** chemically inert cassettes for wet processing.
- **Wafer chuck:** vacuum hold-down on spin-coaters and mask aligners.
- **Cleanroom gloves:** nitrile gloves mandatory; bare hands introduce Na^+, K^+, organics.



(a) Vacuum wand for safe, contamination-free wafer pickup at the wafer edge.



(b) Teflon wafer carrier (boat) used to hold multiple wafers during wet chemical processing.

Figure 5: Standard wafer handling equipment in the cleanroom environment.

Name of the Process Performed

Process: Wet Chemical Wafer Cleaning — $H_2SO_4 : DI\,H_2O$ with N_2 Bubbling (Piranha-type clean)

Category: Pre-process surface preparation / organic contamination removal

The process performed on Day 1 was **wet chemical wafer cleaning** using sulphuric acid diluted in DI water with nitrogen gas agitation. This is a simplified *Piranha-type clean* and is the first critical step in semiconductor fabrication, ensuring the substrate is free of organic contaminants before downstream processing.

Instrument Description: The Wet Processing Bench

Overview

A **wet processing bench** is a self-contained workstation for liquid-phase chemical treatments of semiconductor wafers. It is constructed from chemically resistant materials (polypropylene, PVDF, or PFA) with an exhaust ventilation system to safely remove chemical fumes.

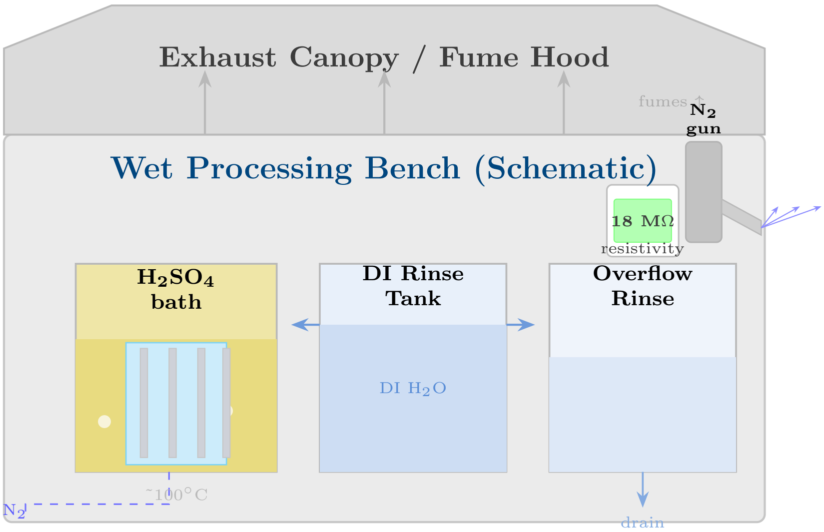


Figure 6: Schematic of a cleanroom wet processing bench showing the H_2SO_4 acid bath with N_2 bubbling and wafer carrier, DI water rinse tanks with overflow, resistivity monitor, N_2 blow-gun, and exhaust fume hood.

Key Components

Table 4: Main components of a wet processing bench.

Component	Function
Exhaust canopy	Draws chemical vapours away from operator; maintains negative pressure.
Quartz/Teflon bath tanks	Hold acid/base solutions; quartz used for hot baths ($\leq$120 °C).
DI water rinse tanks	Cascading overflow tanks remove residual chemicals.
N_2 sparger	Agitates bath with inert gas for uniform cleaning.
Temperature controller	Monitors and regulates bath temperature.
Wafer carrier holders	Teflon racks hold wafers vertically during immersion.
N_2 blow-gun	Dries wafers after rinsing.
Resistivity monitor	Confirms DI water purity (>18 MΩ cm).
Emergency eyewash	Safety requirement for acid/base handling.

Process Description and Parameters

Purpose of Wafer Cleaning

Before any thin-film deposition, photolithography, or thermal oxidation, the wafer surface must be free from organic contaminants (photoresist residues, oils), ionic/metallic impurities (Na^+, Fe^{3+}), native silicon oxide, and particulate matter. Even trace contamination causes gate oxide defects, poor film adhesion, interface trap states, and device reliability failures.

Process Performed: H_2SO_4/DI Water + N_2

Process Parameters / Recipe — Day 1 Cleaning

Chemical:	Concentrated H_2SO_4 in DI water
Mixing ratio:	$H_2SO_4 : DI = 3{:}1$ to $4{:}1$ (v/v)
Agitation:	N_2 gas bubbling via sparger
Temperature:	~80–120 °C (exothermic on mixing)
Immersion time:	10–15 minutes
Rinse:	3× overflow DI water rinse
Drying:	N_2 blow-dry
PPE:	Face shield, acid-resistant gloves, cleanroom suit

5.2.1 Step-by-Step Procedure

Step 1. Gown up. Cleanroom suit, gloves, face shield, and goggles before handling chemicals.

Step 2. Prepare bath. Add H_2SO_4 to DI water (acid to water; never reverse). Solution heats exothermically.

Step 3. Load wafer. Place wafer in Teflon carrier using vacuum wand.

Step 4. Immerse. Lower carrier into hot acid; switch on N_2 bubbling.

Step 5. Soak 10–15 min. H_2SO_4 oxidises and dissolves organic material.

Step 6. Rinse. Transfer to 3 overflowing DI water tanks; confirm resistivity >18 MΩ cm.

Step 7. Dry. N_2 blow-dry, holding wafer at edge.

Step 8. Inspect. Clean surface appears hydrophilic (chemical oxide present).

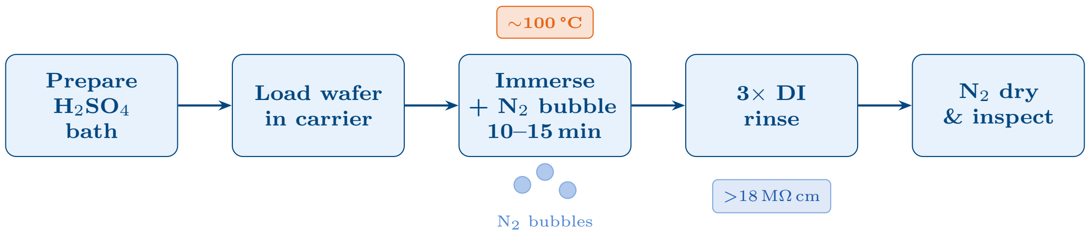


Figure 7: Step-by-step sequence of the Day 1 cleaning process. N_2 bubbling ensures uniform agitation; DI water resistivity confirms complete ionic removal before the wafer is dried and inspected.

Standard Industry Wafer Cleaning Processes

RCA Cleaning

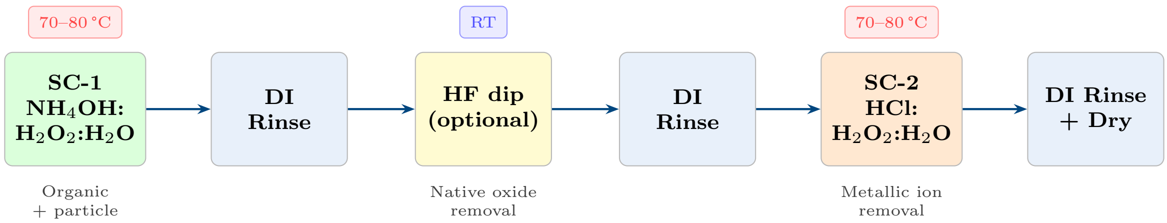
The **RCA clean** (Werner Kern, 1965) is the industry-standard protocol consisting of two sequential chemical baths:

RCA SC-1 — Organics & Particles

Solution: $NH_4OH : H_2O_2 : H_2O = 1{:}1{:}5$
Temp / Time: 70–80 °C, 10 min
Action: Oxidises and lifts organics; slight etching removes particles

RCA SC-2 — Metallic Ions

Solution: $HCl : H_2O_2 : H_2O = 1{:}1{:}6$
Temp / Time: 70–80 °C, 10 min
Action: Forms soluble metal chlorides; removes Na^+, K^+, Fe, Cu



Full RCA Cleaning Sequence

Figure 8: Full RCA cleaning sequence: SC-1 removes organics and particles; an optional HF dip strips native oxide; SC-2 removes metallic ion contaminants. Each chemical step is followed by an overflow DI water rinse.

Table 5: Comparison of common wafer cleaning methods used in the semiconductor industry.

Process	**Chemistry**	**Target**	**Notes**
Piranha (SPM)	H_2SO_4:H_2O_2 4:1	Organics, PR	Very aggressive
H_2SO_4/DI (this lab)	H_2SO_4:DI + N_2	Organics	Simplified Piranha
RCA SC-1	NH_4OH:H_2O_2:H_2O 1:1:5	Organics, particles	Industry standard
RCA SC-2	HCl:H_2O_2:H_2O 1:1:6	Metallic ions	Industry standard
HF dip	1–5% HF in DI	Native SiO_x	Hydrophobic surface
Megasonic	DI + MHz ultrasound	Particles	Physical, no chemistry
O_2 plasma ash	O_2 plasma	Organics, PR	Dry process

Results and Observations

Visual Inspection of Cleaned Wafer

After the H_2SO_4/DI water cleaning and N_2 drying, the wafer was inspected visually and under an optical microscope.

(a) ×50 objective: uniform, featureless surface with no organic residue or particle contamination after cleaning.

(b) ×200 objective: no contamination streaks or particulate deposits confirming effective cleaning across the wafer surface.

Figure 9: Optical microscope images of the silicon wafer surface after H_2SO_4/DI water cleaning. The uniform, featureless surface confirms successful removal of organic contaminants. (Replace with actual lab photographs.)

Observations

- Surface appeared **hydrophilic** after cleaning (DI water spread evenly), indicating a thin chemical oxide and confirming organic removal.
- N_2 bubbling visibly agitated the solution, ensuring uniform circulation so all wafer surfaces contacted fresh acid.
- Solution temperature rose noticeably on mixing (exothermic H_2SO_4 dilution), confirming active chemistry.
- DI rinse tank resistivity returned to $>15\,\mathrm{M\Omega\,cm}$ after rinsing, confirming complete ionic removal.

Justification and Purpose of the Process

Wafer cleaning is the most fundamental step in semiconductor fabrication:

1. **Adhesion:** Organics prevent uniform adhesion of photoresist and dielectrics, producing pinholes, delamination, and lithographic defects.
2. **Electrical integrity:** Metal impurities (Na^+, Fe^{3+}, Cu^{2+}) diffuse into silicon at high temperature, creating deep-level traps that degrade carrier lifetime and gate oxide reliability.
3. **Gate oxide quality:** Sub-monolayer contamination at the Si/SiO_2 interface increases D_{it} and reduces oxide breakdown field.
4. **Process yield:** Particles $> 0.1\,\mu$m nucleate defects in sub-micron features, directly reducing die yield.
5. **Reproducibility:** A standardised cleaning protocol ensures all wafers start in an identical surface state, improving lot-to-lot uniformity.

The H_2SO_4/DI water + N_2 process was used as a safe, accessible introduction to acid-based cleaning, demonstrating the same principles as the industrial Piranha (SPM) clean while familiarising students with wet bench operation, chemical handling, and cleanroom safety.

Safety Considerations

HAZARD NOTICE: Concentrated H_2SO_4 is a *severe corrosive*. Mixing with water is highly exothermic — always add acid to water, never the reverse. Face shield, acid-resistant gloves, and full cleanroom suit are mandatory. Emergency eyewash and safety shower must be immediately accessible.

- All chemicals handled inside the exhaust wet bench with extraction running.
- Acid-resistant apron + nitrile inner gloves + acid gloves + face shield = mandatory PPE.
- Spent acid collected in labelled waste containers; never drain-disposed.
- N_2 pressure <5 psi to prevent aerosol generation.
- Two persons must be present when handling concentrated acids.

Conclusion

Day 1 of the cleanroom training provided a comprehensive introduction to the semiconductor fabrication environment. Key learning outcomes were:

- Understanding the complete 8-step fabrication flow from cleaning through characterisation.
- Identifying silicon wafers by size (6-inch vs. 12-inch), crystal orientation (flats vs. notch), and surface coating (SiO_2 vs. Si_3N_4) by visual inspection.
- Correct wafer handling using vacuum wands and Teflon carriers to prevent contamination.
- Performing H_2SO_4/DI water wet cleaning with N_2 agitation following all safety protocols.
- Understanding how cleaning ensures device quality and process yield, and its relationship to the industry-standard RCA clean.

The successfully cleaned wafer showed a uniform, contamination-free surface under optical microscopy, confirming process effectiveness and readying the substrate for photolithography in Day 2.

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Overview of Day 2 Activities

Day 2 of the cleanroom training programme covered three sequential process steps constituting the core pattern-transfer and etching module of the fabrication flow: **Photolithography**, **Silicon Dry Etching** (using SF_6/Ar plasma), and **Photoresist Removal** (lift-off). Together these steps define a permanent etched pattern in the silicon substrate, which forms the foundation for subsequent metallisation and interconnect steps.

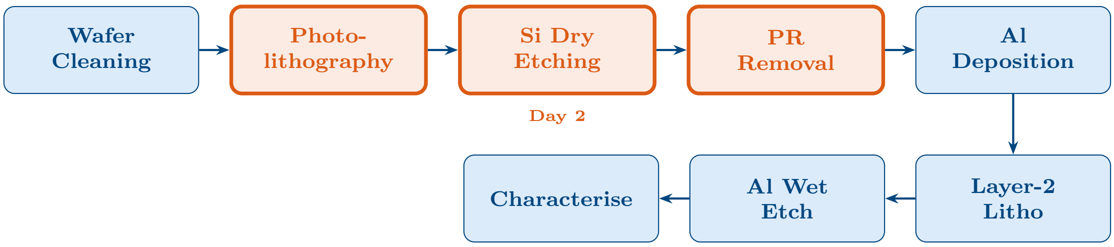


Figure 1: Fabrication flow. Orange highlighted boxes show the three Day 2 steps: photolithography, Si dry etching (SF_6/Ar), and PR removal.

Part 1: Photolithography

Aim

To transfer a predefined circuit pattern from a photomask onto a photoresist-coated silicon wafer using UV light exposure. The patterned resist then acts as a protective mask for the subsequent silicon dry etching step.

Materials and Recipe

Photolithography Recipe — Day 2

Photoresist:	EPG512 (positive-tone)
Developer:	AD-10
Target resolution:	1 µm
Spin speed:	1000 RPM (spread) → 5000 RPM (thickness)
Target thickness:	∼1 µm
Soft bake:	90 °C
UV exposure:	1 minute
Hard bake:	120 °C, 3 minutes

Process Flow

The photolithography sequence followed on Day 2 is summarised in Figure 2 and described in detail below.

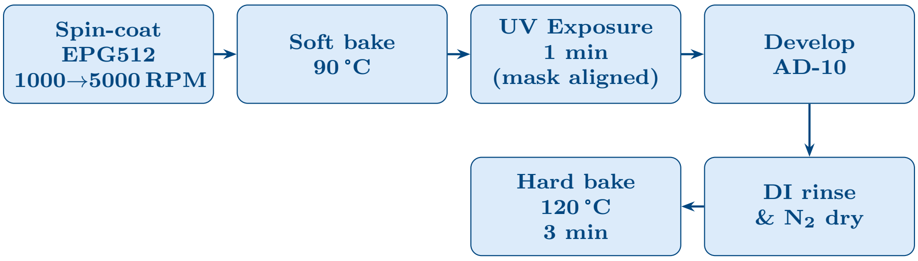


Figure 2: Photolithography process sequence as performed on Day 2.

Step 1: Spin-Coating EPG512 Photoresist

The silicon wafer was coated with **EPG512** positive-tone photoresist using a spin coater. A two-speed spin programme was used:

1. **1000 RPM** (spread phase): distributes the resist uniformly across the wafer surface.
2. **5000 RPM** (thinning phase): thins the resist to the target thickness of approximately 1 µm.

The resist film thickness t scales with spin speed ω as $t \propto 1/\sqrt{\omega}$, so the 5000 RPM final speed determines the film thickness. A 1 µm thick resist provides sufficient etch resistance for the subsequent 300 nm deep Si etch.

(a) Spin coating of EPG512 photoresist onto the Si wafer at 5000 RPM.

(b) Cross-section: 1 µm EPG512 uniformly coated on Si.

Figure 3: Spin coating: lab photograph (left) and cross-sectional schematic (right).

Step 2: Soft Bake

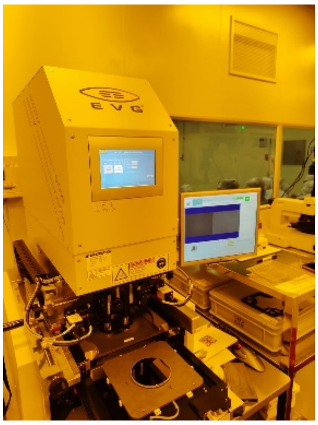
After spin-coating, the wafer was placed on a hotplate at **90 °C** to evaporate the residual casting solvent from the resist. This improves film adhesion to the wafer and prevents solvent bubbling during UV exposure.

Figure 4: Soft bake of the EPG512-coated wafer at 90 °C on a hotplate to remove excess solvent before UV exposure.

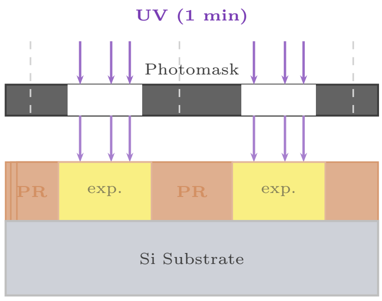
Step 3: UV Exposure and Mask Alignment

The resist-coated wafer was loaded into the UV exposure system. The photomask (carrying the 1 µm resolution pattern) was aligned to the wafer using the alignment marks — two '+' marks are brought into coincidence at two different areas of the wafer to confirm correct overlay before exposure.

UV light was then applied for **1 minute**. In the positive-tone EPG512 resist, UV exposure breaks the photoactive compound (PAC) bonds in the exposed regions, making them soluble in the developer. The unexposed (masked) regions remain insoluble and will protect the underlying Si during etching.



(a) UV exposure of the resist-coated wafer through the photomask for 1 minute.



(b) UV exposure cross-section.

Figure 5: UV exposure and mask alignment: lab photograph (left) and cross-section (right).

Step 4: Development using AD-10

The exposed wafer was immersed in **AD-10** developer solution (TMAH-based alkaline developer). The UV-exposed resist regions dissolve in the alkaline developer while the unexposed regions remain intact, revealing the pattern.

Figure 6: Development of the exposed wafer in AD-10 developer solution. Exposed (solubilised) EPG512 resist is dissolved, leaving the unexposed PR pattern intact.

Step 5: DI Rinse, N_2 Dry, and Hard Bake

After development the wafer was rinsed with DI water to stop the development reaction, then dried with an N_2 gun. A final **hard bake at 120 °C for 3 minutes** crosslinks and hardens the resist, improving its etch resistance for the subsequent SF_6 plasma etch.

Figure 7: Hard bake at 120 °C for 3 minutes to crosslink the photoresist pattern and improve its mechanical and chemical resistance before Si etching.

Part 2: Silicon Dry Etching (SF_6/Ar Plasma)

Aim

To etch the exposed regions of the silicon substrate (areas without photoresist) to a depth of 300 nm using fluorine radicals generated from an SF_6/Ar plasma. The volatile by-product SiF_4 is pumped away, leaving a clean etched profile.

Recipe

Si Dry Etch Recipe — Day 2

Process gas:	SF_6: 10 sccm
Carrier/helper gas:	Ar: 50 sccm
Base pressure:	1 mTorr (achieved by turbo pump)
Working pressure:	5 mTorr (after gas introduction)
RF bias power:	100 W
Etch time:	300 seconds (5 minutes)
Target etch depth:	300 nm
Post-etch:	Wait 1 minute before venting (vacuum clear)

Introduction to SF_6 Dry Etching

Dry etching of silicon using SF_6 is a **reactive ion etching (RIE)** process. The SF_6 gas is dissociated in an RF plasma to generate reactive fluorine radicals ($F^\bullet$), which react spontaneously with silicon to form the volatile by-product silicon tetrafluoride (SiF_4):

$$\mathrm{Si} + 4\,\mathrm{F}^\bullet \longrightarrow \mathrm{SiF_4}\uparrow$$

SiF_4 is a gas at room temperature and is pumped out of the chamber by the vacuum system. The addition of argon (Ar) assists plasma ignition and sustains it, and the Ar^+ ions also contribute a physical bombardment component to the etch, improving anisotropy.

Note on CF_4 alternative: For greater anisotropy and better control over etch depth and sidewall profile, CF_4 can be used instead of SF_6. CF_4 produces a more directional etch with improved dimensional accuracy for sub-micron features. On Day 2, SF_6 was used as specified in the recipe.

Process Flow

Step 1. Load wafer. The patterned (post-lithography) wafer is placed in the RIE chamber. The PR pattern protects the Si below it; exposed Si regions will be etched.

Step 2. Rough pump down. The roughing pump evacuates the chamber to an intermediate pressure. The rough pump is used first (rather than directly engaging the turbo pump) to avoid overloading the turbo pump at atmospheric pressure, which could damage it.

Step 3. Turbo pump to base pressure. The turbomolecular pump is engaged to further reduce the pressure to **1 mTorr**, removing all residual atmospheric gases and moisture. Clean base pressure is essential to prevent oxygen or water from passivating the Si surface or interfering with the fluorine chemistry.

Step 4. Introduce process gases. Ar (50 sccm) and SF_6 (10 sccm) are admitted into the chamber. The chamber pressure rises to the working pressure of **5 mTorr** as the gas flow stabilises.

Step 5. Ignite plasma and etch. An RF bias of **100 W** is applied for **300 seconds**. The RF field accelerates electrons, which ionise the Ar and dissociate the SF_6

molecules into $F^\bullet$ radicals and SF_x^+ ions. The fluorine radicals react with the exposed Si surface, forming volatile SiF_4 that is pumped away.

Step 6. Post-etch pumpdown. After the etch completes, all gas flows are stopped and the chamber is pumped for **1 minute** to clear residual process gases before venting.

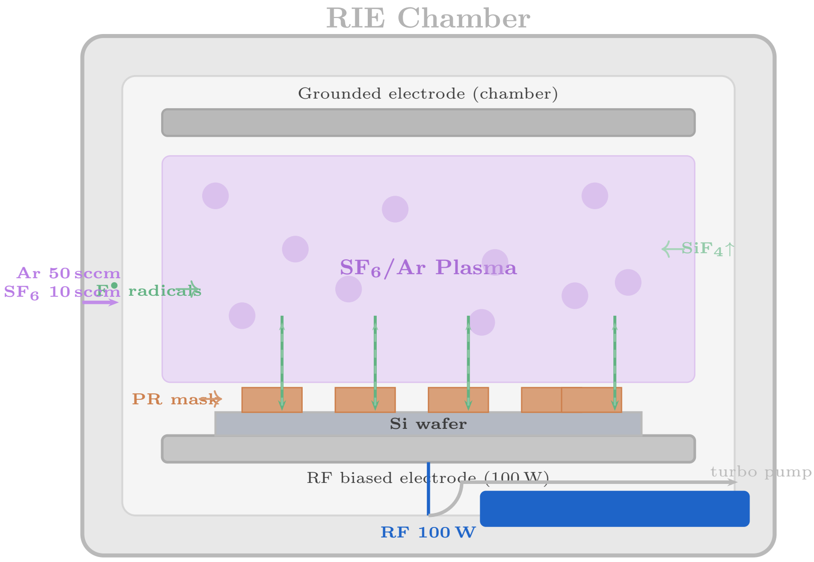
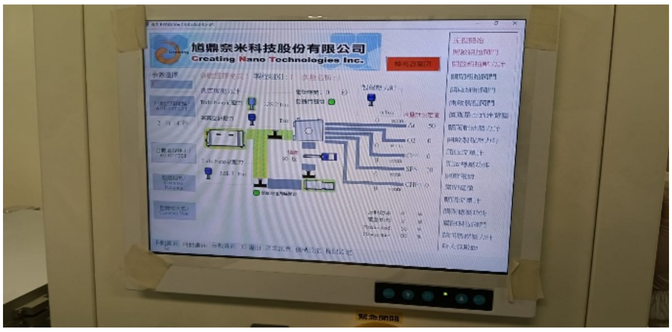


Figure 8: Schematic of the RIE chamber during SF_6/Ar Si etching. $F^\bullet$ radicals generated from SF_6 dissociation react with the exposed Si to form volatile SiF_4, which is pumped away. PR-protected areas are not etched.

Etching Machine and Recipe — Lab Photographs



(a) Recipe of the Si dry etch process displayed on the etching machine control panel (Ar: 50 sccm, SF_6: 10 sccm, RF: 100 W, 300 s).

(b) The Si dry etching machine (RIE system) used on Day 2 for SF_6/Ar plasma etching.

Figure 9: Si dry etching: process recipe (left) and the RIE etching machine (right).

Etch Chemistry and Mechanism

The overall etching chemistry proceeds in two stages:

1. **Plasma dissociation of SF_6:**

$$SF_6 \xrightarrow{\text{RF plasma}} SF_x^+ + F^\bullet$$

   The RF field (100 W) dissociates SF_6 molecules, producing reactive fluorine free radicals ($F^\bullet$) and sulfofluoride ions. Argon assists plasma ignition and provides additional physical bombardment by Ar^+ ions.

2. **Si surface reaction:**

$$Si + 4\,F^\bullet \longrightarrow SiF_4\uparrow$$

   Fluorine radicals attack the exposed Si surface. SiF_4 is a gas at room temperature (bp $= -86\,^\circ$C) and is continuously pumped out of the chamber, driving the reaction forward and leaving a clean etched surface.

The photoresist mask is chemically resistant to fluorine radicals under these conditions, so only the unprotected Si regions are etched.

Etch Depth and Rate

At the recipe conditions (100 W RF, 5 mTorr, 10 sccm SF_6), the Si etch rate is typically ~60 nm/min for fluorine-based RIE. At the 300 s (5 minute) etch time, this yields:

$$\text{Depth} = 60\,\frac{\text{nm}}{\text{min}} \times 5\,\text{min} = 300\,\text{nm}$$

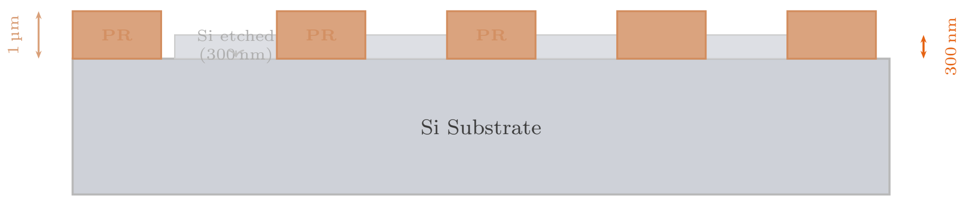


Figure 10: Cross-section of the wafer after Si dry etching (300 nm depth). PR-protected regions are intact; exposed Si regions have been etched to 300 nm.

Part 3: Photoresist Removal (Lift-off)

Aim

After silicon etching, the photoresist mask that protected the Si features during the plasma etch must be removed completely. This step is referred to as **lift-off** or PR stripping. The substrate is processed through a solvent sequence to dissolve and wash away the organic resist without damaging the etched Si features.

Recipe

PR Removal (Lift-off) Recipe — Day 2

Solvent 1:	Acetone (organic solvent — strips resist)
Solvent 2:	Isopropyl alcohol, IPA (intermediate rinse)
Final rinse:	Deionised (DI) water
Agitation:	Sonicator / N_2 gas bubbling in DI water

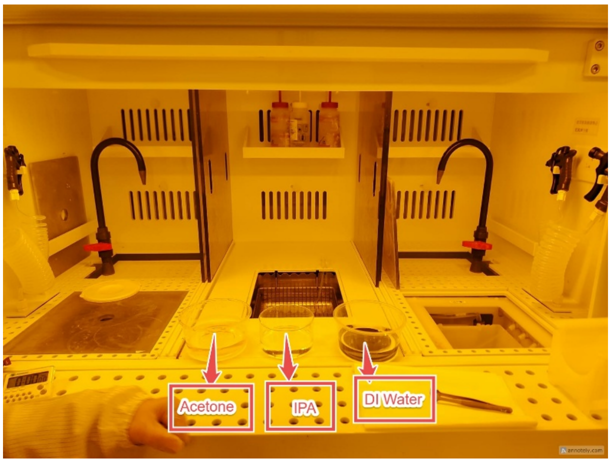
Process Flow and Chemistry

The three-solvent sequence is carefully designed to ensure complete PR removal without redeposition of dissolved resist on the surface:

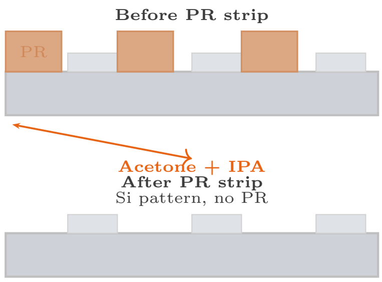
Step 1. Acetone immersion. The wafer is submerged in acetone, an organic solvent that readily dissolves the EPG512 photoresist polymer. Acetone swells the PR film and strips it from the Si surface. However, as acetone is highly volatile, it evaporates rapidly; dissolved PR can therefore *re-precipitate* on the wafer surface if the acetone is not replenished or if the wafer is withdrawn too slowly.

Step 2. Immediate transfer to IPA. To prevent re-precipitation of dissolved PR from evaporating acetone, the wafer is transferred to isopropyl alcohol (**IPA**) *as quickly as possible*. IPA is miscible with acetone and can further dissolve any remaining PR and acetone residues. It acts as an intermediate bridging solvent.

Step 3. DI water rinse with N_2 bubbling. The wafer is finally rinsed with DI water to remove all traces of IPA and residual organic compounds. A sonicator or N_2 gas bubbling through the DI water enhances the mechanical cleaning action, dislodging any loosely adhered particles.



(a) Lift-off process: wafer immersed in the solvent sequence. The photoresist is dissolved and lifted off the Si surface, leaving the etched Si pattern fully exposed.



(b) Cross-section before (top) and after (bottom) PR removal.

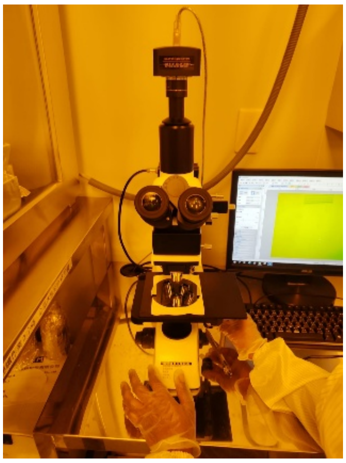
Figure 11: PR lift-off: lab photograph (left) and cross-sectional diagram (right).

Key Observation: On Day 2, the transfer of the wafer from acetone to IPA was performed slightly slowly, resulting in some dissolved PR re-precipitation visible as **white residues** at certain areas on the final wafer surface. This underscores the importance of the rapid transfer between solvents to prevent acetone evaporation and resist re-deposition.

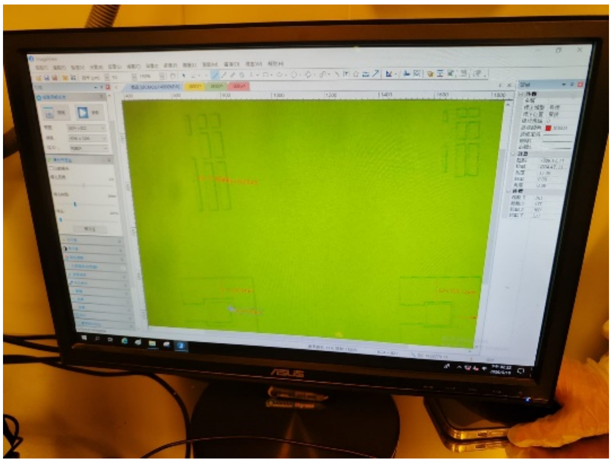
Part 4: Optical Microscope Inspection

Inspection of the Etched Pattern

After PR removal, the wafer was examined under an optical microscope at **5X** and **10X** magnification to verify the quality of the etched silicon pattern.



(a) Optical microscope image of the etched silicon pattern (5X).



(b) Higher magnification view (10X) showing pattern detail and residues.

Figure 12: Optical microscope images of the etched Si pattern after PR removal (Day 2).

Observations and Analysis

- **Successful Si etching:** The silicon substrate was etched successfully in the unprotected regions, confirming that the SF_6/Ar plasma etch at 100 W for 300 seconds achieved the 300 nm target depth.
- **White residues:** Some white residue was observed at certain areas of the wafer. This is attributed to PR re-precipitation during the acetone-to-IPA transfer, consistent with the observation that the transfer was performed too slowly (allowing acetone to partially evaporate with dissolved PR still in solution).
- **Cuts in 2 µm lines:** Some of the narrowest (2 µm) patterned lines showed discontinuities (cuts). These are most likely caused by:
  - Minor mask misalignment during UV exposure.
  - Over-development reducing feature width below the design rule.
  - Physical damage during handling between process steps.
- **Pattern fidelity overall:** The 1 µm resolution features were generally resolved and the bulk of the etched pattern was correctly formed, demonstrating that the full photolithography-etch-strip sequence was successfully executed.

Process Summary and Comparison

Table 1: Summary of all Day 2 process parameters.

Process	**Material/Tool**	**Key Parameter**	**Result**
Spin coating	EPG512 PR	5000 RPM / 1 µm	Uniform film
Soft bake	Hotplate	90 °C	Solvent removed
UV exposure	Photomask (1 µm)	1 minute	Pattern transferred
Development	AD-10 developer	Until clear	PR pattern defined
Hard bake	Hotplate	120 °C / 3 min	Resist hardened
Si dry etch	SF_6:10, Ar:50 sccm	100 W / 300 s / 5 mTorr	300 nm etched
PR removal	Acetone → IPA → DI	Sequential solvent strip	PR removed
Inspection	Optical microscope	5X and 10X	Pattern confirmed

Justification and Importance

1. **Photolithography is the pattern engine of semiconductor fabrication:** Every geometric feature — trenches, contacts, lines, vias — is defined by a lithography step. The 1 µm resolution achieved with EPG512 and the AD-10 developer is the fundamental dimension that all subsequent layers are built upon.

2. **Dry etching provides anisotropic, clean profiles:** Unlike wet etching (which is isotropic and causes undercutting), SF_6 RIE produces a more directional etch profile. The volatile SiF_4 by-product is cleanly pumped away, leaving no liquid residue in the features.

3. **Two-pump sequence protects the turbo pump:** Starting with the rough pump before engaging the turbomolecular pump prevents mechanical stress on the turbo pump at high pressure, extending its operational lifetime — a critical operational detail.

4. **Rapid acetone-to-IPA transfer prevents redeposition:** The re-precipitation observed on Day 2 directly demonstrates why the acetone step must be brief and the IPA transfer immediate. This process knowledge is essential for achieving clean lift-off in production environments.

5. **Hard bake etch resistance:** Without hard bake, the soft (still-solvent-laden) resist would be attacked by the plasma before etching through to the Si surface, causing pattern degradation or resist lifting.

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Overview of Day 3: Electron Beam Evaporation

Day 3 of the cleanroom training programme was dedicated to **Electron Beam (E-beam) Evaporation**, a Physical Vapour Deposition (PVD) technique used to deposit a 200 nm aluminium (Al) thin film on a silicon wafer. This metal film will serve as the interconnect layer in the completed device.

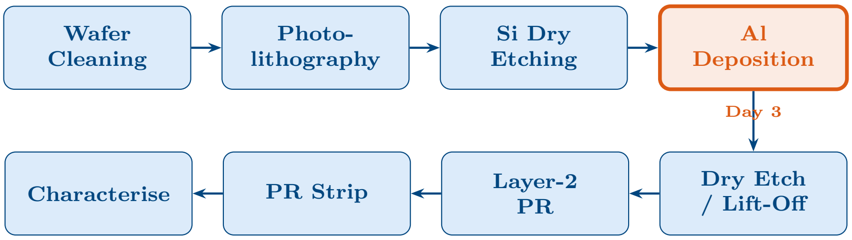


Figure 1: Fabrication flow. The orange highlighted box (Al Deposition, Step 4) is the Day 3 focus, performed by e-beam evaporation.

Introduction to Electron Beam Evaporation

Electron beam evaporation is a PVD process in which a focused, high-energy electron beam (∼5–10 kV) is directed onto a source material (aluminium) held in a crucible inside a high-vacuum chamber. The kinetic energy of the electrons is converted to thermal energy at the point of impact, rapidly heating the Al to its evaporation temperature (>1200 °C). The resulting Al vapour travels in straight ballistic paths through the ultra-high vacuum (UHV) and condenses on the silicon wafer substrate above to form a uniform thin film.

Unlike DC sputtering (which uses plasma momentum transfer), e-beam evaporation is a **thermal PVD** process driven entirely by the heat produced at the beam impact point. The localised heating means only the target material — not the crucible walls — reaches the evaporation temperature, giving exceptionally high film purity.

Instrument Description

Overview

The system used on Day 3 is a high-vacuum thin-film deposition system comprising a stainless-steel UHV chamber, an electron beam gun, a tungsten crucible, a rotating substrate holder, a two-stage vacuum pump system, and a quartz crystal microbalance (QCM) thickness monitor. All components are visible in Figure 2.

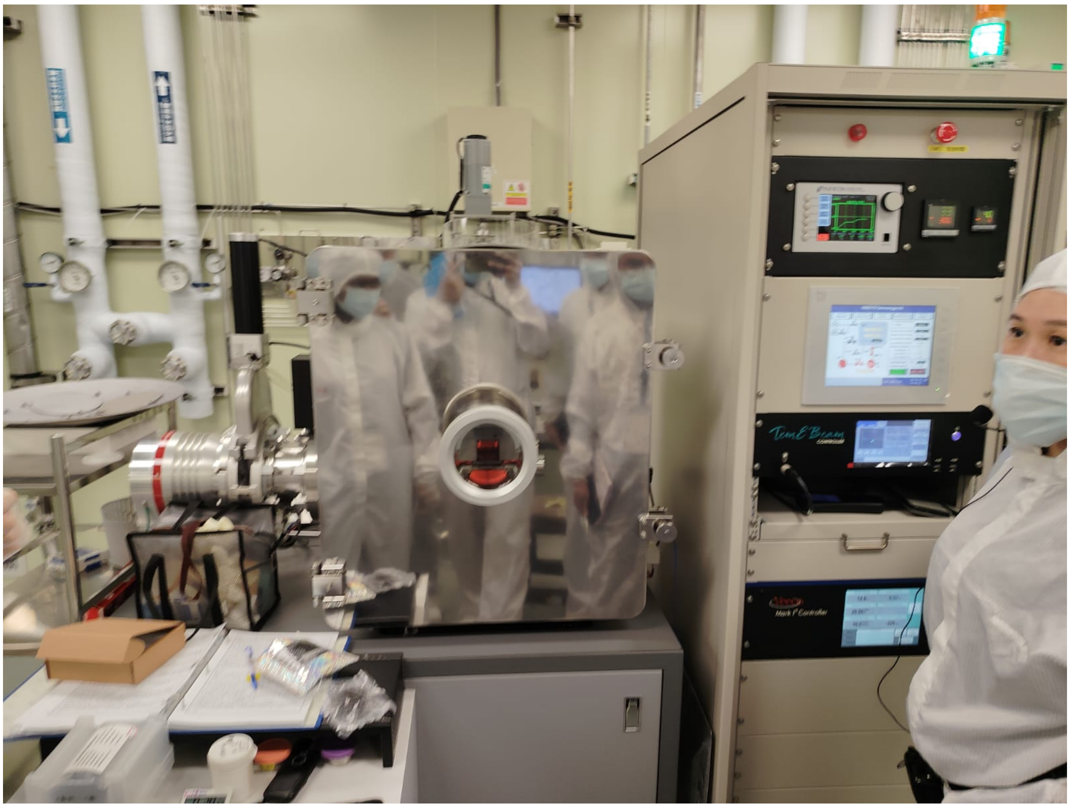


Figure 2: The e-beam evaporation system used on Day 3 at NTHU cleanroom. The main stainless-steel vacuum chamber (centre) with glass viewport is flanked by the electron beam gun assembly (left cylindrical structure) and control panel (right). Vacuum pump connections are visible at the base.

Key Components

1. Vacuum Chamber

The main deposition chamber is a stainless-steel enclosure capable of achieving ultra-high vacuum (UHV). A circular glass viewport allows visual observation of the process. The substrate holder is mounted at the top and the crucible/hearth at the bottom of the chamber interior.

2. Electron Beam Gun

Located on the left side of the system, the e-beam gun is the core of the instrument. It contains a **tungsten filament** that emits electrons by thermionic emission and an **electrostatic/magnetic focusing system** that shapes and steers the beam toward the crucible. Typical accelerating voltages are 5–10 kV.

3. Crucible / Hearth

Located at the bottom of the vacuum chamber, the crucible holds the source material (aluminium on Day 3). The electron beam is focused onto the crucible, locally heating the Al far above its melting point. A **tungsten crucible** was used: its melting point ($\sim$3422 °C) far exceeds that of Al ($\sim$660 °C), so it cannot dissolve into the melt. Graphite crucibles were available but not used to avoid carbon contamination.

4. Substrate Holder

Positioned at the top of the chamber, the substrate holder supports the silicon wafer. It incorporates **rotation** during deposition to ensure uniform film thickness distribution across the full wafer area.

5. Vacuum System

A **roughing pump** followed by a **turbomolecular pump** evacuates the chamber to a base pressure of 3×10^{-8} Torr. At this UHV, the mean free path of Al atoms is on the order of kilometres, guaranteeing straight-line (ballistic) travel without gas-phase collisions.

6. Control Panel

The right-side control panel displays beam current, accelerating voltage, deposition rate (Å/s), and accumulated film thickness monitored by a **quartz crystal microbalance (QCM)**.

System Schematic

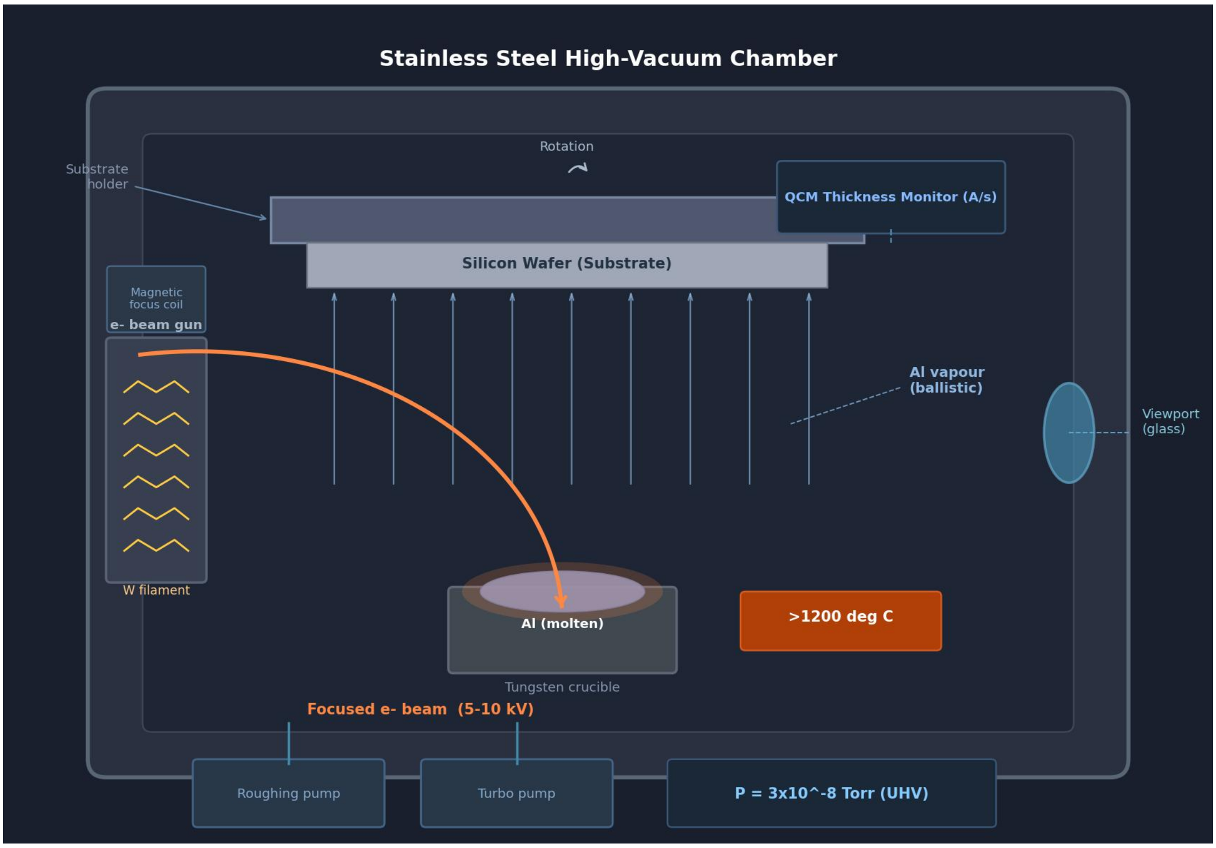


Figure 3: Cross-sectional schematic of the e-beam evaporation system. The electron beam (orange arc) sweeps from the tungsten filament gun, focuses magnetically, and strikes the Al in the tungsten crucible (>1200 °C). Evaporated Al atoms travel upward in ballistic straight lines (blue arrows) and deposit on the rotating wafer. QCM monitors the rate in real time. Base pressure: 3×10^{-8} Torr.

Process Parameters

E-beam Evaporation — Day 3 Process Parameters

Source material:	Aluminium (Al)
Crucible material:	Tungsten (W) — high melting point, no contamination
Crucible alternative:	Graphite available but **not used**
Base pressure:	3×10^{-8} Torr (ultra-high vacuum)
Target film thickness:	200 nm
Deposition rate unit:	Å/second (monitored live by QCM)
E-beam voltage:	5–10 kV
Substrate rotation:	Yes (ensures thickness uniformity)

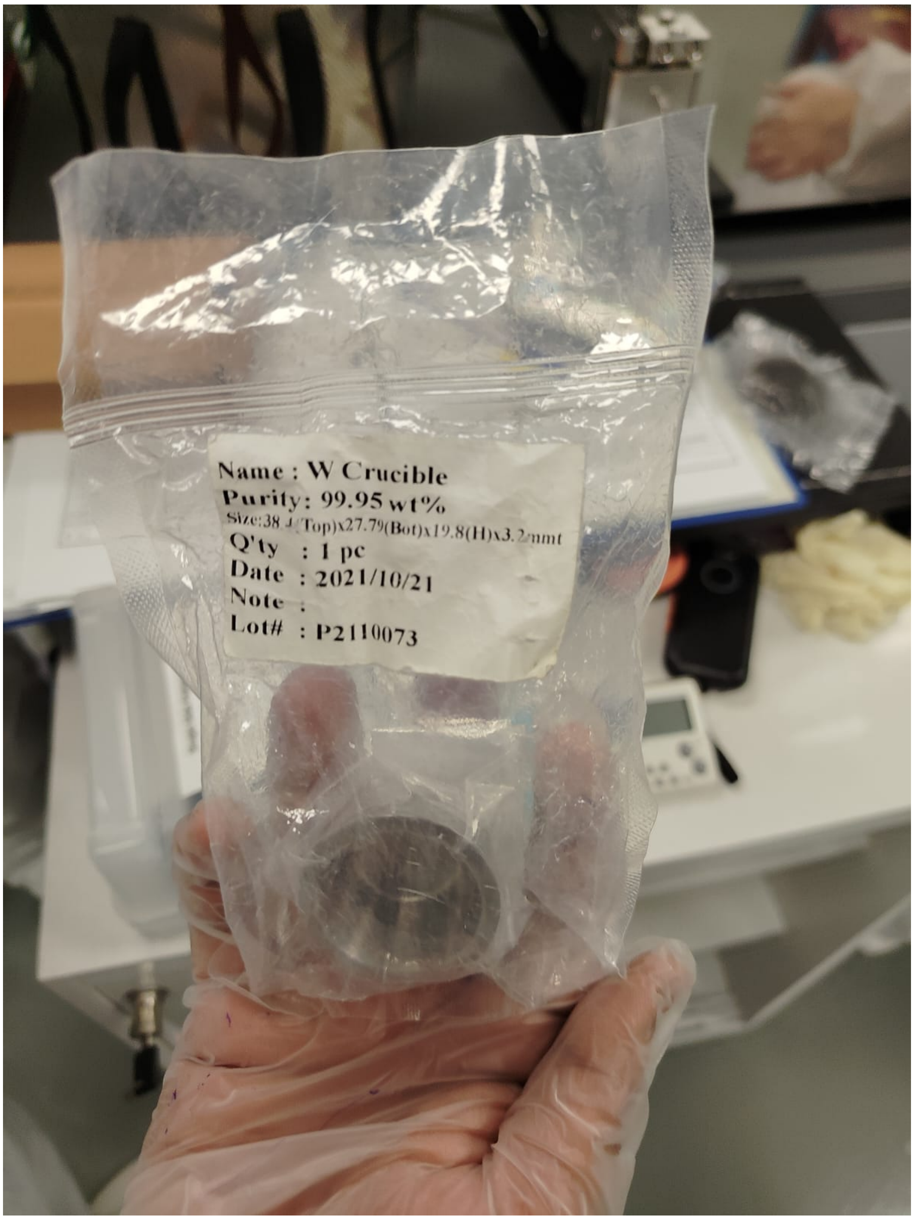


Figure 4: Process parameter display on the e-beam control panel during Day 3 deposition. The QCM readout (in Å/s), accumulated thickness, pressure gauge reading (3×10^{-8} Torr), and beam power settings are all visible.

Process Steps

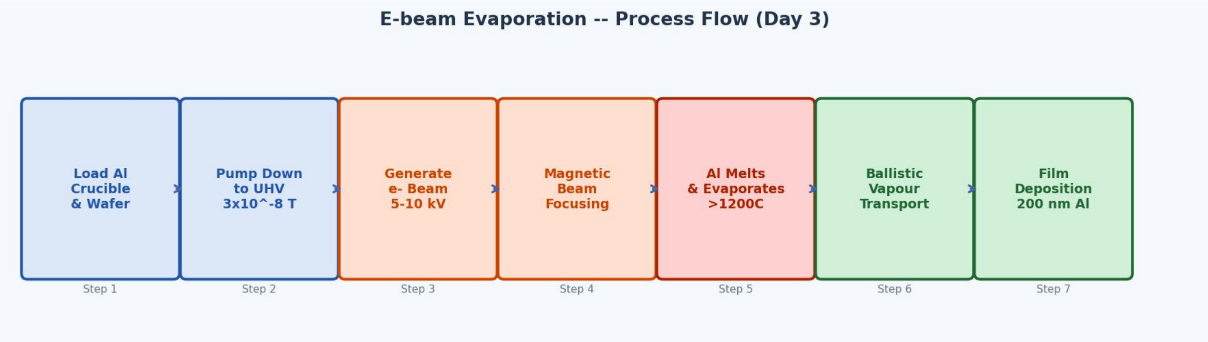


Figure 5: E-beam evaporation 7-step process flow as performed on Day 3.

Step 1. Load Material and Substrate. Aluminium source material is loaded into the tungsten crucible at the bottom of the chamber. The silicon wafer is mounted on the substrate holder at the top. The tungsten crucible is selected because its melting point (∼3422 °C) far exceeds that of aluminium (∼660 °C), ensuring it does not melt or contaminate the deposited film.

Step 2. Create High Vacuum. The roughing pump first reduces the chamber pressure to an intermediate level, after which the turbomolecular pump takes over to achieve the base pressure of 3×10^{-8} Torr. High vacuum is essential for two reasons: (a) to prevent Al vapour from reacting with residual oxygen or moisture, which would form an Al_2O_3 insulating barrier in the film, and (b) to ensure atoms travel in straight-line ballistic paths without gas-phase collisions.

Step 3. Generate Electron Beam. The tungsten filament is resistively heated until it emits electrons by **thermionic emission**. These electrons are accelerated by the high voltage (5–10 kV) between the filament and the anode plate inside the gun.

Step 4. Beam Focusing. Magnetic fields generated by focusing coils shape the electron stream into a tight, steerable beam. The beam can be electromagnetically swept across the crucible surface to ensure uniform heating of the entire Al charge.

Step 5. Material Heating and Evaporation. The focused electron beam strikes the Al source material. Kinetic energy of the electrons is converted to thermal energy at the impact point, rapidly raising the local temperature above 1200 °C. The Al melts and evaporates from the surface of the molten pool. The evaporation rate is controlled by beam power.

Step 6. Vapour Transport. In the UHV environment, evaporated Al atoms travel in straight lines (ballistic motion) from the crucible to the substrate. Because the mean free path at 3×10^{-8} Torr is on the order of kilometres, atoms experience essentially zero collisions during transit.

Step 7. Film Deposition. Al atoms arrive at the rotating wafer surface, condense, and form a continuous thin film. The wafer rotation ensures uniform thickness distribution. The QCM monitors the deposition rate in Å/s in real time and stops the shutter when the target thickness of 200 nm is reached.

Figure 6: E-beam evaporation process in progress on Day 3. The glow visible through the viewport confirms the electron beam is actively striking the Al crucible. The thermal radiation from the melt pool and the Al vapour flux rising toward the substrate above are characteristic of active deposition.

Physics of the Process

Thermionic Emission

Electrons are generated in the filament by thermionic emission. When the tungsten filament is heated to ~2500 K, electrons gain sufficient thermal energy to overcome the work function of tungsten and escape into the vacuum. The current density follows the Richardson–Dushman equation:

$$J = AT^2 \exp\left(-\frac{\phi}{k_B T}\right)$$

where ϕ is the work function of tungsten, k_B is Boltzmann's constant, T is filament temperature, and A is a material constant.

Energy Conversion at the Crucible

When the accelerated electron beam (5–10 kV) strikes the aluminium, the kinetic energy $E_k = eV$ is converted to thermal energy at the impact zone. This highly localised heating creates a molten pool at the Al surface from which evaporation proceeds. The power density at the impact point can exceed $10^4\,\mathrm{W/cm^2}$, sufficient to evaporate virtually any material including tungsten and molybdenum.

Ballistic Transport in UHV

At 3×10^{-8} Torr, the mean free path λ of Al atoms is:

$$\lambda = \frac{k_B T}{\sqrt{2}\,\pi d^2 P} \approx 10^5\,\mathrm{m}$$

This is vastly larger than the chamber dimensions ($\sim$0.5 m), so atoms travel in perfectly straight lines from crucible to substrate with no scattering — producing a highly directional, line-of-sight deposition pattern.

Results and Observations

- UHV base pressure of 3×10^{-8} Torr confirmed before shutter open.
- Visible glow through the viewport confirmed active beam impact on the Al crucible (Figure 6).
- QCM readout in Å/s showed a stable, controlled deposition rate throughout.
- A uniform, shiny, reflective 200 nm Al film was deposited on the wafer, consistent with a low-roughness, high-purity metallic film.
- Wafer rotation throughout deposition produced uniform thickness across the full 6-inch wafer diameter.
- Tungsten crucible showed no visible degradation; no contamination of the Al melt was observed.

Advantages, Limitations and Justification

Table 1: Advantages and limitations of e-beam evaporation for Al deposition.

Advantages	**Limitations**
Very high film purity (only target heated)	Line-of-sight deposition; poor step coverage
High deposition rates achievable	X-ray emission can damage sensitive device layers
Works for high-melting-point materials (W, Mo)	Complex system; UHV maintenance required
Precise real-time thickness control via QCM	Film stress can be high
UHV environment prevents oxidation	Crucible choice critical to avoid contamination

1. **UHV prevents Al oxidation:** Al oxidises instantly in air. Even ppm-level O_2 during deposition would form an insulating Al_2O_3 layer, destroying the film's conductivity. The 3×10^{-8} Torr UHV guarantees a pure metallic film.
2. **Tungsten crucible is critical:** A lower-melting-point crucible would dissolve into the Al melt and co-deposit foreign atoms, degrading film conductivity and device reliability. Graphite was explicitly excluded for this reason.

3. **Precise thickness control determines device performance:** The 200 nm Al thickness directly sets the sheet resistance of the metal interconnects. The QCM feedback ensures the target is met to within a few nanometres.
4. **The deposited Al film is the device interconnect layer:** Without a high-quality, low-resistance Al film, transistors and contacts cannot be electrically connected. The quality of this step directly determines whether the final device is functional.

Overview of Day 4 Activities

Day 4 of the cleanroom training programme covered two major sequential process steps: **Rapid Thermal Processing (RTP/RTA)** and **Photolithography**. RTP was performed first, following the ion implantation carried out in a previous session, to electrically activate the implanted dopants. The wafer was then transferred to the photolithography room for pattern definition using UV photoresist exposure.

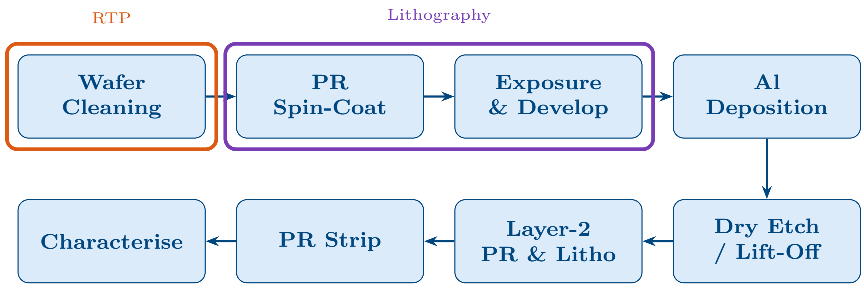


Figure 1: Fabrication flow: orange = RTP (post ion implant); violet = Photolithography (both Day 4).

Part A: Rapid Thermal Processing (RTP / RTA)

Introduction to RTP

Rapid Thermal Processing (RTP), also referred to as Rapid Thermal Annealing (RTA), is a semiconductor manufacturing technique that uses high-intensity tungsten-halogen lamp heating to rapidly raise a single wafer to temperatures exceeding 1000 °C within seconds, rather than the minutes or hours required by conventional batch furnaces. The principal advantage is the minimisation of unwanted dopant diffusion during post-implant annealing, which is critical for fabricating the shallow junctions needed in modern CMOS devices.

Key Principle of RTP

Fast ramp-up (>200 °C/s) combined with a short dwell time (seconds) activates implanted dopants and repairs crystal lattice damage while keeping the total thermal budget low enough to prevent lateral and vertical diffusion of dopants beyond the designed junction depth.

RTP System Description

The RTP tool uses a **cold-wall single-wafer chamber**: only the wafer is heated by radiation from tungsten-halogen lamps above and below, while the chamber walls remain cool. This contrasts sharply with conventional tube furnaces where the entire furnace interior reaches the set temperature, creating a large thermal budget.

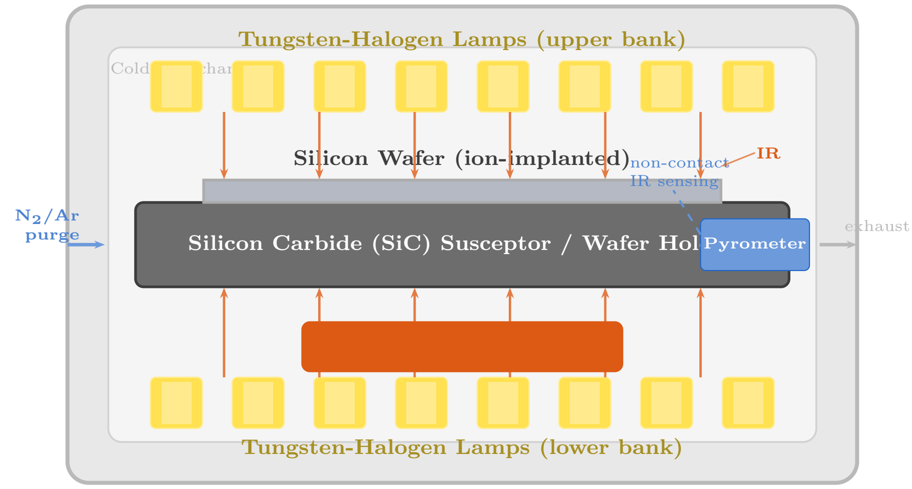


Figure 2: Cross-sectional schematic of the RTP cold-wall chamber used on Day 4. Tungsten-halogen lamps above and below deliver intense IR radiation to the wafer. A SiC susceptor supports the wafer; a pyrometer provides real-time temperature feedback.

Temperature Measurement: Pyrometer

A **pyrometer** (optical/infrared thermal sensor) was used to measure wafer temperature in real time. The pyrometer detects the infrared radiation emitted by the hot wafer surface and converts it into a temperature reading without physical contact, enabling closed-loop lamp power control. Key points:

- Physical thermocouples are not practical for single-wafer fast-ramp processing due to contamination risk and thermal lag at >200 °C/s ramp rates.
- Pyrometry allows real-time temperature feedback at very high ramp rates.
- Emissivity of the wafer surface must be correctly calibrated for accuracy.

Wafer Holder Selection

Two holder materials were available depending on the required temperature and gas ambient, as shown in Table 1.

Table 1: Wafer holder (susceptor) selection in the RTP system.

Holder	**Max Temperature**	**Typical Use**	**Gas Restriction**
Graphite (glass-coated)	<500 °C	Low-temp anneals	N_2/Ar only — **no O_2**
Silicon Carbide (SiC)	~1200 °C	High-temp processes	N_2, Ar, O_2 all compatible

Critical Rule — Graphite Holder & O_2: Graphite must **never** be used when O_2 is introduced. At high temperature, oxygen reacts with carbon to form CO_2, destroying the holder and contaminating the wafer with carbon:

$$C_{(\text{graphite})} + O_2 \longrightarrow CO_2 \uparrow$$

The SiC holder is fully compatible with all gases including O_2.

Process Parameters Used in the Lab

For the Day 4 dopant activation anneal, the **SiC holder** was used (high temperature, compatible with all gases).

Day 4 RTP Recipe

Holder:	Silicon Carbide (SiC)
Temperature:	1000 °C
Dwell time:	30 seconds
Ramp rate:	>200 °C/s
Purge gas:	N_2 or Ar; O_2 also used at 1000 °C
Temperature sensor:	Pyrometer (non-contact IR)
Purpose:	Activate ion-implanted dopants; repair lattice damage
Junction control:	Short anneal limits dopant out-diffusion depth

Temperature–Time Profile

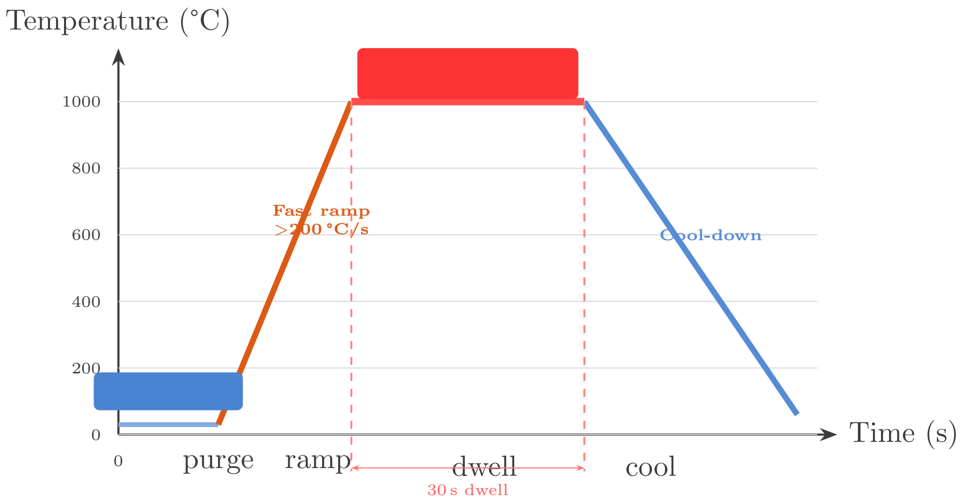


Figure 3: RTP temperature–time profile: N_2/Ar purge, fast ramp-up (>200 °C/s), 30 s dwell at 1000 °C, then rapid cool-down.

What Happens Inside the RTP Chamber

During the 30-second anneal at 1000 °C, three key physical processes occur simultaneously inside the chamber:

1. **Dopant activation:** Ion-implanted dopant atoms (e.g. boron, phosphorus, arsenic) that sit at interstitial lattice positions (electrically inactive) gain sufficient thermal

energy to migrate to substitutional lattice sites, where they become electrically active donors or acceptors.

2. **Crystal damage repair:** The ion bombardment during implantation creates a damaged, amorphous region near the surface. At 1000 °C, solid-phase epitaxial regrowth (SPER) recrystallises this amorphous layer, restoring the single-crystal structure.

3. **Diffusion control:** Because the dwell time is only 30 s, the total thermal budget (temperature × time) is low. The diffusion length $L = \sqrt{Dt}$ remains small (where D is the diffusivity), preserving the shallow junction profile implanted in the previous step.

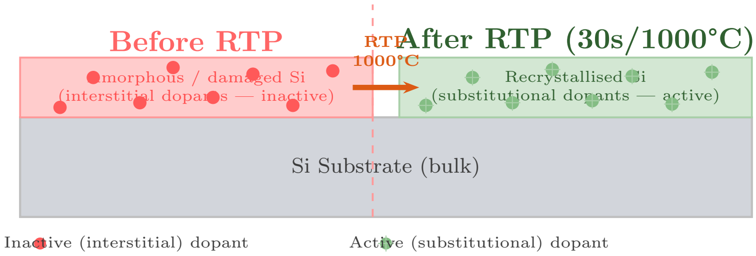


Figure 4: Atomic-scale diagram of what happens inside the RTP chamber: implant damage is repaired by solid-phase epitaxial regrowth and dopant atoms move from interstitial (inactive) to substitutional (active) lattice positions.

RTP Machine — Lab Photograph

Figure 5: RTP system at NTHU cleanroom facility (lab photograph).

Applications and Limitations of RTP

Table 2: Primary RTP applications in semiconductor manufacturing.

Application	Abbreviation	Temperature
Post-implant dopant activation	RTA	900–1100 °C
Rapid thermal gate oxidation	RTO	900–1050 °C
Metal silicide formation (Ti, Co, Ni)	SALICIDE	600–900 °C
Dielectric glass reflow (BPSG)	RTG	700–900 °C
Oxynitride gate dielectric	RTN	900–1000 °C

Table 3: RTP advantages and limitations compared to conventional batch furnace.

	RTP (single wafer)	Batch Furnace
Thermal budget	Very low (seconds)	High (minutes–hours)
Dopant diffusion	Minimal	Significant
Junction control	Excellent (shallow)	Poor for <100 nm junctions
Wafer uniformity	Challenging at fast ramps	Excellent
Throughput	Low (1 wafer at a time)	High (25–150 wafers/run)
Temperature range	Up to ∼1200 °C	Up to ∼1200 °C

Part B: Photolithography

Introduction

After the RTP anneal, the wafer was transferred to the **photolithography room** (illuminated with yellow/amber light to prevent accidental UV exposure of the photoresist). Photolithography uses a light-sensitive polymer film (photoresist) coated on the wafer and selectively exposed through a patterned mask or by a direct-write laser, to transfer circuit features onto the wafer surface.

The complete Day 4 photolithography sequence is shown in Figure 6.

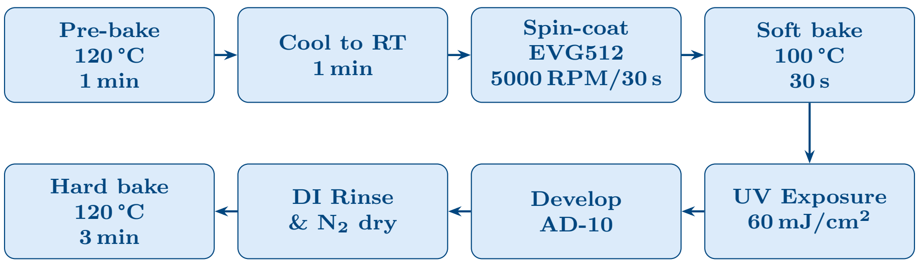


Figure 6: Complete photolithography sequence performed on Day 4.

Step 1: Pre-Bake (Dehydration Bake)

Before applying photoresist, the wafer was baked on a hotplate at **120 °C for 1 minute** to evaporate surface moisture. Adsorbed water molecules on the silicon surface would otherwise prevent adhesion of the photoresist. After the pre-bake the wafer was allowed to cool at **room temperature for 1 minute** before spin-coating, to prevent thermal shock to the resist.

Step 2: Photoresist Spin-Coating

3.3.1 Photoresist: EVG512 (Positive-Tone PR)

The photoresist used was **EVG512**, a positive-tone photoresist in which UV-exposed regions become soluble in the developer. The material is composed of three components:

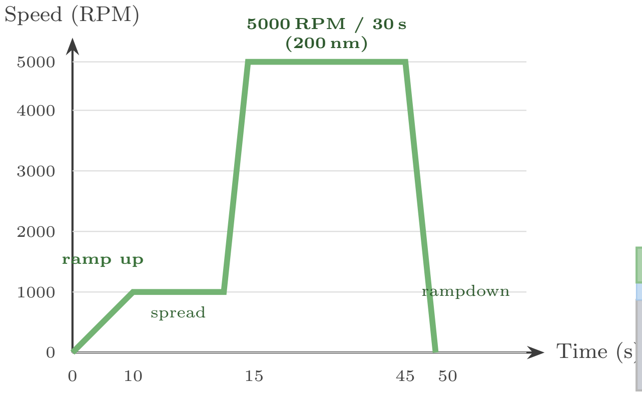
- **Polymer (Novolac resin):** base matrix providing film cohesion and mechanical stability.
- **Photoactive compound (PAC — diazonaphthoquinone, DNQ):** absorbs UV and undergoes photolysis on exposure, converting to an alkali-soluble ketene product (Wolff rearrangement).
- **Casting solvent:** controls viscosity during spin-coat and evaporates completely during the soft bake.

3.3.2 Spin-Coat Recipe

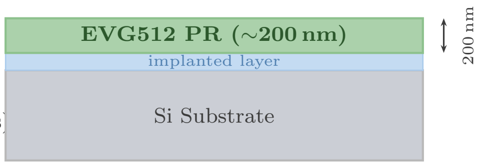
Spin-Coat Recipe — EVG512 PR, 200 nm target

Step 1 – Ramp up:	$0 \rightarrow 1000$ RPM in 10 seconds (resist spreads)
Step 2 – Final speed:	5000 RPM for 30 seconds (thickness control)
Step 3 – Ramp down:	Gradual deceleration to stop
Target thickness:	∼200 nm uniform coating
Thickness relation:	$t \propto 1/\sqrt{\omega}$ (higher RPM = thinner film)

The slow initial ramp prevents the resist bead from flying off the wafer edge before it has spread. The 5000 RPM final speed produces the 200 nm target thickness uniformly across the 6-inch wafer surface.



(a) Spin-coat RPM profile.



(b) Cross-section: 200 nm EVG512 on wafer.

Figure 7: Spin-coat RPM profile (left) and resulting photoresist cross-section (right).

3.3.3 Spin Coater Machine — Lab Photograph

Figure 8: EVG spin coater machine used for photoresist deposition (lab photograph).

Step 3: Soft Bake

After spin-coating, the wafer was baked on a hotplate at **100 °C for 30 seconds**. This drives off the residual casting solvent from the PR film, improves adhesion to the wafer surface, and stabilises the film without damaging the photosensitive PAC compound.

Step 4: UV Exposure (Laser Writer)

Pattern transfer was performed using a **UV laser writer** at an exposure dose of **60 mJ/cm^2**. The UV light causes photolysis of the DNQ photoactive compound in the positive resist at the exposed positions only, converting the insoluble DNQ to a soluble indene-carboxylic acid via the Wolff rearrangement. Unexposed areas retain the insoluble DNQ and remain as the mask.

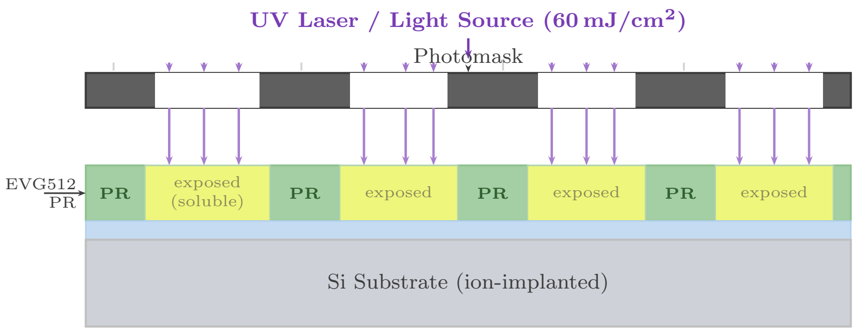


Figure 9: UV exposure cross-section: UV passes through mask openings and breaks PAC bonds in EVG512 (positive PR) at the exposed positions. Dose: 60 mJ/cm^2.

3.5.1 Laser Writer / Mask Aligner — Lab Photograph

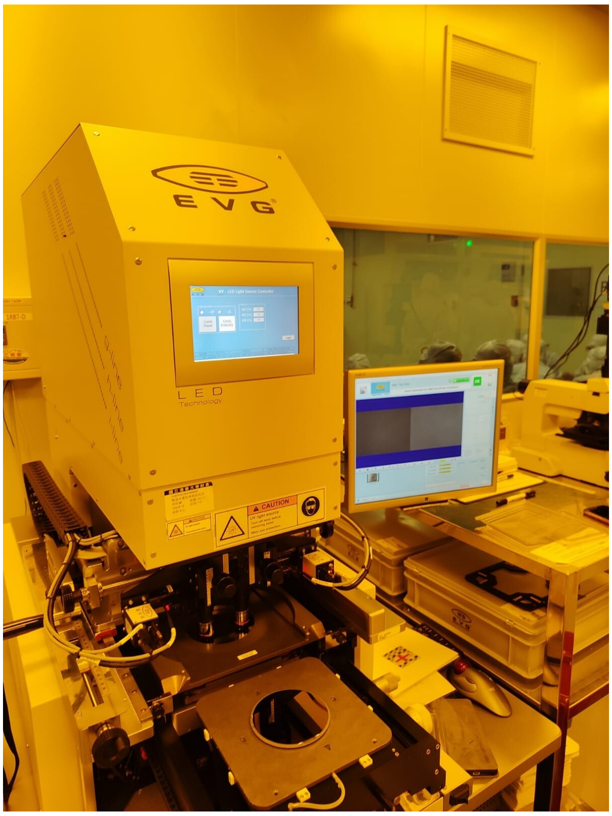


Figure 10: UV laser writer used for photoresist exposure and pattern definition (lab photograph).

Step 5: Development (AD-10 Developer)

After UV exposure, the wafer was immersed in **AD-10 developer solution**, an aqueous alkaline (TMAH-based) developer. The UV-exposed positive-resist regions, now converted to the soluble ketene/indene-carboxylic acid form, dissolve rapidly in the alkaline solution. Unexposed regions remain insoluble and intact on the surface.

Development Parameters

Developer:	AD-10 (aqueous alkaline, TMAH-type)
Method:	Immersion of wafer in developer solution
Result:	UV-exposed PR dissolves; unexposed PR remains as etch mask
Rinse:	DI water overflow rinse (stops development)
Dry:	N_2 blow-gun

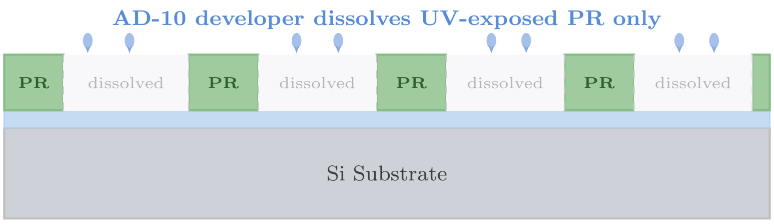


Figure 11: After AD-10 development: exposed PR removed, unexposed PR pattern remains as protective mask for subsequent etch or lift-off.

3.6.1 Developer Solution — Lab Photograph

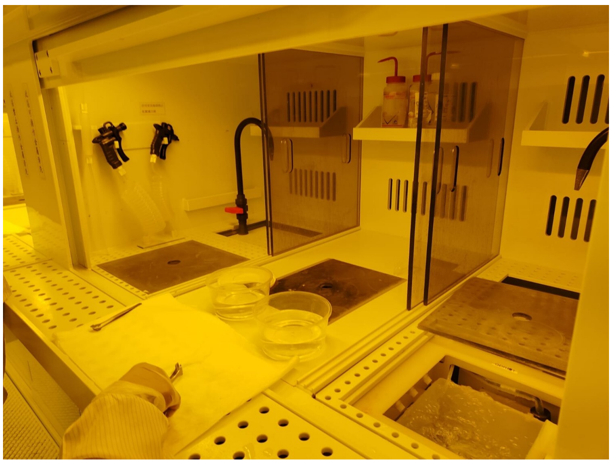


Figure 12: AD-10 developer solution and wet bench (lab photograph).

Step 6: DI Water Rinse and N_2 Dry

Immediately after development, the wafer was thoroughly rinsed with **deionised water** to stop the development reaction completely and wash away all residual AD-10. Residual developer left on the surface would continue to attack the PR, causing over-development and loss of pattern fidelity. The wafer was then dried using a **nitrogen (N_2) blow-gun**, which removes water droplets without introducing ionic contamination.

Step 7: Hard Bake

The final lithography step was a **hard bake at 120 °C for 3 minutes**. This step:

- Removes all remaining solvent from the photoresist.
- Thermally crosslinks and hardens the resist for improved etch resistance.

- Improves PR adhesion to the underlying layer.
- Slightly rounds the PR sidewall profile, which can improve etch uniformity.

After the hard bake, the patterned photoresist is ready to serve as the etch/lift-off mask for the next fabrication step.

Results and Observations

RTP Observations

- Wafer reached 1000 °C in approximately 5 seconds; fast ramp confirmed by pyrometer display.
- 30-second dwell completed with stable temperature; lamp power modulated by pyrometer feedback loop.
- Post-anneal wafer surface appeared clean and unoxidised (N_2/Ar ambient prevented oxidation).
- SiC holder showed no degradation; no CO_2 evolution observed.

Photolithography Observations

- EVG512 PR appeared uniform in colour after spin-coating (consistent thickness).
- After UV exposure and AD-10 development, the designed pattern was clearly visible: exposed regions showed the underlying layer; unexposed PR remained intact.
- DI rinse confirmed developer removal; N_2 drying left no water marks.
- After hard bake, the PR pattern showed well-defined edges with no cracking or delamination.

Justification and Importance

1. **RTP is essential after ion implantation:** Implanted dopants are electrically inactive at interstitial sites; annealing at 1000 °C activates them. Without RTP, the device would show no useful electrical behaviour.
2. **Low thermal budget protects junction depth:** Short anneal time (30 s) limits diffusion length, preserving the shallow junction profile critical for short-channel transistors.
3. **Holder selection is safety-critical:** Graphite + O_2 produces CO_2, destroying the tool and contaminating the wafer. Correct holder choice is both a safety and process-quality requirement.
4. **Photolithography defines all device features:** Every transistor gate, contact, and metal line is defined by lithography. Resist thickness, exposure dose, and development time directly control feature dimensions and device yield.
5. **Correct bake temperatures matter:** Under-baking leaves solvent that causes bubbling during exposure; over-baking destroys the photoactive compound. Precise temperature control at each bake step is essential.

Conclusion

Day 4 of the NTHU cleanroom training programme successfully covered two critical fabrication steps:

- **RTP/RTA:** The ion-implanted wafer was annealed at 1000 °C for 30 seconds using a SiC susceptor in an N_2/Ar ambient, monitored by pyrometer. Dopants were electrically activated and crystal lattice damage was repaired while preserving the shallow junction depth. The critical importance of holder selection (graphite below 500 °C with no O_2; SiC for all temperatures and gas ambients) was understood and applied.
- **Photolithography:** A complete lithography cycle was executed using EVG512 positive photoresist: pre-bake (120 °C/1 min), spin-coat at 5000 RPM for 200 nm thickness, soft bake (100 °C/30 s), UV exposure at 60 mJ/cm^2, development in AD-10, DI rinse, and hard bake (120 °C/3 min). The resulting patterned resist provides the template for the subsequent etch or lift-off steps.

Together, these two steps advance the wafer from a post-implant substrate to a fully patterned, process-ready device platform.

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Overview of Day 5: Al Wet Etching and Photoresist Stripping

Day 5 of the cleanroom training programme focused on **aluminium (Al) wet chemical etching** using a three-component acid etchant, followed by **photoresist stripping** to reveal the final patterned metal features on the silicon wafer. This session represents a pivotal step in the device fabrication sequence: it is the point at which the Al metal layer is selectively removed from designated areas to define the interconnect pattern, leaving well-defined Al lines or contacts on the Si substrate.

The Day 5 process sits within a larger, previously established fabrication context, as described in the following section.

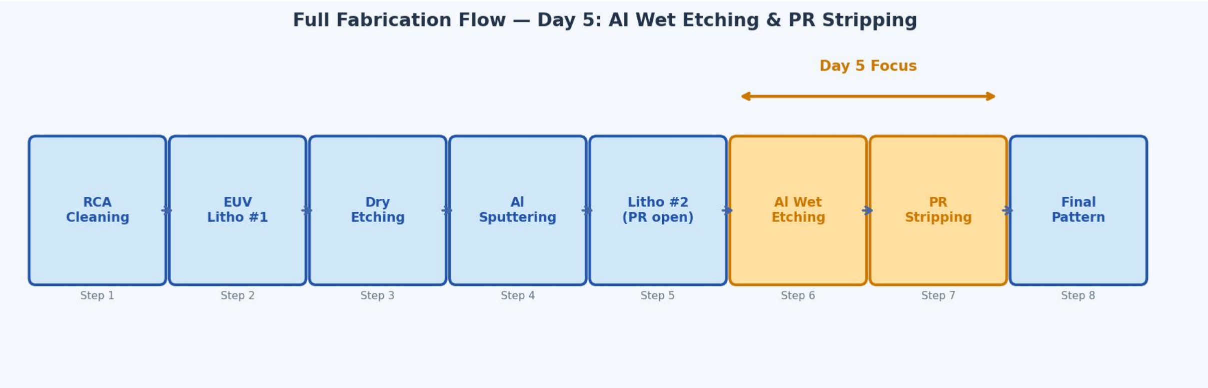

Figure 1: Full 8-step fabrication flow. Steps 6 (Al wet etching) and 7 (PR stripping) are highlighted in orange as the Day 5 focus.

Context: Steps Performed Before Day 5

Before the wet etching performed on Day 5, the following sequential processes had already been completed on the wafer:

1. **RCA cleaning:** The silicon wafer was cleaned using the standard RCA protocol (SC-1 followed by SC-2) to remove organic contamination, particles, and ionic metallic impurities from the surface.
2. **EUV photolithography (Mask 1):** A first photoresist layer was deposited and patterned using extreme ultraviolet (EUV) lithography to define the regions of the substrate that would be permanently etched.
3. **Dry etching:** The exposed regions of the silicon substrate were etched by a dry (plasma) etch process, leaving a permanent topographic feature (trenches or recesses) in the Si surface at the Mask 1 positions.
4. **Al deposition by DC sputtering:** A 300 nm aluminium film was deposited uniformly across the entire wafer surface, including over the etched features, by DC sputtering (as covered in Day 3).
5. **Photolithography (Mask 2):** A second photoresist layer was applied and patterned using a second photomask. This second patterning step opened windows in the photoresist *precisely over the regions where Al was to be removed* (i.e., the previously dry-etched Si regions). The remaining PR acts as a protective mask covering the Al lines to be retained.

After these five preparatory steps, the wafer presents the following cross-sectional structure going into Day 5:

- Si substrate with dry-etched recesses.
- 300 nm Al coating on top.
- Patterned PR mask: windows open over the Al areas to be removed, PR intact over the Al to be retained.

The Day 5 task was therefore to selectively remove the Al exposed in the PR windows by wet chemical etching.

Introduction to Wet Etching

What is Wet Etching?

Wet etching is a material removal process in which a substrate is immersed in a liquid chemical solution (the etchant) that selectively dissolves the target material through chemical reactions. It is one of the oldest and most widely used techniques in microfabrication, valued for its simplicity, high throughput, and high etch rates.

In semiconductor processing, wet etching is used to remove specific thin films — metals, oxides, nitrides — from defined areas of the wafer, using a photoresist (PR) or hard mask to protect regions that should be kept intact.

Key Characteristics of Wet Etching

Etch mechanism:	Chemical reaction between etchant and material
Selectivity:	High — etchant chosen to attack only the target material
Directionality:	Isotropic (etches equally in all directions)
Mask material:	Photoresist or SiO_2 (must be resistant to etchant)
By-products:	Soluble salts + gases (e.g. NO_x bubbles)
End-point:	Timed etch (rate known), or visual inspection

Isotropic Nature of Wet Etching

A critical property of wet etching is that it is **isotropic**: the chemical reaction proceeds at the same rate in all directions — vertically into the film and laterally beneath the mask edge. This causes **undercutting** of the photoresist mask, as illustrated in Figure 2. For a 300 nm-thick Al film, the lateral undercut is approximately equal to the film thickness (1:1 undercut ratio for isotropic wet etching), meaning features must be drawn slightly oversized to account for this.

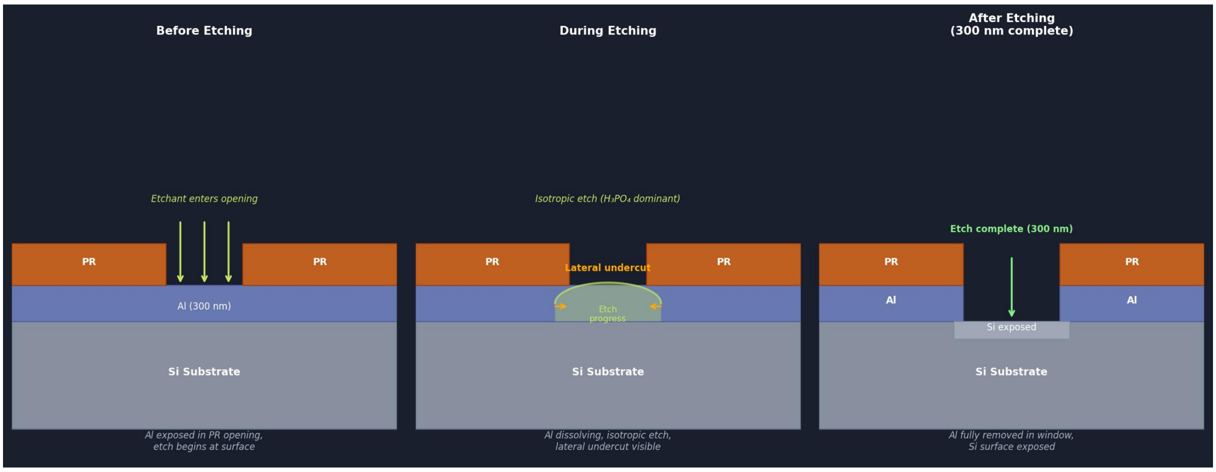

Figure 2: Cross-sectional progression of isotropic Al wet etching. Left: etchant enters the PR opening and attacks the Al surface. Centre: Al dissolves isotropically, creating lateral undercut beneath the PR edges. Right: 300 nm etch complete, Si surface fully exposed in the window.

Comparison: Wet Etching vs. Dry Etching

Table 1: Wet etching compared to dry (plasma) etching for metal patterning.

Property	Wet Etching	Dry Etching (RIE/Plasma)
Mechanism	Chemical (liquid phase)	Physical + chemical (plasma)
Directionality	Isotropic (undercut)	Anisotropic (vertical walls)
Selectivity	Very high	Moderate
Etch rate	Fast (nm/s range)	Slower but controllable
Equipment cost	Low	High
Feature size	>1 µm (undercut limits)	Sub-100 nm capable
By-products	Liquid waste, NO_x gas	Gas-phase (easier disposal)
Uniformity	Dependent on agitation	Excellent (plasma uniformity)

The Aluminium Wet Etchant: Composition and Chemistry

Etchant Composition

The aluminium wet etchant used on Day 5 is a three-component mixture known as **phosphoric–nitric–acetic acid (PAN etch)**. The composition used was:

Al PAN Etchant — Recipe Used on Day 5

Component	Concentration	Role
Phosphoric acid (H_3PO_4)	70–80%	Primary etchant
Nitric acid (HNO_3)	1–3%	Oxidiser
Acetic acid (CH_3COOH)	1–12%	Surfactant / wetting agent
Deionised water (H_2O)	Balance	Dilutent / reaction medium

Role of Each Component

Each acid plays a distinct and necessary role in the etching process:

Phosphoric Acid (H_3PO_4) — 70–80%

Phosphoric acid is the **primary etchant**. It dissolves the aluminium oxide (Al_2O_3) layer that forms on the Al surface through the following reaction:

$$Al_2O_3 + 2\,H_3PO_4 \longrightarrow 2\,AlPO_4 + 3\,H_2O$$

$AlPO_4$ is soluble in the aqueous solution, so it is carried away from the surface. The high concentration of H_3PO_4 ensures a sufficient supply of acid to sustain the dissolution reaction throughout the etch.

Nitric Acid (HNO_3) — 1–3%

Nitric acid is the **oxidiser**. Aluminium metal (Al^0) is not directly soluble in phosphoric acid; it must first be oxidised to Al_2O_3. HNO_3 performs this oxidation:

$$Al + 3\,HNO_3(\text{dilute}) \longrightarrow Al(NO_3)_3 + \tfrac{3}{2}\,H_2\uparrow$$

The freshly oxidised surface is then immediately dissolved by H_3PO_4. The combination of HNO_3 (oxidation) and H_3PO_4 (dissolution) forms a self-sustaining etch cycle. The concentration of HNO_3 is kept low (1–3%) to avoid excessive gas evolution and maintain a controlled etch rate. The NO_x gas produced is responsible for the bubbles observed rising from the wafer surface during etching.

Acetic Acid (CH_3COOH) — 1–12%

Acetic acid acts as a **surfactant and wetting agent**. It:

- Reduces the surface tension of the etchant solution, improving contact between the liquid and the Al surface in the PR window openings.
- Helps to produce a smooth, uniform etch front.
- Moderates the etch rate to prevent too-rapid etching that would cause non-uniform results across the wafer.
- Prevents the formation of etch residues (Al hydroxide) that could re-deposit on the surface.

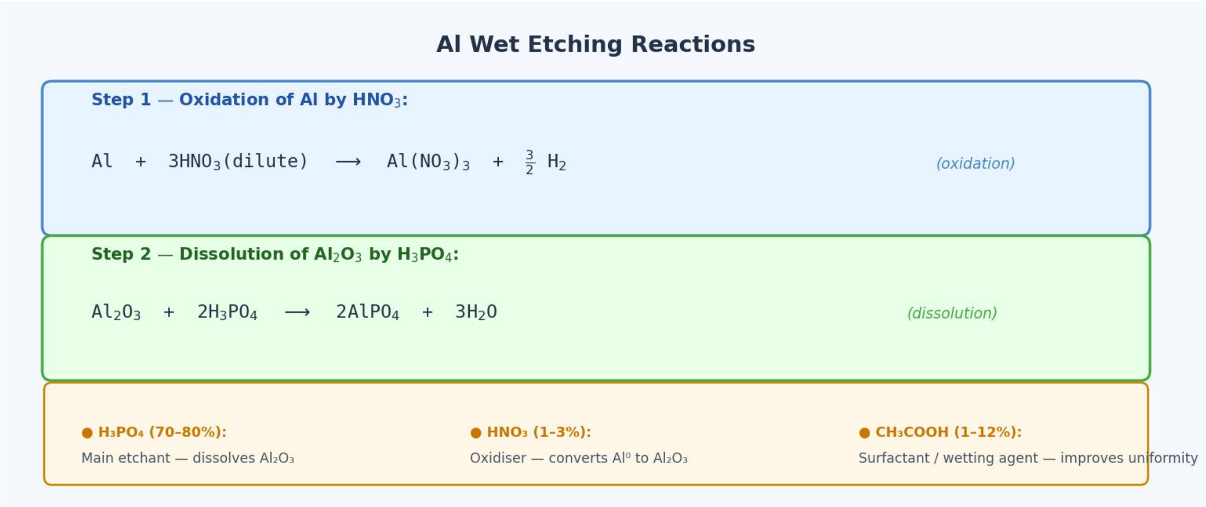

Figure 3: Summary of the two-step Al etching reactions and the role of each etchant component. HNO_3 oxidises Al metal to Al_2O_3; H_3PO_4 dissolves the oxide; CH_3COOH improves etch uniformity.

Wet Etching Process: Step-by-Step

Process Parameters

Day 5 Al Wet Etching Recipe

Etchant:	H_3PO_4 (70–80%) + HNO_3 (1–3%) + CH_3COOH (1–12%)
Temperature:	Room temperature (∼25 °C) or slightly elevated
Etch time:	1 minute
Etch depth:	300 nm (full Al film thickness)
Method:	Immersion (wafer dipped in etchant beaker)
Agitation:	Gentle (NO_x gas bubbling provides natural agitation)
End-point:	Timed (rate pre-characterised); visual: bubbles cease when Al gone
Rinse:	DI water overflow rinse immediately after etch
Dry:	N_2 blow-dry

Step-by-Step Procedure

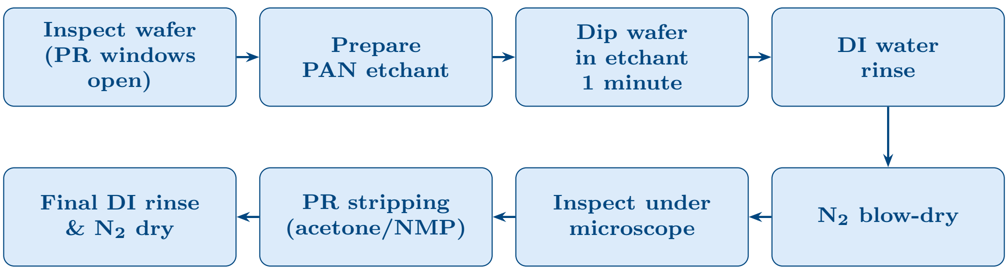

Figure 4: Day 5 wet etching and PR stripping process flow.

Step 1. Pre-etch inspection. Before etching, the wafer was inspected under an optical microscope to confirm that the Mask 2 photolithography had been successful: the PR windows must be open and correctly aligned over the Al regions to be removed. Any misalignment or incomplete development at this stage would cause pattern defects in the final device.

Step 2. Prepare the PAN etchant. The three-component etchant (H_3PO_4, HNO_3, CH_3COOH) is mixed in the correct proportions in a chemical-resistant beaker inside the wet bench fume hood. All PPE (face shield, acid-resistant gloves, cleanroom suit) must be worn.

Step 3. Immerse wafer in etchant (1 minute). The wafer is lowered into the PAN etchant solution using Teflon tweezers or a wafer carrier and held for exactly **1 minute**. Upon immersion, the HNO_3 immediately begins oxidising the exposed Al, and H_3PO_4 dissolves the resulting oxide. Visible **NO_x gas bubbles** form at the Al surface and rise through the solution — these bubbles indicate active etching.

The etch is **isotropic**: the Al is removed at the same rate in both the vertical direction (into the film) and the lateral direction (beneath the PR mask edges). For the 300 nm target depth, some lateral undercut occurs at the mask edges.

Step 4. DI water rinse. Immediately after 1 minute, the wafer is transferred to the DI water rinse tank to stop the etch reaction. The rinse must be performed promptly because continued immersion beyond the target time will over-etch the Al (excessive undercut or punch-through). The DI water dilutes and displaces all etchant from the wafer surface.

Step 5. N_2 blow-dry. The rinsed wafer is dried with a nitrogen gun to prevent water spots or contamination from the rinse water.

Step 6. Post-etch inspection. The wafer is inspected under the optical microscope to verify that the Al has been fully removed in the etched regions. In the correctly etched areas, the underlying Si surface (grey-silver) should be visible, while Al (slightly darker metallic) remains in the protected regions.

Wet Etching Observation — Lab Photograph

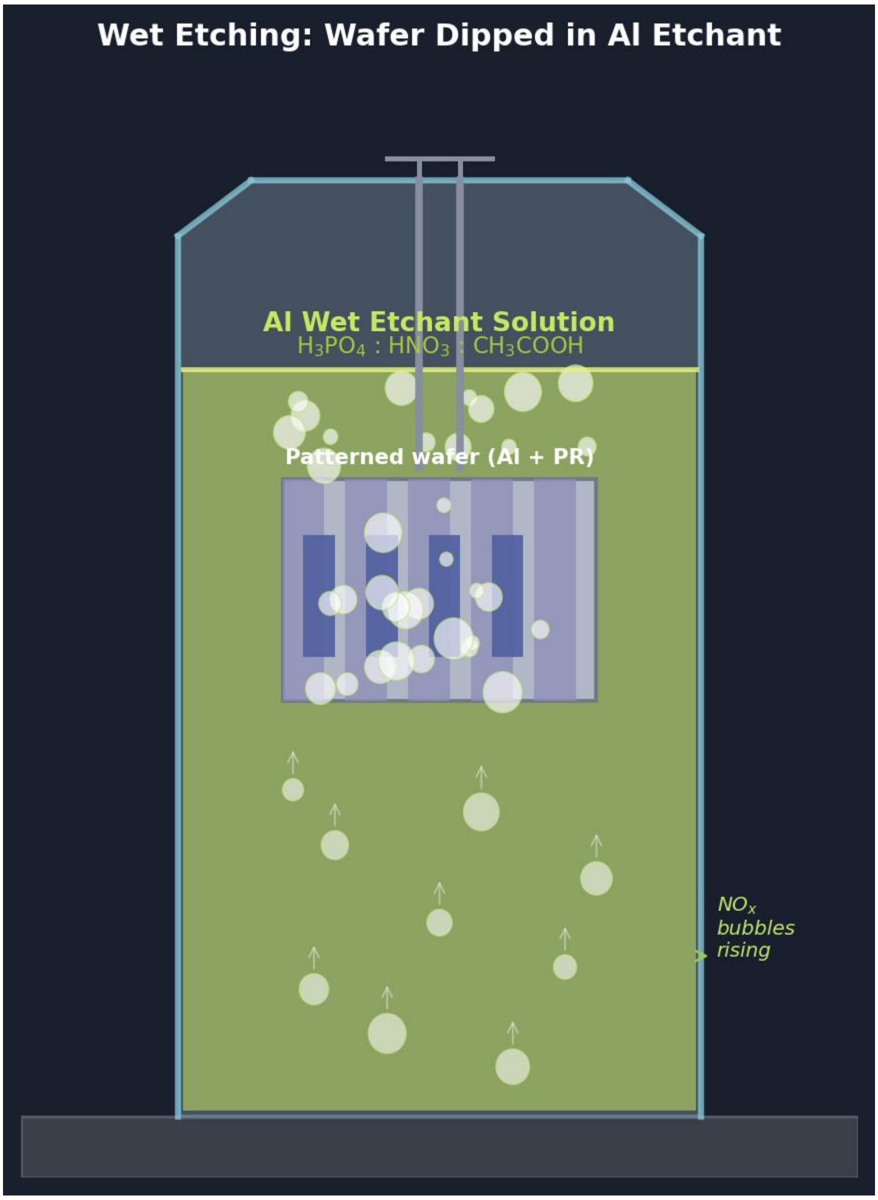

Figure 5: Wet etching in progress: the patterned wafer is dipped in the PAN Al etchant (yellow-green solution). NO_x bubbles are visible rising from the wafer surface through the liquid, indicating active aluminium dissolution at the exposed Al regions. This gas evolution confirms the HNO_3 oxidation reaction is proceeding.

Etch Rate and Depth Control

Etch Rate Characterisation

The etch rate of the PAN etchant on aluminium depends on:

- Etchant temperature (higher temperature = faster etch)
- Exact acid concentrations and freshness of the solution
- Degree of agitation (the NO_x bubbles provide natural convection)

For the recipe used on Day 5 (H_3PO_4:HNO_3: CH_3COOH at room temperature), a typical Al etch rate is approximately 300 nm/min, which conveniently matches the 300 nm target depth at the **1 minute** etch time specified.

Etch Depth Calculation

The relationship between etch time and depth is:

$$\text{Etch depth} = \text{Etch rate} \times \text{Time}$$
$$300\,\text{nm} = 300\,\frac{\text{nm}}{\text{min}} \times 1\,\text{min}$$

The etch is designed to be a **full film etch**: the entire 300 nm Al layer is removed in the PR window areas, stopping on the underlying Si surface. Because the Si substrate is essentially chemically inert to the PAN etchant (which is selective for Al), minor over-etch time does not damage the substrate, giving a small process window.

Etch Selectivity

The PAN etchant has excellent selectivity for Al over Si and SiO_2:

- **Al etch rate:** ~300 nm/min
- **Si etch rate:** Negligible (<1 nm/min)
- **SiO_2 etch rate:** Very low
- **PR etch rate:** Very low (PR resistant to dilute phosphoric acid)

This high selectivity ensures the photoresist mask remains intact and the Si substrate is not attacked during the 1-minute etch.

Safety Note — PAN Etchant Handling: H_3PO_4 and HNO_3 are corrosive acids. HNO_3 is additionally an oxidiser. The NO_x gas evolved during etching is toxic — all operations must be performed inside the exhaust wet bench fume hood. Face shield, acid-resistant gloves, and cleanroom suit are mandatory. Spent etchant must be collected in labelled acid waste containers.

Photoresist Stripping

Purpose of PR Stripping

After the wet Al etch is complete, the photoresist mask that protected the Al lines during etching is no longer needed. If left on the device, the PR would interfere with subsequent processing steps (electrical contacts, passivation, characterisation). **PR stripping** removes this remaining photoresist completely from the wafer surface, leaving the final device pattern: bare Al metal lines and exposed Si in the etched windows.

Stripping Methods

Photoresist can be removed by two main approaches:

1. **Wet chemical stripping:** Immersion in an organic solvent (acetone, NMP — N-methyl-2-pyrrolidone, or commercial PR stripper such as Remover PG or Microposit 1165). The solvent swells and dissolves the PR polymer. This is the most common method for positive-tone resists and was used on Day 5.

2. **Dry plasma ashing:** O_2 plasma oxidises the organic PR to CO_2 and H_2O gas, which are pumped away. Used when metal contamination from solvents must be avoided, or for fully crosslinked hard-baked resists.

PR Stripping Procedure (Day 5)

Step 1. Prepare stripper bath. Acetone or a commercial alkaline PR stripper is prepared in a clean chemical beaker. The stripper must not attack Al — acetone is safe for Al; strong alkaline strippers (such as TMAH-based developers) may etch Al and must be avoided.

Step 2. Immerse wafer. The wafer is immersed in the stripper solution. The PR begins to swell and peel from the surface within seconds. Gentle agitation (or sonication) helps dislodge loosened PR fragments.

Step 3. Visual confirmation. Complete stripping is indicated when no coloured or opaque residue is visible on the wafer surface under yellow cleanroom lighting.

Step 4. DI water rinse. The wafer is transferred to a DI water rinse to remove all traces of stripper and dissolved PR.

Step 5. N_2 dry. The wafer is dried with a nitrogen gun.

PR Stripping Observation

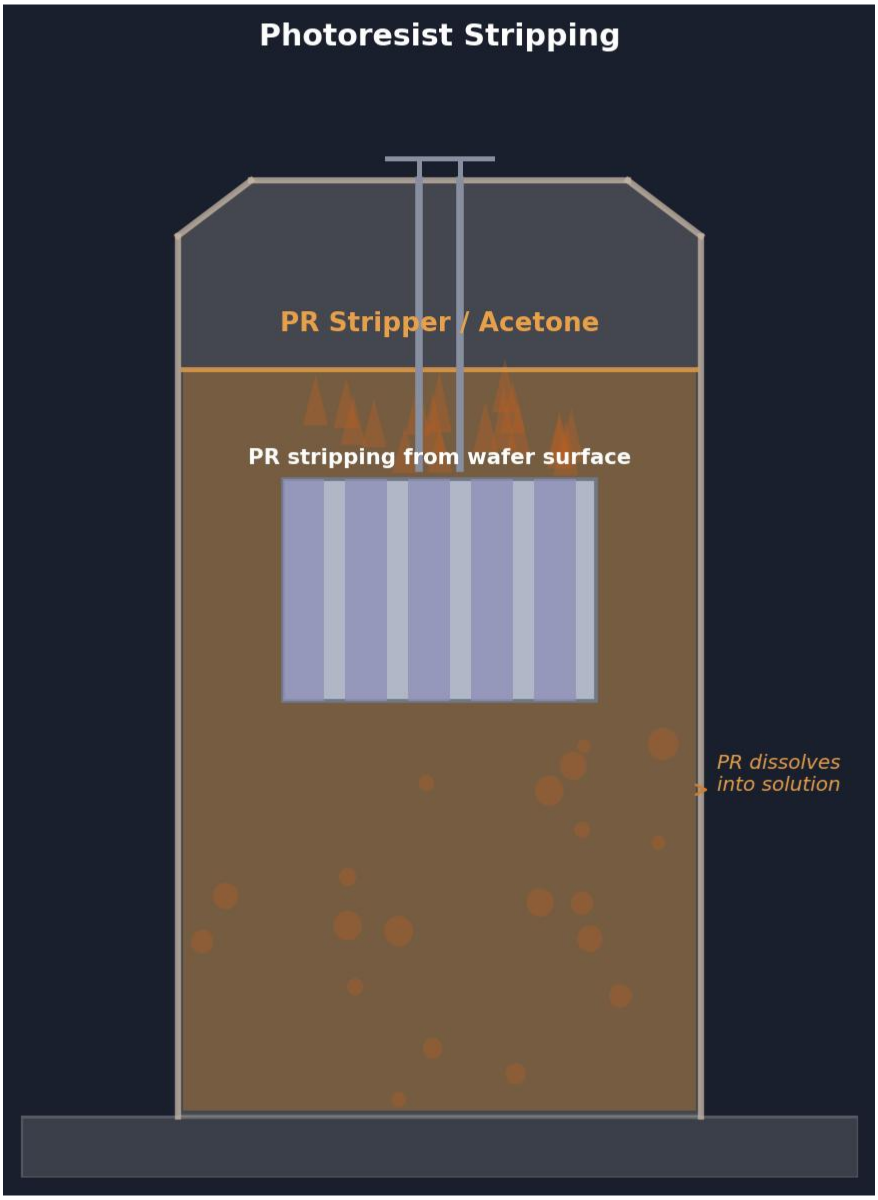

Figure 6: Photoresist stripping in progress. The wafer is immersed in the PR stripper (orange-brown solution). Dissolved photoresist material is visible as wisps and particulates separating from the wafer surface and dispersing into the stripper solution, confirming active PR removal.

Final Patterned Wafer: Results and Analysis

Expected Final Structure

After Al wet etching (1 min, 300 nm) and PR stripping, the wafer presents the following final cross-sectional structure:

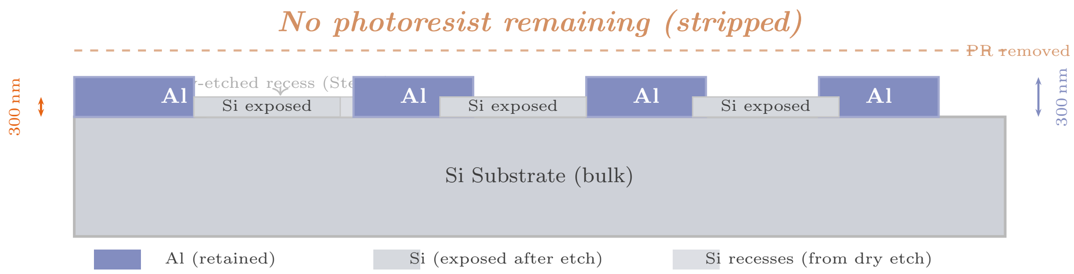

Figure 7: Final cross-sectional structure after Al wet etching (300 nm depth, 1 min) and PR stripping. Al metal lines remain on the substrate surface; Si is exposed in the etched windows; the earlier dry-etched recesses are visible in the Si substrate. No photoresist remains on the device.

Final Patterned Wafer — Photograph and Top View

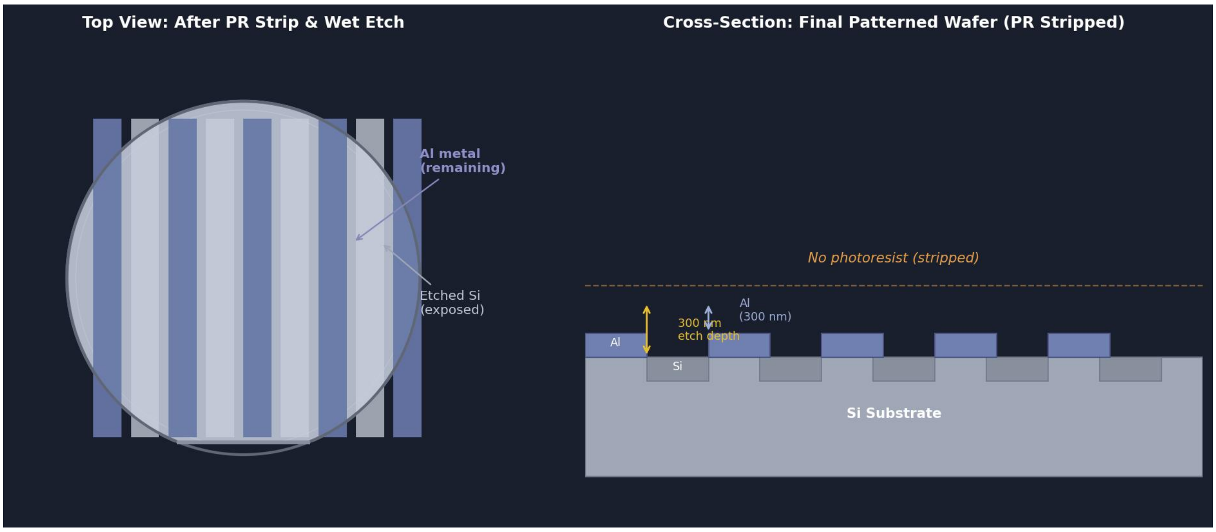

Figure 8: Left: Top-view illustration of the patterned wafer after Al wet etching and PR stripping. Dark blue-grey stripes are the retained Al metal lines; lighter grey stripes are the exposed Si surface where Al was etched away. Right: Cross-section confirming the 300 nm Al thickness and the Si exposed in the etched windows with no remaining photoresist.

Observations

- **During etching:** NO_x gas bubbles were clearly visible rising from the wafer surface in the PR openings, confirming active Al oxidation by HNO_3. Bubble generation ceased in the etched areas after approximately 1 minute, indicating complete Al removal.
- **After etch, before PR strip:** Optical microscope inspection showed the Al present in PR-protected areas and the Si substrate visible in the etched windows. Slight lateral undercutting (≈300 nm) at PR edges was observed, consistent with isotropic wet etching.
- **After PR strip:** The wafer surface was clean, with well-defined Al lines and smooth, uncontaminated Si in the etched regions. No PR residue or etch staining was observed.

- **Film integrity:** The retained Al lines showed a reflective metallic surface with no pitting, confirming the etch had not attacked the protected Al.

Justification and Importance of Al Wet Etching

1. **Metal patterning defines device interconnects:** Wet etching of Al is the process by which a blanket metal film is converted into the conductive tracks and contact pads of the device. The precision and selectivity of the etch directly determines whether the device will be electrically functional.
2. **High selectivity protects the substrate:** The PAN etchant attacks Al orders of magnitude faster than Si or SiO_2, which means the etch naturally stops once the Al is gone, without damaging the underlying layers. This built-in selectivity provides a robust process with a wide process window.
3. **Etch rate control is critical:** At 300 nm/min, even a 30-second over-etch would cause significant additional lateral undercut ($\sim$150 nm) that could disconnect narrow Al lines. Precise timing and consistent etchant concentration are therefore essential.
4. **NO_x evolution as process indicator:** The visible bubble generation during etching provides a qualitative real-time indicator of etch progress. The cessation of bubbling in a specific region signals that all the Al has been removed there, giving the operator a visual endpoint signal to supplement the timed etch.
5. **PR stripping must not attack Al:** The choice of PR stripper is critical. Strong alkali strippers (e.g., KOH or TMAH solutions) will etch Al and must be avoided. Acetone or neutral organic solvents are compatible with Al and were therefore used. Incomplete stripping leaves a dielectric PR film on the device surface, which would prevent electrical contact at subsequent testing.
6. **Sequential mask logic:** The two-mask approach (dry etch pattern with Mask 1, then Al deposition followed by wet etch with Mask 2) allows the Si recesses and the metal contacts to be independently defined at different length scales, demonstrating the power of multi-step photolithographic patterning in semiconductor device fabrication.

Results Summary

Table 2: Summary of Day 5 process results.

Process Step	Parameter	Target	Result
Al wet etch	Etch time	1 min	1 min (confirmed)
	Etch depth	300 nm	300 nm (full film)
	Etchant	PAN	H_3PO_4:HNO_3:CH_3COOH
	Bubble generation	YES (active)	Observed
	Si surface exposed	YES	Confirmed by inspection
PR stripping	Residue	None	Clean surface confirmed
	Al integrity	Intact	No attack on Al lines
	Method	Wet (solvent)	Acetone/PR stripper

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Overview of Day 6: Plasma Enhanced Chemical Vapour Deposition (PECVD)

Day 6 of the cleanroom training programme was dedicated to **Plasma Enhanced Chemical Vapour Deposition (PECVD)**, a thin-film deposition technique used to grow dielectric layers — primarily silicon dioxide (SiO_2) and silicon nitride (Si_3N_4) — at significantly lower temperatures than conventional thermal CVD, by using an RF-generated plasma to drive the chemical reactions.

On Day 6, a **100 nm SiO_2** film was deposited on a silicon wafer using silane (SiH_4) and nitrous oxide (N_2O) as process gases, with the substrate held at 300 °C.

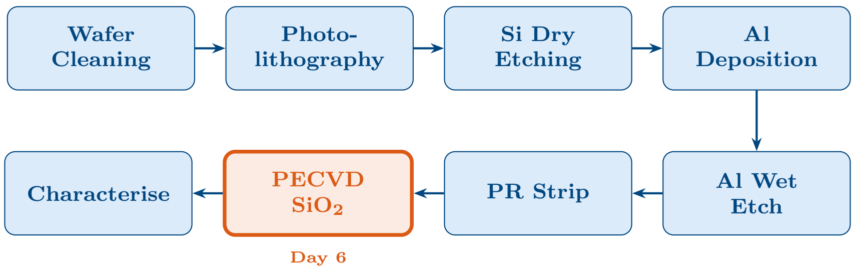


Figure 1: Fabrication flow: PECVD SiO_2 deposition (orange, Day 6) deposits a dielectric passivation or interlayer dielectric film on the processed wafer.

Introduction to PECVD

What is PECVD?

Chemical Vapour Deposition (CVD) is a process in which gaseous precursors react chemically at or near a heated substrate surface to deposit a solid thin film. In **conventional thermal CVD**, the substrate must be heated to 700–900 °C to provide the activation energy for the chemical reaction. This high temperature is incompatible with devices that already contain aluminium metal interconnects (Al melts at ∼660 °C) or temperature-sensitive polymer layers.

PECVD solves this problem by using a high-frequency RF plasma (13.56 MHz) to generate the reactive species (radicals and ions) needed to drive the deposition reaction, *without* relying on substrate heat for activation energy. This allows film deposition at temperatures as low as 200–400 °C while still achieving high-quality, dense dielectric films.

PECVD Key Advantages

Low temperature:	Deposits at ∼300 °C vs. ∼800 °C for thermal CVD
Plasma activation:	RF plasma provides reaction energy; no need for extreme heat
Fast deposition:	∼100 nm deposited in only 1 min 40 s
Wide material range:	SiO_2, Si_3N_4, a-Si, SiN_x
CMOS compatible:	Safe for Al and other metal layers already on the wafer

Where PECVD Fits in the Fabrication Flow

PECVD is used at stages where a dielectric film must be deposited after temperature-sensitive materials have already been processed. Typical applications in device fabrication include:

- **Interlayer dielectric (ILD):** SiO_2 between metal interconnect layers, preventing short circuits.
- **Passivation layer:** Final protective SiO_2 or Si_3N_4 film over the completed device.
- **Gate dielectric:** In certain thin-film transistor technologies.
- **Etch stop layer:** Si_3N_4 between oxide layers to act as a selective etch barrier.

Instrument Description: The PECVD System

Overview

The PECVD system is a parallel-plate capacitively coupled plasma (CCP) reactor. The two major components are a stainless-steel vacuum chamber and an RF power supply connected to the chamber electrodes. Unlike UHV systems (e-beam, 10^{-8} Torr), PECVD operates at a much higher working pressure ($\sim$68.5 Pa) using controlled gas flows through mass flow controllers (MFCs).

Figure 2: The PECVD system used on Day 6. Key visible elements include the main stainless-steel vacuum chamber (centre), gas supply lines with MFCs on the left, the RF power supply and control system on the right, and the vacuum pump connections at the base.

Key System Components

1. Vacuum Chamber

The central stainless-steel enclosure where deposition occurs. It operates at low pressure (not ultra-high vacuum like e-beam), typically $\sim$0.1 Pa base pressure rising to $\sim$68.5 Pa during deposition. The chamber contains two parallel plate electrodes: the showerhead (top, grounded) and the substrate holder (bottom, RF-biased).

2. RF Power System (13.56 MHz)

An RF generator supplies power at the industry-standard frequency of **13.56 MHz**. A minimum of **50 W** is required to ignite and sustain the plasma between the electrodes. The RF field accelerates electrons in the gas phase, causing ionisation and dissociation of the precursor molecules.

3. Gas Inlet System and Mass Flow Controllers (MFCs)

Process gases are supplied from cylinders through dedicated stainless-steel lines fitted with **mass flow controllers (MFCs)**, which regulate the flow rate with high precision. On Day 6, two gases were used:

- **SiH_4 (Silane):** ≈65 sccm — silicon source.
- **N_2O (Nitrous oxide):** ≈300 sccm — oxygen source.

4. Showerhead (Top Electrode)

A perforated metal plate at the top of the chamber that distributes process gases uniformly over the entire wafer surface. Uniform gas distribution is essential for achieving a uniform film thickness across the wafer. The showerhead also acts as the grounded electrode of the RF circuit.

5. Substrate Holder / Wafer Chuck (Bottom Electrode)

The wafer is placed on the bottom electrode, which is both **RF-biased** and **heated to 300 °C**. The resistive heating ensures sufficient surface mobility for the depositing species to form a dense, adherent film. The bottom electrode is also responsible for attracting ions from the plasma for physical bombardment of the growing film, which improves film density.

6. Plasma Region

The region between the two electrodes where the RF field ionises the gas. It contains a mixture of electrons, positive ions, and reactive neutral radicals. The visible *glow discharge* (purple-violet in appearance) is the visible signature of the plasma.

7. Vacuum Pump System

A combination of a roughing pump and a dry/turbomolecular pump maintains the chamber pressure. The pump also continuously removes gaseous by-products (H_2, N_2) during deposition to drive the reaction forward.

8. Control Panel

The front control panel displays and regulates: chamber pressure, RF power level, MFC gas flow rates, substrate temperature, and deposition time.

Figure 3: PECVD system control panel and gas delivery section. The MFC flow rates (SiH_4: 65 sccm, N_2O: 300 sccm), RF power (>50 W), chamber pressure, and deposition time are all monitored and controlled from this panel.

System Schematic

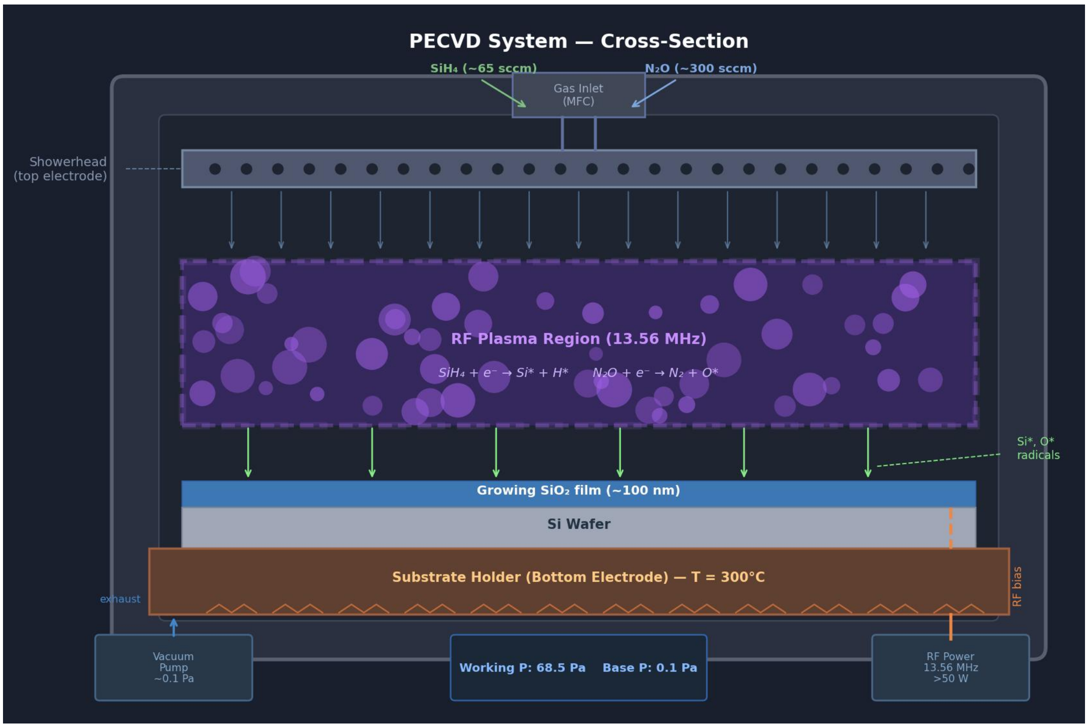


Figure 4: Detailed cross-sectional schematic of the PECVD system used on Day 6. Gas enters through the showerhead (top electrode), the RF plasma forms between the electrodes, reactive Si* and O* radicals migrate to the heated wafer surface and react to form SiO_2. Unreacted gas and by-products are removed by the vacuum pump.

Process Parameters

Day 6 PECVD Recipe — SiO_2 Deposition

Deposited film:	Silicon dioxide (SiO_2)
Precursor gas 1:	SiH_4 (Silane) — $\approx$65 sccm
Precursor gas 2:	N_2O (Nitrous oxide) — $\approx$300 sccm
RF frequency:	13.56 MHz
RF power:	>50 W (to generate and sustain plasma)
Substrate temperature:	$\sim$300 °C
Base pressure:	$\sim$0.1 Pa (before gas introduction)
Working pressure:	$\sim$68.5 Pa (after gas flow stabilised)
Target film thickness:	$\sim$100 nm
Deposition time:	$\sim$1 min 40 s ($\approx$100 s)
Deposition rate:	$\approx$60 nm/min (1 nm/s)

Figure 5: PECVD process parameter display showing temperature (300 °C), pressure readings, gas flow rates (SiH_4: 65 sccm, N_2O: 300 sccm), and RF power settings as set for the Day 6 SiO_2 deposition.

Working Principle of PECVD: Step-by-Step

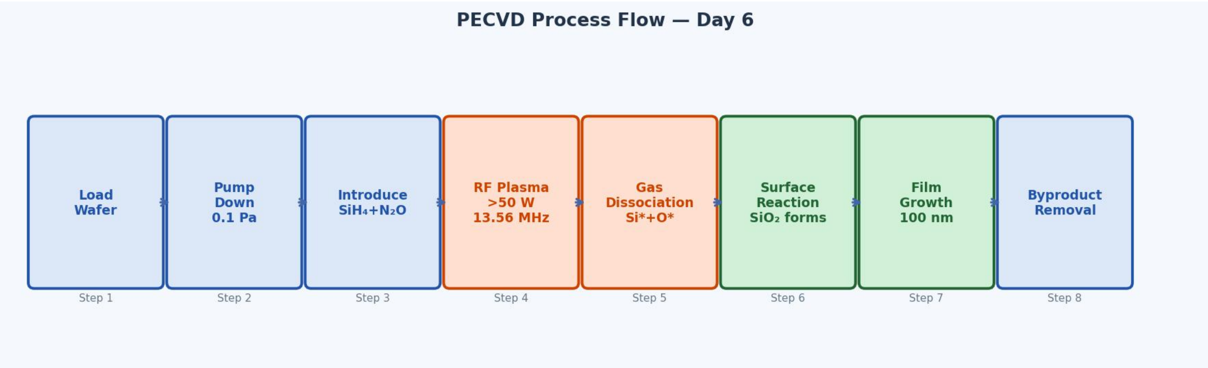


Figure 6: PECVD 8-step process flow from wafer loading through plasma generation, SiO_2 deposition, and by-product removal.

Step 1. Load Wafer. The chamber is opened and the silicon wafer is placed on the bottom electrode (substrate holder). The chamber is then closed and sealed.

Step 2. Pump Down to Base Pressure. The vacuum pump evacuates the chamber to a base pressure of $\approx$**0.1 Pa**. This removes atmospheric gases, moisture, and any residual contaminants that could incorporate into the growing film or interfere with the plasma chemistry. The substrate heater also ramps up to 300 °C during this step.

Step 3. Introduce Process Gases. SiH_4 (65 sccm) and N_2O (300 sccm) are admitted into the chamber through the MFCs and distributed uniformly through the showerhead. As the gases flow in, the chamber pressure rises to the working pressure of **68.5 Pa**. The N_2O:SiH_4 ratio ($\approx$4.6:1) is chosen to ensure a stoichiometric SiO_2 film rather than a silicon-rich sub-oxide.

Step 4. Plasma Generation. The RF generator applies >50 W at 13.56 MHz between the electrodes. The oscillating electric field rapidly accelerates electrons (which are very light and easily gain energy). These energetic electrons collide with gas molecules, causing:

- **Ionisation:** $SiH_4 \rightarrow {SiH_3}^+ + e^-$
- **Dissociation:** $SiH_4 + e^- \rightarrow Si^* + H^*$ (reactive radicals)
- **Excitation:** $N_2O + e^- \rightarrow N_2 + O^*$ (oxygen radicals)

A visible *glow discharge* (purple plasma) fills the inter-electrode space, confirming active plasma formation.

Step 5. Gas Dissociation into Reactive Radicals. The plasma continuously generates highly reactive neutral radicals — primarily Si* (silicon radicals from silane dissociation) and O* (oxygen radicals from nitrous oxide dissociation). These species have very short lifetimes and high chemical reactivity.

Step 6. Surface Reaction and Film Deposition. The reactive radicals diffuse from the plasma to the heated wafer surface, where the surface reaction occurs:

$$Si^* + 2\,O^* \longrightarrow SiO_2$$

The SiO_2 film nucleates on the wafer surface and grows continuously as more Si* and O* radicals arrive. The substrate temperature of 300 °C provides sufficient surface mobility for the depositing atoms to form a dense, pin-hole-free film rather than a porous, low-density microstructure.

Step 7. Thin Film Growth to Target Thickness. The deposition continues for **≈1 minute 40 seconds** (≈100 s), at which point ≈100 nm of SiO_2 has been deposited. The deposition rate under these conditions is approximately ∼1 nm/s (60 nm/min).

Step 8. By-product Removal. Gaseous by-products (H_2, N_2, and unreacted precursor fragments) are continuously removed by the vacuum pump during deposition, preventing film contamination. After deposition is complete, all gas flows are stopped, the RF is switched off, and the chamber is pumped down before the wafer is unloaded.

Plasma Chemistry and Deposition Mechanism

Plasma Dissociation Reactions

The RF plasma at 13.56 MHz dissociates the precursor gases into reactive species through electron-impact reactions:

$$\mathrm{SiH_4} + e^- \longrightarrow \mathrm{Si}^* + 4\,\mathrm{H}^* \tag{1}$$

$$\mathrm{N_2O} + e^- \longrightarrow \mathrm{N_2} + \mathrm{O}^* \tag{2}$$

The asterisk (*) denotes a reactive radical species — an atom or molecule with an unpaired electron and high chemical reactivity. Because radicals do not require further thermal activation to react, the film can grow at the relatively low substrate temperature of 300 °C.

Film Formation Reaction

At the heated substrate surface, the Si* and O* radicals undergo recombination:

$$\mathrm{Si}^* + 2\,\mathrm{O}^* \longrightarrow \mathrm{SiO_2} \tag{3}$$

The resulting SiO_2 is a stable, electrically insulating dielectric with excellent properties as an interlayer dielectric or passivation layer. By-products (H_2, N_2) are gases that are pumped out of the chamber continuously.

Atomic-Level Deposition Mechanism

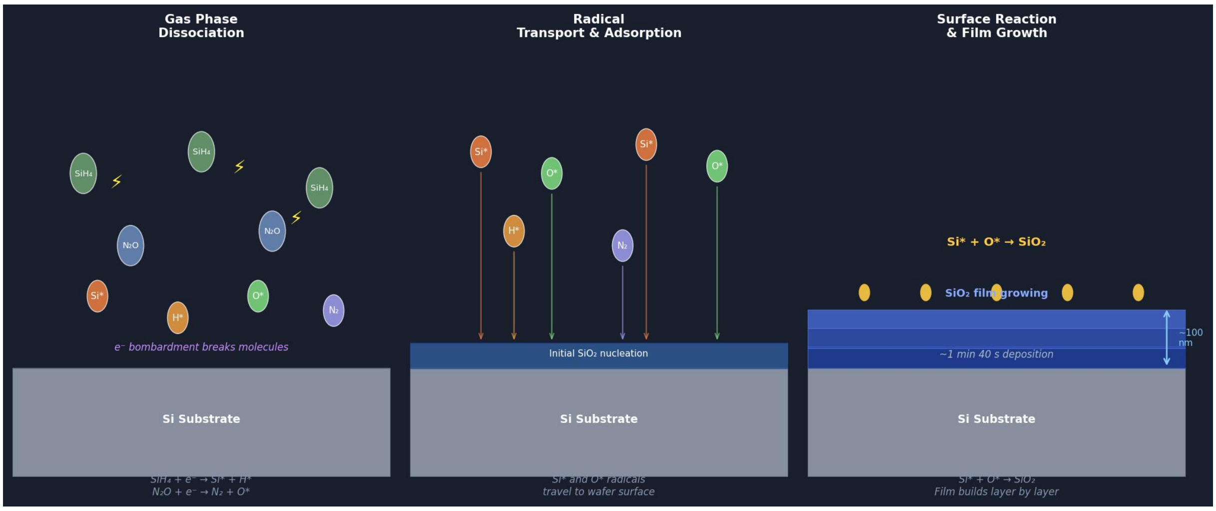


Figure 7: Atomic-level deposition mechanism in three stages. Left: plasma dissociates SiH_4 and N_2O into Si* and O* radicals. Centre: radicals transport from plasma to the wafer surface by diffusion. Right: Si* + O* $\rightarrow$ SiO_2 at the heated surface, building the 100 nm film in $\approx$100 s.

Role of Substrate Temperature (300 °C)

Although 300 °C is low compared to thermal CVD, it still plays an important role:

- Provides surface energy for adsorbed radicals to migrate to energetically favourable bonding sites, producing a denser, less porous film.
- Promotes desorption of weakly adsorbed H atoms, reducing hydrogen incorporation in the SiO_2 film (Si–H bonds degrade electrical properties).
- Enhances adhesion of the SiO_2 film to the Si substrate.

PECVD vs. Other CVD Techniques

Table 1: Comparison of PECVD with thermal CVD and LPCVD for SiO_2 deposition.

Parameter	**Thermal CVD**	**LPCVD**	**PECVD (Day 6)**
Temperature	700–900 °C	550–700 °C	**~300 °C**
Pressure	~10^4 Pa (atm)	~100 Pa	**68.5 Pa**
Energy source	Thermal (heat)	Thermal (heat)	**RF plasma (13.56 MHz)**
Deposition rate	Moderate	Slow	**Fast (~1 nm/s)**
Film conformality	Good	Excellent	Moderate
Film density	High	High	Moderate (H incorporation)
CMOS compatible	No (too hot)	Marginal	Yes
Metal-layer safe	No (Al melts)	No	Yes

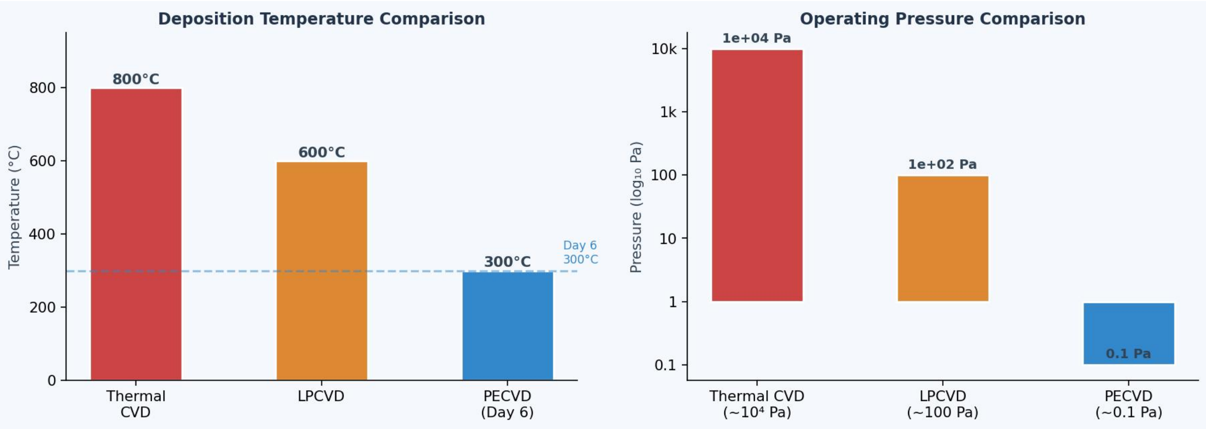


Figure 8: Bar charts comparing deposition temperature (left) and operating pressure (right) for thermal CVD, LPCVD, and PECVD. PECVD operates at significantly lower temperature (300 °C) and moderate vacuum, making it uniquely compatible with Al metallisation layers already on the wafer.

Advantages and Limitations of PECVD

Table 2: Advantages and limitations of PECVD for SiO_2 deposition.

Advantages	**Limitations**
Low deposition temperature (∼300 °C) — compatible with Al and other metals	Non-uniform deposition at large film thicknesses (>500 nm)
Plasma activation replaces thermal energy — fast deposition rate	Poor step coverage compared to LPCVD (isotropic plasma, but directional ions cause shadowing)
CMOS backend compatible — can be applied after full metallisation	Hydrogen incorporation from SiH_4: Si–H bonds can cause reliability issues
Wide range of films (SiO_2, Si_3N_4, a-Si, SiN_x)	No in-situ thickness measurement — timed deposition only
Low stress can be tuned by RF power / pressure	Particle contamination if plasma conditions are unstable

Note on SiH_4 Safety: Silane (SiH_4) is a *pyrophoric* gas — it ignites spontaneously on contact with air. All gas handling, line connections, and MFC calibration must be performed by trained personnel using inert gas purge protocols. Never expose silane lines to atmosphere without prior purging with N_2 or Ar.

Results and Observations

Deposition Results

- Chamber base pressure of ≈0.1 Pa was confirmed before gas introduction.
- On admitting SiH_4 (65 sccm) and N_2O (300 sccm), the working pressure stabilised at ≈68.5 Pa.

- Plasma ignited at RF power >50 W; a visible purple glow discharge was observed between the electrodes through the chamber viewport, confirming active plasma.
- After ≈1 min 40 s, the deposition was stopped. A ∼**100 nm SiO_2** film was deposited on the wafer surface.
- Post-deposition visual inspection: the wafer surface showed a slight blue-tint interference colour consistent with a ∼100 nm oxide layer on silicon.

Film Cross-Section: Final Structure

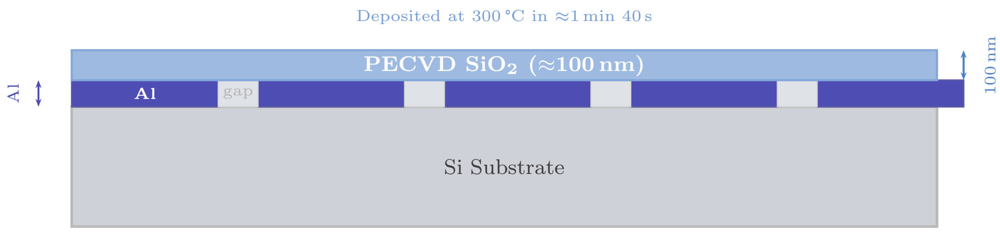


Figure 9: Cross-section of the wafer after Day 6 PECVD: a uniform 100 nm SiO_2 film conformally covers the Al patterned layer and exposed Si surface, acting as an interlayer dielectric or passivation layer.

Justification and Importance

1. **Low-temperature deposition preserves previous layers:** By the time PECVD is performed, the wafer already carries patterned Al interconnects. Any deposition process exceeding ∼450 °C would damage or melt these metal lines. PECVD at 300 °C is the only practical CVD method for adding dielectrics at this stage of fabrication.
2. **Plasma activation is the enabling technology:** The RF plasma converts stable, low-reactivity precursor molecules (SiH_4, N_2O) into highly reactive radicals (Si*, O*) without requiring high temperature, fundamentally decoupling the *reaction activation* from *thermal budget*. This is the core innovation of PECVD.
3. **N_2O:SiH_4 ratio controls film stoichiometry:** The ≈4.6:1 ratio (300:65 sccm) ensures oxygen-rich conditions that produce stoichiometric SiO_2 with good dielectric properties. A lower ratio would produce silicon-rich SiO_x with higher leakage current.
4. **Precise time control determines film thickness:** Since no in-situ thickness monitor is used, the deposition rate must be pre-characterised and the process stopped at exactly 1 min 40 s. Accurate timing at the known rate of ∼1 nm/s gives the 100 nm target.
5. **SiO_2 serves multiple functions:** The deposited film can act simultaneously as a passivation layer (protecting the device from environmental moisture and ionic contamination), an interlayer dielectric (electrically isolating metal layers), and a planarisation layer for subsequent processing steps.

Process Summary Table

Table 3: Summary of Day 6 PECVD process parameters and results.

Parameter	Target	Actual	Notes
Deposited film	SiO_2	SiO_2	From SiH_4+N_2O
Film thickness	100 nm	≈100 nm	Timed deposition
Deposition time	≈100 s	1 min 40 s	Confirmed
SiH_4 flow	65 sccm	65 sccm	Via MFC
N_2O flow	300 sccm	300 sccm	Via MFC
RF power	>50 W	Applied	Plasma confirmed (glow)
RF frequency	13.56 MHz	13.56 MHz	Industry standard
Temperature	300 °C	300 °C	Bottom electrode heater
Base pressure	0.1 Pa	≈0.1 Pa	Pre-deposition
Working pressure	—	68.5 Pa	After gas introduction

Chapter 7

Day 7 — Semiconductor Device Characterisation

Overview of Day 7

Day 7 marked the transition from *fabrication* to *measurement*, providing hands-on experience in the electrical characterisation of semiconductor devices using the **Keysight (Agilent) B1500A Semiconductor Device Parameter Analyser** and a **probe station**. The session covered three device families — MOSFETs, AlGaN/GaN HEMTs, and BJTs — and focused on extracting the key figures of merit that determine device performance in circuit applications.

The measurements performed were:

- I_D–V_D output characteristics of the VN1OLP n-channel MOSFET.
- I_D–V_G transfer characteristics including subthreshold behaviour.
- Extraction of threshold voltage V_T, transconductance g_m, on/off current ratio, and subthreshold swing S.
- Observation of BJT (I_C–V_{CE} and I_C–V_{BE}) characteristics and current gain β.

Device Background: MOSFETs

MOSFET Structure

A Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) consists of a gate electrode separated from the semiconductor by a thin gate oxide (SiO_2), with n^+ source and drain regions diffused into a p-type substrate. Applying a positive gate voltage ($+V_{GS}$) attracts electrons to the semiconductor surface, forming an inversion channel between source and drain through which current flows.

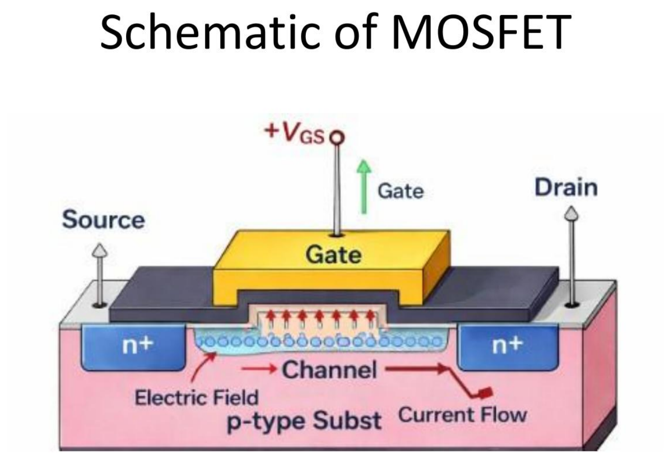


Figure 7.1: Schematic cross-section of an n-channel MOSFET. A positive gate voltage $+V_{GS}$ creates an electric field that induces a conducting electron channel (2DEG) between the n^+ source and drain regions in the p-type substrate. Current flows from drain to source through this channel.

Types of MOSFETs

MOSFETs are classified into four types based on channel polarity (n- or p-channel) and operating mode (enhancement = normally off, depletion = normally on). The cross-section, output characteristics (I_D–V_D), and transfer characteristics (I_D–V_G) differ accordingly, as summarised in Figure 7.2.

Types of MOSFETs

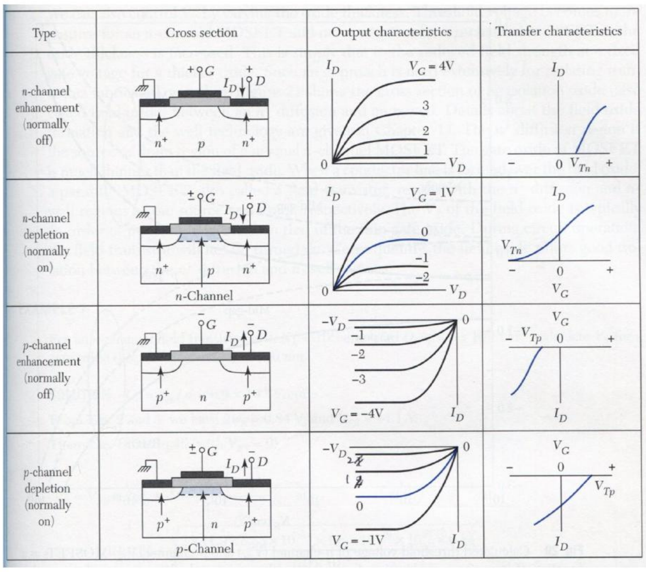


Figure 7.2: The four MOSFET types and their characteristic curves. Enhancement-mode devices (rows 1 and 3) require a gate voltage exceeding V_T to form a channel; depletion-mode devices (rows 2 and 4) conduct at zero gate bias and require a reverse gate voltage to switch off.

Current-Voltage Characteristics and Operating Regions

The I_D–V_D (output) and I_D–V_G (transfer) curves together define all MOSFET operating regions. Figure 7.3 shows the idealised characteristics including the linear region, saturation region, and the subthreshold region visible in the semi-log transfer plot.

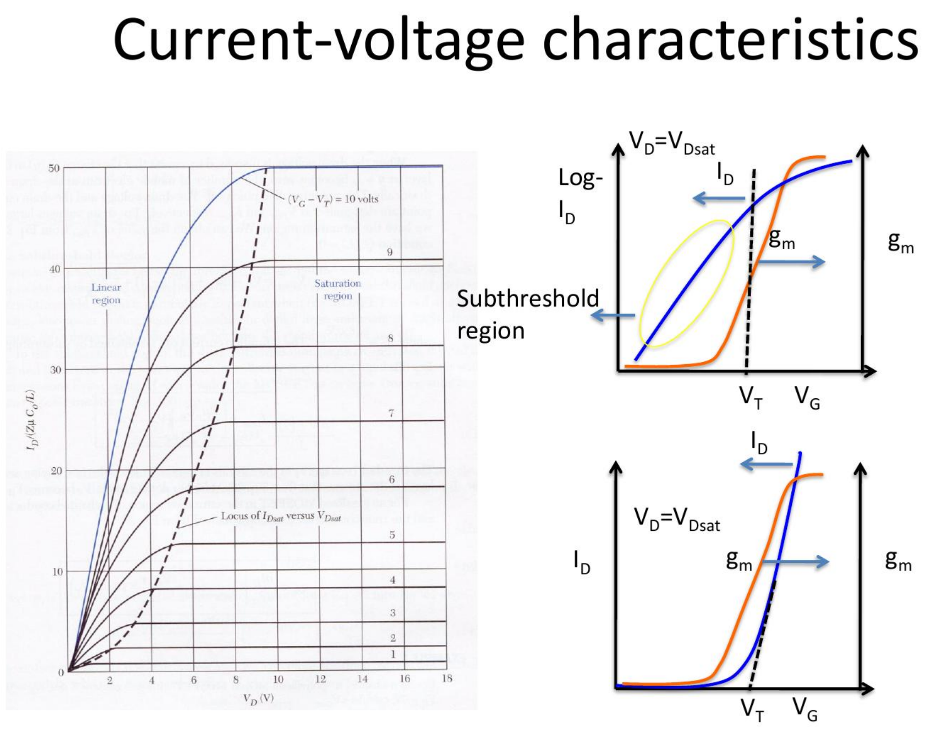


Figure 7.3: MOSFET current-voltage characteristics. Left: I_D–V_D output curves showing the linear region (low V_D) and saturation region (high V_D) separated by the locus of $I_{D,\text{sat}}$ vs. $V_{D,\text{sat}}$ (dashed). Right: Transfer curves on semi-log (top) and linear (bottom) scales showing the subthreshold region, threshold voltage V_T, and transconductance g_m.

AlGaN/GaN HEMT

An AlGaN/GaN High Electron Mobility Transistor (HEMT) is a semiconductor device used in high-frequency and high-power applications (microwave and RF ranges). Unlike a silicon MOSFET, the HEMT relies on a *two-dimensional electron gas* (2DEG) formed spontaneously at the AlGaN/GaN heterojunction due to spontaneous and piezoelectric polarisation fields — no gate-induced inversion is needed. The layer structure consists of a Si(111) 1 mm substrate, a GaN buffer layer, the GaN channel, the AlGaN barrier, a passivation layer, and final metal contacts (Source, Gate, Drain).

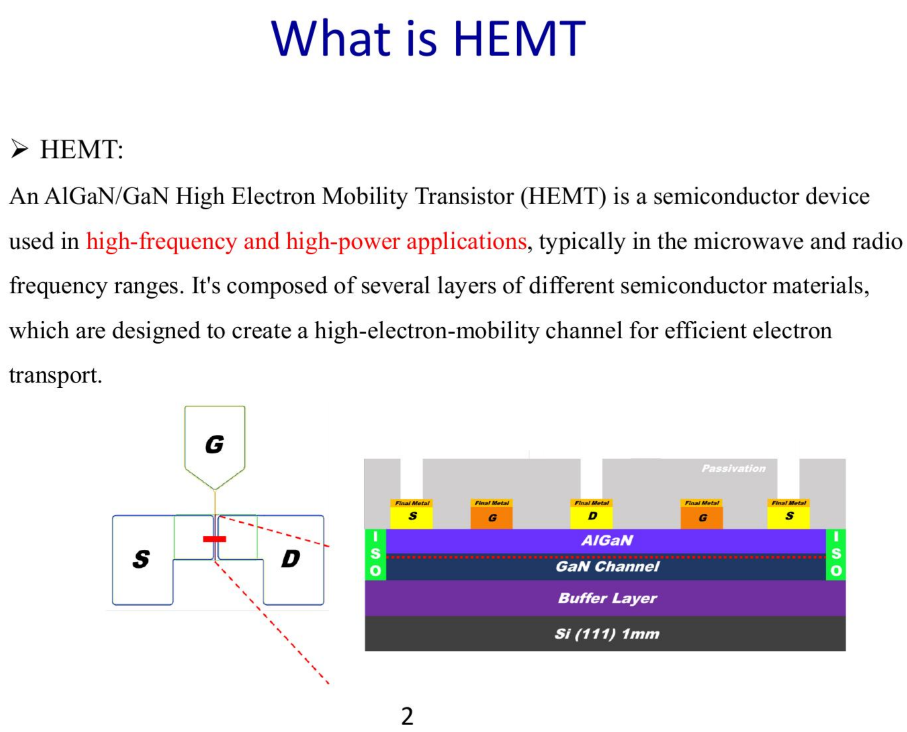


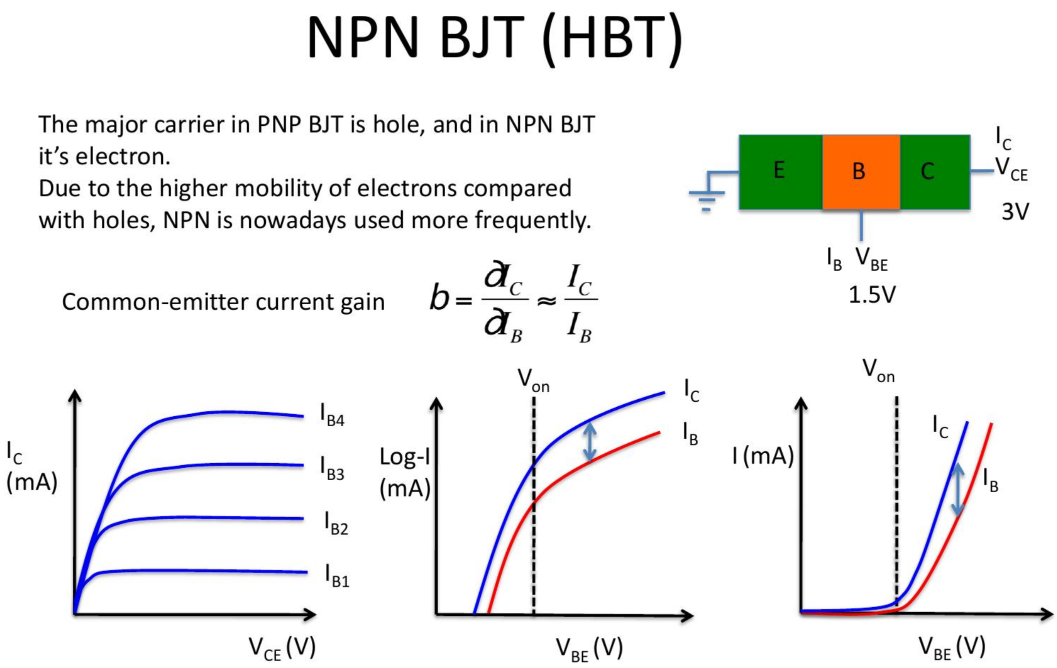
Figure 7.4: AlGaN/GaN HEMT structure. Left: top-view layout showing the Gate (G), Source (S), and Drain (D) pads. Right: cross-sectional layer stack — Si(111) substrate / Buffer / GaN channel / AlGaN barrier / Passivation / Final metal contacts. The 2DEG (dotted line) forms at the AlGaN/GaN interface without requiring gate bias, making this a normally-on (depletion-mode) device.

BJT Background

A Bipolar Junction Transistor (BJT) uses both electron and hole currents. The NPN BJT (Figure 7.5) is preferred over PNP because electrons have higher mobility than holes, enabling faster operation. In the common-emitter configuration the key metric is the **current gain**:

$$\beta = \frac{\partial I_C}{\partial I_B} \approx \frac{I_C}{I_B} \tag{7.1}$$

β is extracted from the vertical spacing between I_C–V_{CE} curves taken at different base currents I_B.



$$b = \frac{\partial I_C}{\partial I_B} \approx \frac{I_C}{I_B}$$

Figure 7.5: NPN BJT (HBT) characteristics. Left: I_C–V_{CE} family of curves for increasing base currents I_{B1}–I_{B4}, showing the saturation and active regions. Centre: semi-log I_C, I_B vs. V_{BE}, illustrating the exponential turn-on and the separation between I_C and I_B that defines β. Right: linear I_C, I_B vs. V_{BE} showing the turn-on voltage V_{on}.

Instrument Description: Keysight B1500A

Overview and Lab Photograph

The **Keysight B1500A** (Figure 7.6) is a modular semiconductor parameter analyser capable of sourcing and measuring current and voltage simultaneously over a dynamic range from femtoampere (fA) to ampere (A) and from microvolt (µV) to volt (V). It houses multiple **Source/Measure Units (SMUs)** that can independently source voltage or current and measure the complementary quantity at each device terminal. The **EasyEXPERT** software provides a graphical interface for configuring sweeps, running measurements, and visualising results.

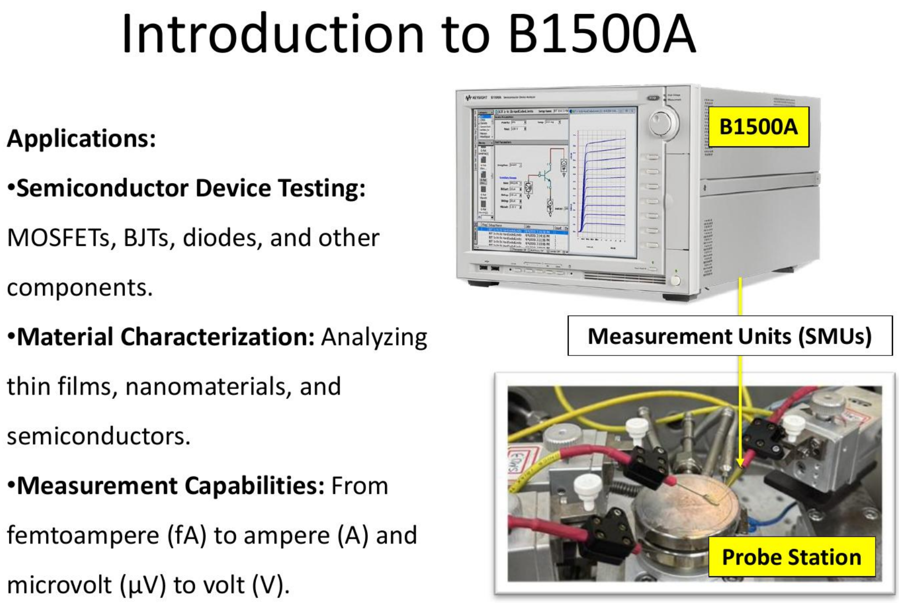


Figure 7.6: Keysight B1500A Semiconductor Device Parameter Analyser (top right) with the EasyEXPERT display showing a measured I_D–V_D curve. The SMU modules are located inside the mainframe. Below: the probe station used for on-wafer and packaged device testing; SMU triaxial cables connect to the tungsten needle probes that contact the device terminals (Gate, Drain, Source).

System Components

Table 7.1: Key components of the Keysight B1500A measurement system.

Component	Function
Source/Measure Units (SMUs)	Independently source voltage or current and measure the complementary quantity at each device terminal.
EasyEXPERT software	Graphical interface for configuring sweep parameters, running measurements, and exporting data.
Probe station	Mechanical platform with micro-positioners that bring tungsten needle probes into electrical contact with device pads.
Triaxial cables	Shielded cables that minimise leakage current, enabling femtoampere-level measurements.

Devices Characterised

Three device types were available for measurement at **Engineering Building 2, Room 801**. Figure 7.7 shows all device types and their on-wafer microscope images.

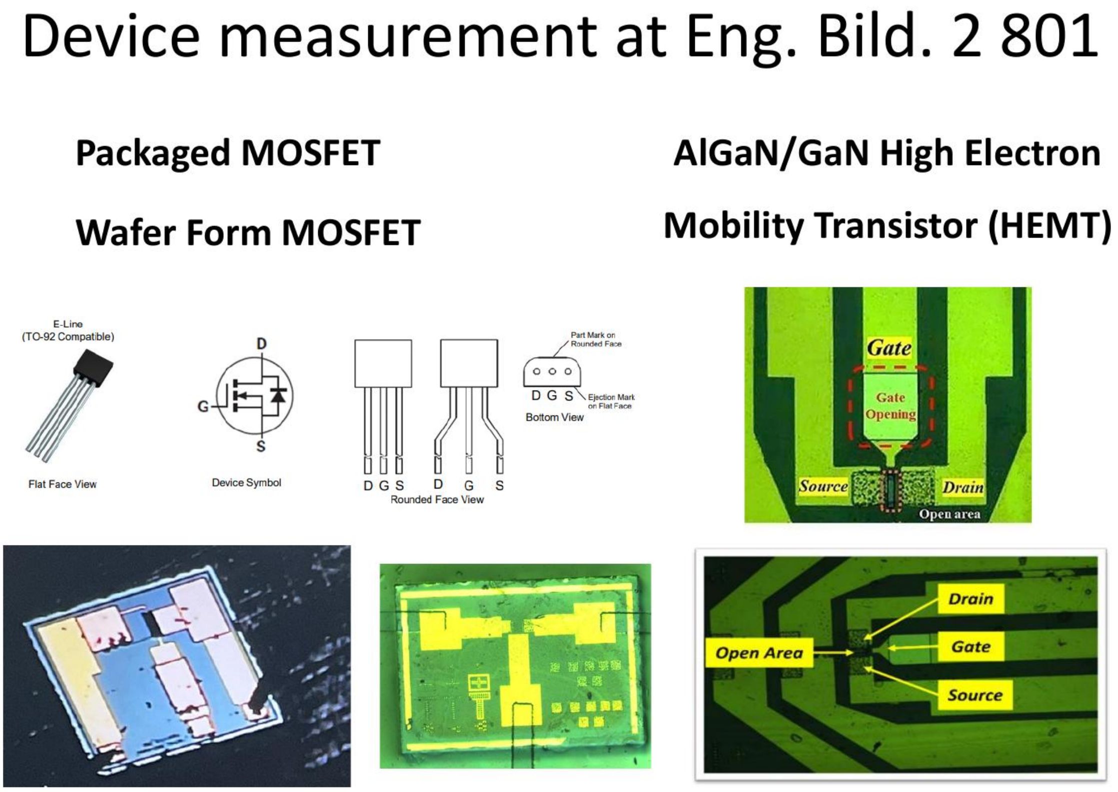


Figure 7.7: Devices available for characterisation on Day 7. Left column: packaged VN1OLP MOSFET (TO-92, flat-face view) with device symbol and pin-out (D — G — S). Bottom centre: wafer-form MOSFET chips. Right column: AlGaN/GaN HEMT — microscope images showing the gate opening, source, and drain metal pads; the bottom-right image clearly labels Drain, Gate, and Source with the open recessed gate area.

The channel connections used for all FET measurements:

Drain → SMU1 — **Gate** → SMU2 — **Source** → SMU3 (ground)

Measurement 1: I_D–V_D Output Characteristics

Aim

To measure drain current (I_D) as a function of drain voltage (V_D) for a series of fixed gate voltages (V_G), mapping the output characteristics and identifying the linear and saturation operating regions.

EasyEXPERT Setup

FET I_D–V_D Measurement Recipe — Day 7

Library:	Category: Discrete → FET Id-Vd
Polarity:	Nch (n-channel enhancement mode)
Primary sweep (Drain):	$V_D = 0 \to 10\,\text{V}$, step $= 100\,\text{mV}$
Secondary sweep (Gate):	$V_G = 1.7\,\text{V}$ to $2.7\,\text{V}$, step $= 0.1\,\text{V}$
Current compliance:	$I_{D,\max} = 10\,\text{mA}$
Integration time:	SHORT
Channel mapping:	Drain: SMU1 — Gate: SMU2 — Source: SMU3

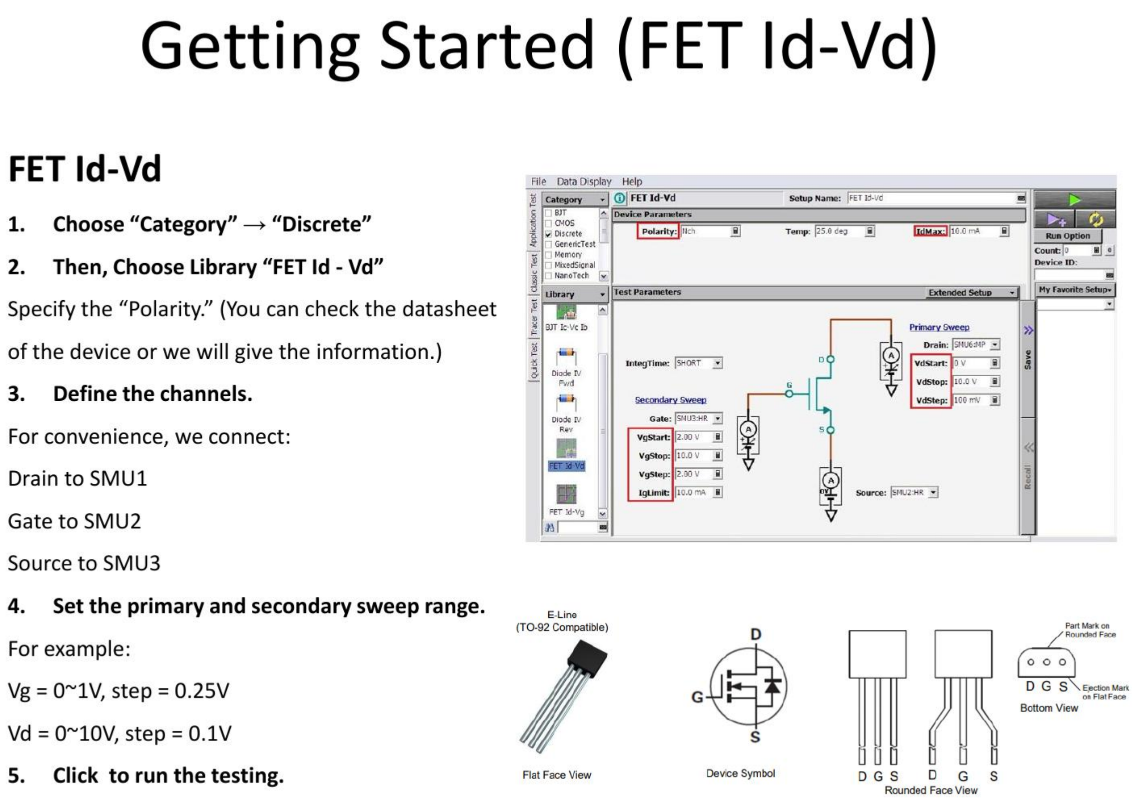


Figure 7.8: EasyEXPERT FET Id-Vd setup screen. Left side: step-by-step procedure for selecting the library (Category: Discrete → FET Id-Vd), setting polarity, defining channel connections, and configuring the primary (V_D) and secondary (V_G) sweep ranges. Bottom: the VN1OLP device pinout (flat face, device symbol, rounded face, and bottom views) used to correctly connect the probe station arms to Drain, Gate, and Source.

Results and Analysis

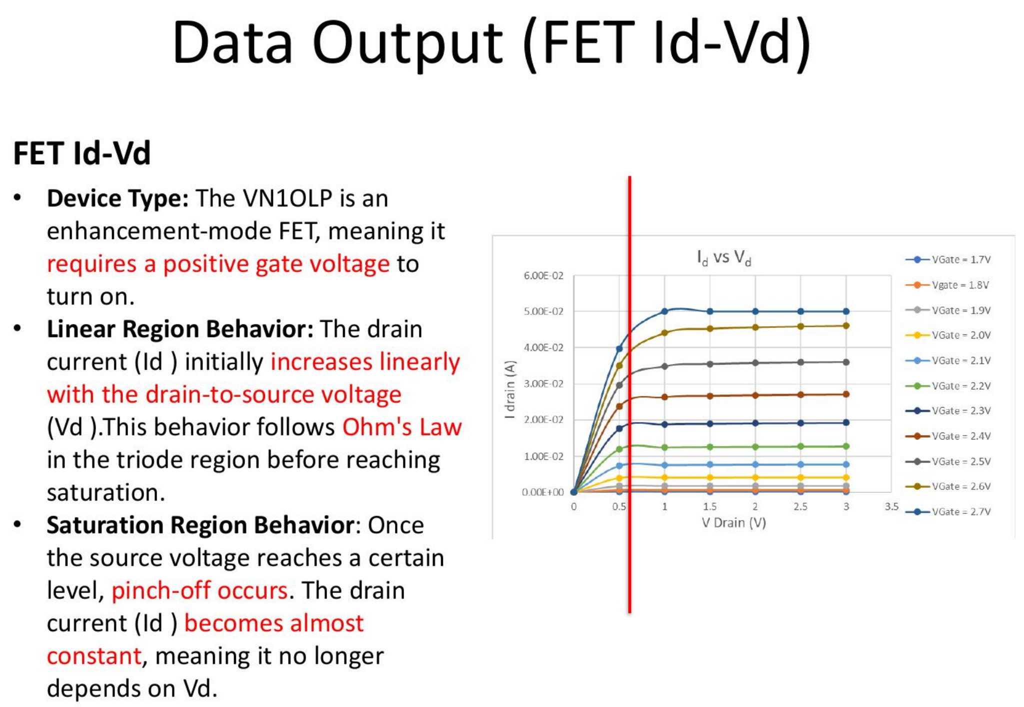


Figure 7.9: Measured I_D–V_D output characteristics of the VN1OLP n-channel MOSFET. Each curve corresponds to a fixed V_G from 1.7 V to 2.7 V (step 0.1 V). The red dashed vertical line at $V_D \approx 0.5$ V marks the boundary between the linear region (left, I_D rises with V_D) and the saturation region (right, I_D is approximately constant at pinch-off). At $V_G = 2.7$ V, $I_{D,\text{sat}} \approx 50$ mA.

Key observations from Figure 7.9:

- **Enhancement-mode device.** No current flows for $V_G < V_T \approx 1.5$ V; a positive gate voltage is required to turn on the device on.
- **Linear region** ($V_D < V_G - V_T$): I_D follows Ohm's law, rising linearly with V_D:

$$I_D \approx \mu_n C_{\text{ox}} \frac{W}{L}\left[(V_G - V_T)V_D - \tfrac{V_D^2}{2}\right]. \tag{7.2}$$

- **Saturation region** ($V_D \geq V_G - V_T$): Pinch-off occurs; I_D becomes approximately constant:

$$I_{D,\text{sat}} \approx \tfrac{1}{2}\,\mu_n C_{\text{ox}} \frac{W}{L}(V_G - V_T)^2. \tag{7.3}$$

- Higher V_G raises $I_{D,\text{sat}}$ quadratically, consistent with the square-law model.

Measurement 2: I_D–V_G Transfer Characteristics

Aim

To extract the threshold voltage (V_T), transconductance (g_m), on/off current ratio, and subthreshold swing (S) by sweeping gate voltage at fixed drain bias.

EasyEXPERT Setup

FET I_D–V_G Measurement Recipe — Day 7

Library:	Category: Discrete → FET Id-Vg
Polarity:	Nch
Primary sweep (Gate):	$V_G = 0 \to 10\,\text{V}$, step $= 100\,\text{mV}$
Secondary sweep (Drain):	$V_D = 2\,\text{V}, 4\,\text{V}, 6\,\text{V}, 8\,\text{V}, 10\,\text{V}$
Current compliance:	$I_{D,\max} = 10\,\text{mA}$
Integration time:	MEDIUM
Channel mapping:	Drain: SMU1 — Gate: SMU2 — Source: SMU3

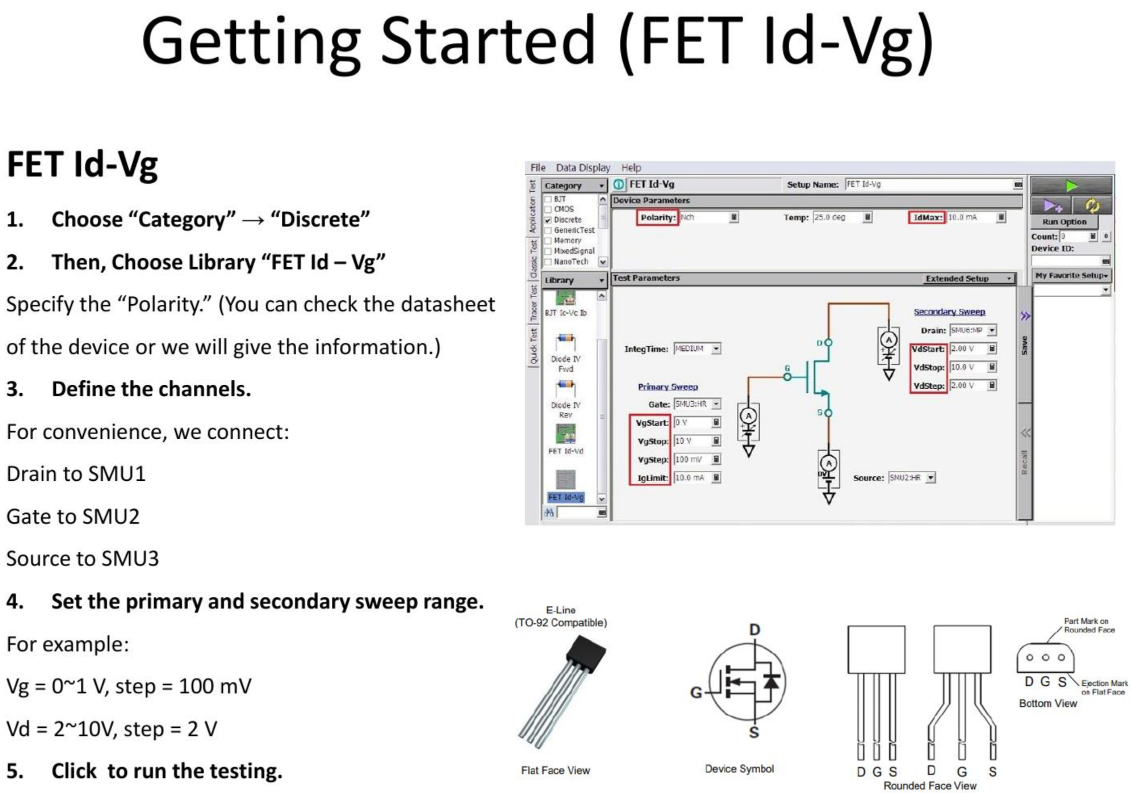


Figure 7.10: EasyEXPERT FET Id-Vg setup screen. The primary sweep is the Gate voltage ($V_G = 0 \to 10\,\text{V}$, step $100\,\text{mV}$) and the secondary sweep steps the Drain voltage ($V_D = 2 \to 10\,\text{V}$, step $2\,\text{V}$). Integration time is set to MEDIUM for improved accuracy in the subthreshold region. The device pinout is shown at the bottom for correct probe connection.

Key Parameters Extracted

Table 7.2: Device parameters extracted from the I_D–V_G transfer characteristics.

Parameter	**Definition**	**Typical**
Threshold voltage (V_T)	Gate voltage at which the inversion channel forms.	$\approx 1.5\,\mathrm{V}$
Transconductance (g_m)	$g_m = \partial I_D/\partial V_G\|_{V_{DS}}$; voltage-to-current conversion efficiency.	Peak in sat.
On/off current ratio	$I_{D,\mathrm{on}}/I_{D,\mathrm{off}}$; determines noise margin and leakage power.	$> 10^4$
Subthreshold swing (S)	$[\partial(\log I_D)/\partial V_G]^{-1}$; switching speed below V_T.	mV/decade

Subthreshold Region

Below threshold ($V_G < V_T$), a small diffusion current persists:

$$I_D \sim \exp\left[\frac{q(V_G - V_T)}{kT}\right], \qquad V_G < V_T. \tag{7.4}$$

The **subthreshold swing** S quantifies how rapidly the device switches:

$$S \equiv \left[\frac{\partial(\ln I_D)}{\partial V_G}\right]^{-1} \quad [\mathrm{mV/decade}]. \tag{7.5}$$

The ideal room-temperature lower limit is $60\,\mathrm{mV/decade}$. A smaller S gives faster switching and lower standby power — critical for digital logic.

Discussion: Operating Regions and Applications

Table 7.3: MOSFET operating regions and circuit applications.

Region	**Condition**	**Application**
Cut-off	$V_G < V_T$; $I_D \approx 0$	Digital "OFF" state; power-saving standby.
Linear (triode)	$V_D < V_G - V_T$	Variable resistor; analog switches; pass gates.
Saturation	$V_D \geq V_G - V_T$	Amplifiers; current source; digital "ON" state.
Subthreshold	$V_G < V_T$; exponential I_D	Ultra-low-power logic; IoT/wearable electronics.

Results and Conclusion

Summary of Day 7 Measurements

Table 7.4: Summary of all Day 7 measurements performed.

Measurement	Device	Parameters	Outcome
I_D–V_D output	VN1OLP n-MOSFET	$V_G = 1.7$–$2.7\,\mathrm{V}$; $V_D = 0$–$10\,\mathrm{V}$	Linear & saturation regions confirmed.
I_D–V_G transfer	VN1OLP n-MOSFET	$V_D = 2$–$10\,\mathrm{V}$; $V_G = 0$–$10\,\mathrm{V}$	V_T, g_m, on/off ratio, S extracted.
BJT characteristics	NPN BJT / HBT	I_C–V_{CE}; I_C–V_{BE}	Current gain β extracted.
HEMT observation	AlGaN/GaN HEMT	On-wafer probing	2DEG channel structure observed.

Conclusion

Day 7 provided hands-on experience in the electrical characterisation of semiconductor devices using an industry-standard parameter analyser. The I_D–V_D measurements confirmed the expected linear and saturation regimes of the n-channel enhancement-mode VN1OLP MOSFET, while the I_D–V_G transfer curves enabled extraction of V_T, g_m, on/off ratio, and subthreshold swing.

These figures of merit are essential for evaluating device quality and for selecting the correct operating regime: the linear region for low-loss switching, saturation for amplification, and the subthreshold region for ultra-low-power logic. The AlGaN/GaN HEMT demonstrated how heterojunction engineering dramatically enhances carrier mobility beyond silicon limits, while the NPN BJT characterisation showed how current gain β is extracted from the common-emitter I_C–V_{CE} family of curves.

Together, Days 1–7 formed a complete end-to-end semiconductor manufacturing and characterisation cycle — from cleaning a bare silicon wafer to measuring finished transistors — providing a solid foundation for advanced work in semiconductor technology.

Report compiled by Group 3 — Indo-Taiwan Advanced Semiconductor Manufacturing Program, NTHU Taiwan.

Overview of Day 9: BioFET Aptasensor for Mercury Ion Detection

Day 9 of the cleanroom training programme introduced a fundamentally different class of semiconductor device: the **BioFET** (Biological Field-Effect Transistor). Unlike the purely inorganic process steps covered in Days 1–8, a BioFET integrates biological recognition elements — in this case an **aptamer** specific to the Hg^{2+} ion — directly onto the gate surface of a field-effect transistor, enabling label-free, real-time detection of chemical or biological targets in solution.

The specific application on Day 9 was detection of **mercury (Hg)** ions dissolved in a phosphate-buffered saline (PBS) solution, as a model for environmental water quality monitoring.

Fabrication context: The BioFET sensing experiment represents a post-fabrication characterisation step, performed after the device chip has been fabricated through the standard cleanroom process flow (Days 1–8) and surface-functionalised with MMP-7 aptamers. On Day 9, the functionalised chip was connected to the portable BioFET readout instrument and tested with synthesised PBS+Hg solutions of varying concentration.

Background: BioFETs and Biomarker Detection

Limitations of Conventional Detection: ELISA

The conventional gold-standard method for detecting biomarkers such as **MMP-7** (Matrix Metalloproteinase-7) — a protein biomarker used in cancer diagnostics — is the Enzyme-Linked Immunosorbent Assay (ELISA). ELISA requires bulky, expensive laboratory equipment, involves multiple time-consuming steps, and is typically restricted to hospital or centralised laboratory settings. It cannot easily be adapted for point-of-care or field deployment. Detection of MMP-7 by ELISA involves the PRISM 8 optical platform, which further adds to the complexity and cost.

The CG-FET BioFET: A Portable Alternative

The **CG-FET** (Common-Gate Field-Effect Transistor) BioFET is a biosensor that overcomes these limitations. Its key advantages are:

- **Sensitivity:** Capable of detecting analyte concentrations down to 10^{-9} M (nanomolar) and below.
- **Selectivity:** Achieved through surface functionalisation with target-specific aptamers that bind only the analyte of interest.
- **Portability:** The sensing chip integrates with a compact, portable readout device connected via USB to a laptop — no bulky laboratory infrastructure required.
- **Real-time output:** The device produces a continuous drain current (I_D) vs. time trace; analyte binding shifts I_D measurably within seconds.

The BioFET chip used on Day 9 was a **sensing chip** (see Figure 1a), with a flex-cable connector providing electrical contact to the portable readout unit shown in Figure 1b.

Aptamers as Biorecognition Elements

An **aptamer** is a short, synthetic oligonucleotide (DNA or RNA) sequence — effectively a sequence of protein-like molecules — selected to bind a specific target with high affinity. Aptamers are generated through the **SELEX** process (*Systematic Evolution of Ligands by Exponential enrichment*), an iterative in-vitro selection procedure that identifies oligonucleotide sequences with the highest binding affinity for the target from a large random library.

On Day 9, MMP-7 aptamers were immobilised on the extended gate surface of the CG-FET. These aptamers capture Hg^{2+} ions from the test solution, causing a change in surface charge that is transduced into a measurable shift in drain current (ΔI_D).

Surface Functionalisation and KPFM Analysis

The BioFET gate surface was prepared through a multi-step **surface functionalisation** process:

1. **Bare gold (Au) surface** on the extended gate.
2. **Aptamer immobilisation:** MMP-7 specific aptamers are chemically attached (thiol–gold bonding) to the gold gate surface.
3. **Target capture:** When the test solution is introduced, Hg^{2+} ions bind to the immobilised aptamers, inducing a local charge change that modulates the FET gate potential.

KPFM (Kelvin Probe Force Microscopy) analysis was used to verify the surface modification at each stage — bare gold, aptamer-immobilised, and after Hg^{2+} capture — by measuring the surface potential (work function) change at nanometre resolution.

Name of the Process Performed

Process: BioFET Aptasensor Characterisation — CG-FET Drain Current (I_D) vs. Time Measurement with Hg^{2+} Analyte Introduction

Category: Post-fabrication biosensing / device characterisation

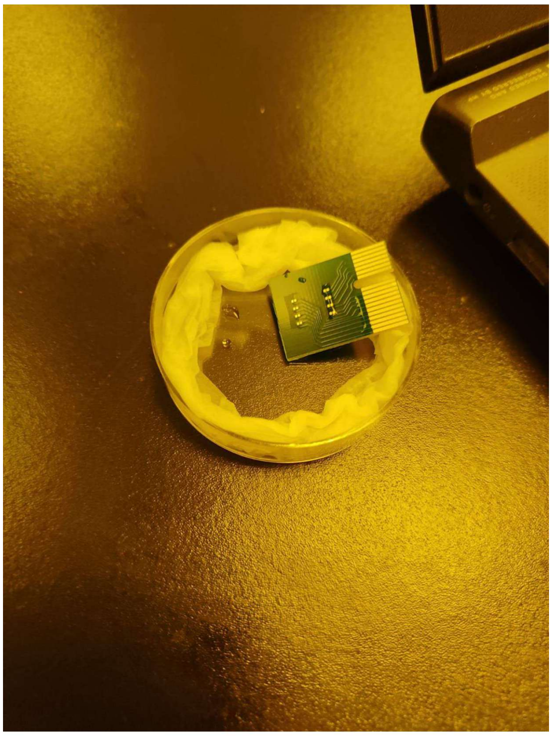
The process performed on Day 9 was the real-time electrical characterisation of an aptamer-functionalised CG-FET BioFET chip by recording the drain current (I_D) as progressively higher concentrations of Hg^{2+} (dissolved in PBS buffer) were pipetted onto the extended gate. Each concentration step produced a measurable step increase in I_D, enabling a calibration curve to be constructed.

Instrument Description

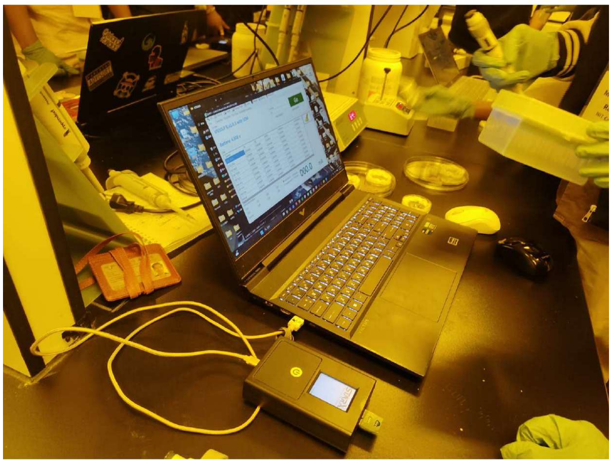
The BioFET Portable Readout System

The measurement system consists of three main components:

1. **BioFET sensing chip:** A multi-channel CG-FET chip with an extended gold gate, surface-functionalised with MMP-7 aptamers. The chip is housed in a circular dish holder (Figure 1a) with a flex-cable providing 8 channels of electrical contact.
2. **Portable readout unit (VN10LP B, v1.0.3 with LCM):** A compact, battery-powered instrument (battery: 4.302 V during the experiment) that applies the gate and drain bias voltages, measures the drain current on up to 8 channels (CH1–CH8) simultaneously, and communicates with the laptop via USB. The unit includes a small built-in display for status monitoring.
3. **Laptop with BioFET control software:** The software (Figure 2a) displays real-time I_D (mA) readings, logs raw data, and provides a live chart of the scheduled-loop measure-



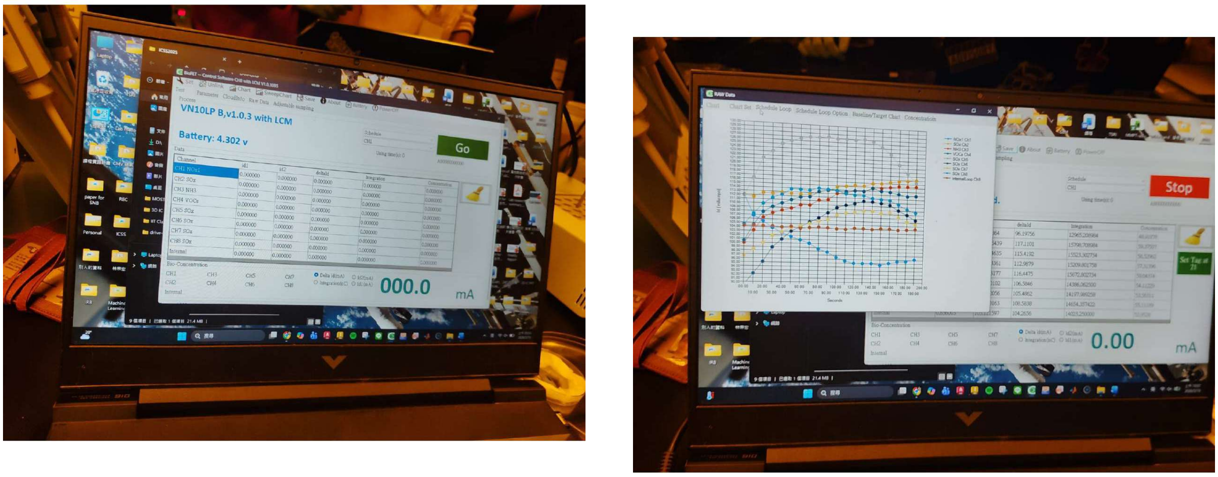
(a) The BioFET sensing chip mounted in its circular Petri-dish holder with flex-cable connector for electrical contact.



(b) Full experimental setup: the BioFET chip (connected via USB to the portable readout unit) and the laptop running the BioFET control software (VN10LP B, v1.0.3 with LCM).

Figure 1: BioFET aptasensor hardware used on Day 9.

ments. The y-axis is set to Delta I_D (mA) mode, with manual scaling (min: 90, max: 120 as noted).



(a) BioFET software main screen showing 8-channel current readings (all CH initialised to 0.000 mA), battery voltage, and **Go** button for starting the scheduled measurement loop.

(b) Live RAW Data chart during measurement: multi-channel I_D vs. time traces showing the initial ramp-up and stabilisation before analyte addition.

Figure 2: BioFET control software and live data display on the laptop.

Process Parameters / Recipe

Day 9 BioFET Measurement Recipe

Buffer solution:	0.02× PBS (phosphate-buffered saline)
Analyte:	Hg^{2+} dissolved in PBS (synthesised sample — no actual blood)
Analyte concentrations:	10^{-9} M (baseline) → 10^{-8} M → 10^{-7} M → 10^{-6} M
Sample volume (each):	6 µL (pipetted onto BioFET extended gate)
Measurement mode:	Schedule loop, Delta I_D (mA)
Sampling interval:	Every 10 s (one I_D data point per 10 s)
Total tests per run:	300 schedule loop cycles
y-axis range:	Min: 90, Max: 120 (manual, check use manual before accepting)
Stabilisation:	Wait until chart stabilises at $\sim 10^{-9}$ baseline before first addition
Hg preparation:	Hg solution heated to 65 °C for 10 s (spin once) before pipetting
Safety note:	Hg is toxic; can be used for fish data (environmental monitoring)

Step-by-Step Procedure

The measurement procedure followed the numbered steps recorded during the session:

Step S1 — Set measuring baseline (0.02× PBS): Place 50 µL of 0.02× PBS buffer onto the BioFET extended gate using a pipette, then drop a further 50 µL again to fully wet the gate and establish the ionic environment required for FET operation. This sets the electrolytic gate reference and measurement baseline.

Step S2 — Press Go: Press the **Go** button in the BioFET software to begin the scheduled measurement loop.

Step S3 — Go to Chart → Schedule Loop: Navigate to the Chart view and select the Schedule Loop display to monitor the real-time I_D trace across all active channels.

Step S4 — After it opens, select Delta I_D (mA): In the chart settings, select **Delta I_D (mA)** as the displayed quantity to observe relative current changes induced by analyte binding, rather than the absolute drain current.

Step S5 — Set y-axis: Manually set the y-axis to: Min = 90, Max = 120. Check the use manual scaling option and press accept/confirm.

Step S6 — Stabilise at 10^{-9} M baseline: Run 300 measurement cycles (one data point every 10 s, i.e. 3000 s total) while waiting for the chart to stabilise, indicating a stable I_D baseline at the lowest analyte concentration (10^{-9} M).

Step S7 — Add Hg at 10^{-7} M: Take the Hg solution, spin it once, heat at 65 °C for 10 s. Take 6 µL of the 10^{-7} M Hg solution and pipette it directly onto the BioFET gate. A small step-up in ΔI_D will be observed as Hg^{2+} ions bind to the aptamers, modulating the gate charge.

Note: Hg is toxic and can be used as a model analyte for fish/environmental contamination studies.

Step S8 — Repeat for higher concentrations (10^{-8}, 10^{-6} M): Repeat the analyte addition process with progressively higher Hg concentrations (10^{-8} M, then 10^{-6} M). At higher concentrations, a larger step-up in ΔI_D will be observed, consistent with increased surface charge from greater aptamer occupancy.

Step S9 — Immobilised → After adding DNA: The final measurement point corresponds to the fully Hg^{2+}-loaded aptamer surface, confirming successful biorecognition.

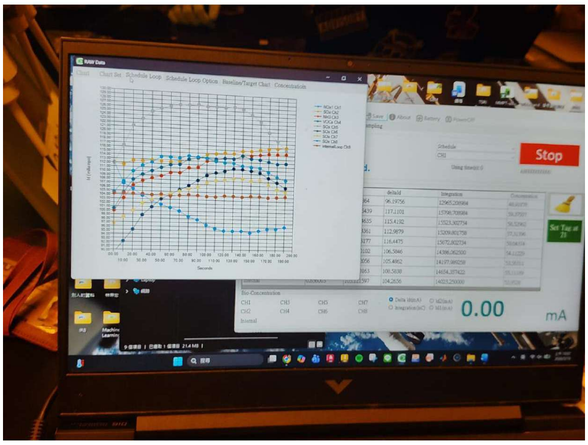
On Day 9, the sample used was a **synthesised PBS+Hg solution** (not actual blood or biological fluid). This is a controlled experimental condition used to validate the BioFET sensor response and construct a calibration curve across a known concentration range (10^{-9} to 10^{-6} M). Real blood samples would require additional sample pre-processing (dilution, filtration) before measurement.

Results and Observations

Live Data Charts

KPFM Surface Characterisation

(a) SCHEDULE Data chart: multi-channel ΔI_D (mA) vs. time (seconds). The traces show gradual stabilisation of all 8 channels toward their respective baseline values over the first 300 measurement cycles, with the separated blue channel (CH1) showing a distinct drift — indicative of the aptamer-functionalised sensing channel responding differently to the PBS environment.

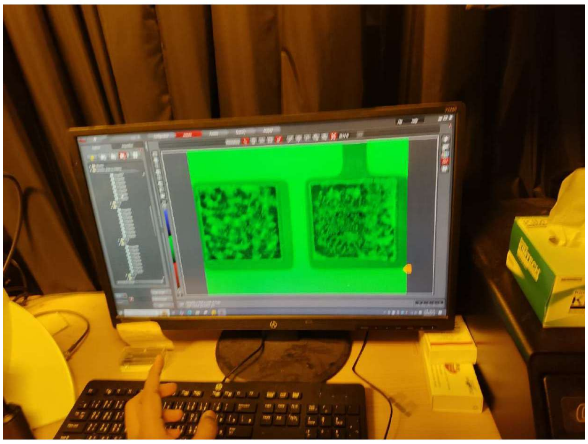


(b) RAW Data chart showing the initial I_D transient for all channels immediately after PBS introduction and Go initiation. The multi-channel traces rise from zero and converge toward their steady-state values, with one data point collected every 10 s.

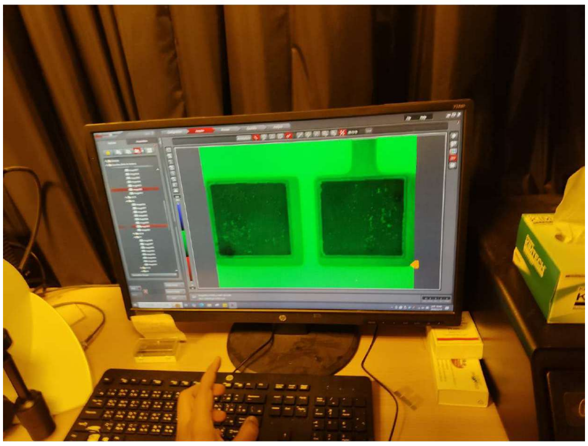
Figure 3: Real-time I_D measurement results on Day 9. Left: Schedule loop data after 300 cycles showing baseline stabilisation. Right: Raw data chart from measurement start.

Summary of Observations

- **Baseline stabilisation:** After initiating the schedule loop with 0.02× PBS, the ΔI_D traces for all channels gradually stabilised over the 300-cycle run, approaching a steady-state baseline, as expected for a stable electrolyte-gate interface.
- **Step-up response on Hg addition:** Upon pipetting 6 µL of 10^{-7} M Hg^{2+} onto the gate, a small but measurable step increase in ΔI_D was observed on the aptamer-functionalised channel, consistent with Hg^{2+} binding to the immobilised aptamers and increasing the positive charge density at the gate–electrolyte interface.
- **Concentration-dependent response:** At higher Hg concentrations (10^{-8}, 10^{-6} M), larger step-up responses were observed, confirming that ΔI_D increases monotonically with analyte concentration — the expected behaviour for a calibrated biosensor.
- **KPFM confirmation:** The KPFM maps (Figures 4a and 4b) show a clear change in surface potential distribution between the bare gold gate and the aptamer-loaded gate after Hg^{2+} capture, providing independent nanoscale confirmation of the biofunctionalisation.



(a) KPFM surface potential map of the BioFET extended gate: **before aptamer functionalisation** (bare gold surface). The relatively featureless potential map confirms a clean, uniform Au surface prior to biofunctionalisation.



(b) KPFM surface potential map **after aptamer immobilisation and Hg^{2+} capture**. The marked change in surface potential distribution across the gate area confirms successful aptamer attachment and analyte binding, verifying the functionalisation process.

Figure 4: KPFM (Kelvin Probe Force Microscopy) analysis of the BioFET extended gate surface. The two maps demonstrate the change in surface potential induced by aptamer immobilisation and subsequent Hg^{2+} capture.

- **Multi-channel readout:** The 8-channel simultaneous readout allowed comparison between the sensing channel (aptamer-loaded) and reference channels, improving the reliability of the differential ΔI_D measurement.

Justification and Importance of the BioFET Process

1. **Replaces bulky laboratory assays:** The ELISA gold standard requires large instruments, skilled operators, and long processing times. The CG-FET BioFET delivers comparable sensitivity in a portable, low-cost platform that can be deployed in the field for real-time environmental monitoring (e.g. Hg in river water).
2. **Aptamer selectivity eliminates matrix interference:** The SELEX-derived aptamer is engineered to bind Hg^{2+} specifically, even in a complex ionic matrix (PBS), suppressing false positives from other metal ions.
3. **Extended gate architecture enables liquid-phase sensing:** The CG-FET extended gate design physically separates the electrolyte solution from the FET channel, preventing corrosion and short circuits, while the gate charge change is still faithfully transduced to the FET drain current.

4. **ΔI_D mode suppresses drift:** By measuring the *change* in drain current (ΔI_D) rather than the absolute I_D, the measurement is insensitive to slow baseline drift, improving signal-to-noise ratio and long-term stability.
5. **Portable form factor enables point-of-care deployment:** The complete measurement system (chip + portable readout unit + laptop) fits on a benchtop and can be battery-operated, enabling use in resource-limited or remote settings — a key advantage for environmental and clinical screening.
6. **Synthesised sample validates sensor before clinical use:** Using a synthesised PBS+Hg solution (rather than actual blood) on Day 9 allowed the sensor response to be validated under controlled conditions, confirming the calibration before the device is applied to real biological matrices.

Mercury (Hg) is a highly toxic heavy metal. Even at the nanomolar concentrations used (10^{-9} to 10^{-6} M), all Hg-containing solutions must be handled with nitrile gloves, and all waste must be collected in labelled heavy-metal waste containers. Never pour Hg solutions down the drain. The 65 °C heating step must be performed in a ventilated area.

Process Summary Table

Table 1: Summary of Day 9 BioFET aptasensor measurement parameters and results.

Parameter	Target	Actual	Notes
Sensing chip	CG-FET (8-ch)	CG-FET BioFET	Au extended gate
Biorecognition elem.	MMP-7 aptamer	Aptamer-loaded	SELEX-derived
Buffer	0.02× PBS	0.02× PBS	50 µL baseline load
Analyte	Hg^{2+} in PBS	Synthesised sample	Not actual blood
Conc. range	10^{-9}–10^{-6} M	4 steps	$10^{-9}, 10^{-7}, 10^{-8}, 10^{-6}$ M
Sample volume	6 µL	6 µL	Pipetted on gate
Sampling rate	1 point/10 s	1 point/10 s	300 cycles total
Measurement mode	ΔI_D (mA)	ΔI_D (mA)	Schedule loop
Hg heating	65 °C, 10 s	Applied	Prior to pipetting
Step response	Step-up at each conc.	Observed	Larger at higher conc.
KPFM verification	Surface potential shift	Confirmed	Bare vs. functionalised

Conclusion

Day 9 of the NTHU cleanroom training programme provided hands-on exposure to BioFET technology — an advanced integration of semiconductor device physics and bionanotechnology. The key outcomes were:

- Understanding of the CG-FET BioFET architecture and its advantage over conventional ELISA-based biomarker detection.
- Appreciation of aptamer design (SELEX) and surface functionalisation of the extended gold gate with MMP-7 aptamers specific to Hg^{2+}.
- Successful execution of the measurement protocol: PBS baseline loading, schedule-loop initiation, concentration-step addition of Hg^{2+}, and real-time ΔI_D recording.

- Observation of concentration-dependent step responses in ΔI_D, confirming the BioFET's ability to quantitatively detect Hg^{2+} across the 10^{-9}–10^{-6} M range.
- KPFM analysis confirming the change in gate surface potential after aptamer immobilisation and analyte capture.

The BioFET experiment illustrates how the semiconductor fabrication skills developed throughout Days 1–8 (cleaning, lithography, etching, deposition, characterisation) can be extended to create intelligent biosensing platforms with real-world applications in environmental monitoring, clinical diagnostics, and point-of-care testing.

Chapter 8

Day 8 — MEMS Fabrication and Characterisation

Overview of Day 8

Day 8 combined both **microfabrication** and **device characterisation**, focusing on Micro-Electro-Mechanical Systems (MEMS). The fabrication activity involved creating a **silicon cantilever** using thin-film deposition, photolithography, and a sequence of dry and wet etching steps. The characterisation activity introduced five advanced instruments used at NTHU's MEMS laboratory to measure the structural, optical, and dynamic properties of fabricated devices.

The day covered:

Fabrication: Cantilever fabrication on a 4-inch Si wafer using PVD-deposited silicon nitride, photolithography (EVG512 positive resist), silicon nitride dry etching, isotropic silicon release etching, and front-side photoresist removal.

Characterisation: Open-loop and closed-loop RF/network analysis (Agilent E5071C VNA + Keithley 2612B source meter), White Light Interferometry (Zygo), Laser Doppler Vibrometry, Confocal Microscopy, and Digital Holographic Microscopy.

Part A: MEMS Cantilever Fabrication

Overview of the Fabrication Sequence

The cantilever was fabricated on a 4-inch silicon wafer using a multi-step surface-micromachining process. The full fabrication sequence is summarised in Table 8.1 and described step by step below.

Table 8.1: Day 8 MEMS cantilever fabrication process flow.

Step	Process	Purpose
1	Si_3N_4 deposition (PVD)	Deposit the structural / hard-mask layer on the Si wafer.
2	Inline characterisation	Verify film thickness and uniformity before lithography.
3	Photoresist spin-coating	Apply EVG512 positive resist at 5000 RPM → 200 nm.
4	Soft bake	Remove casting solvent; improve adhesion.
5	UV exposure	Transfer cantilever pattern at 60 mJ/cm^2 dose.
6	Development (AD-10)	Dissolve exposed resist; reveal Si_3N_4 in open areas.
7	Hard bake	Crosslink resist for etch resistance.
8	Si_3N_4 dry etch (anisotropic)	Pattern the nitride hard mask.
9	Si isotropic dry etch	Release the cantilever beam (undercut).
10	PR removal	Strip residual photoresist in AD-10 solvent.

Step 1: Silicon Nitride Deposition (PVD)

The fabrication began with **Physical Vapour Deposition (PVD)** of silicon nitride (Si_3N_4) on a clean 4-inch silicon wafer. Silicon nitride was chosen as the structural material for the cantilever because of its excellent mechanical stiffness, chemical inertness, and high selectivity against silicon etchants. The PVD process deposits the nitride layer conformally over the wafer surface.

Following deposition, **inline characterisation** was performed to confirm the film thickness and uniformity before advancing to photolithography.

Steps 2–4: Photoresist Spin-Coating and Soft Bake

EVG512 Photoresist Spin-Coat Recipe — Day 8

Photoresist:	EVG512 (positive tone)
Composition:	Polymer, resin, and sensitiser compound
Ramp up:	0 → 1000 RPM in 10 s (spread phase)
Final speed:	5000 RPM for 30 s (thinning phase)
Target thickness:	∼200 nm
Ramp down:	Controlled deceleration
Soft bake:	100 C for 30 s (hotplate)

The 4-inch Si wafer with the deposited Si_3N_4 film was taken to the **lithography room**. A thin coat of EVG512 positive photoresist was applied using a spin coater at a two-step programme: a low-speed spread phase (1000 RPM, 10 s) followed by a high-speed thinning phase (5000 RPM, 30 s) to achieve a uniform 200 nm film. EVG512 is a diazoquinone (DQ)/novolak positive resist: before exposure, the photoactive compound (PAC) renders the film insoluble; UV exposure breaks the PAC bonds and converts the exposed regions to a base-soluble indene-carboxylic acid via the Wolff rearrangement.

After spin-coating, the wafer underwent a **soft bake** at 100 C for 30 s to evaporate residual casting solvent, improve film adhesion, and stabilise the PAC without damaging its photosensitivity.

Steps 5–7: UV Exposure, Development, and Hard Bake

UV Exposure and Development Recipe — Day 8

UV dose:	60 mJ/cm^2
Exposure tool:	UV laser writer
Developer:	AD-10 (aqueous alkaline, TMAH-type)
Development method:	Immersion in developer solution
Rinse:	Deionised (DI) water overflow rinse
Drying:	N_2 blow-gun
Hard bake:	120 C for 3 min (crosslink resist)

The cantilever pattern was transferred using **UV light** at a dose of 60 mJ/cm^2. The UV photons break the PAC bonds only in the exposed regions, rendering them soluble in the alkaline developer AD-10. The wafer was then immersed in AD-10 developer: the exposed resist dissolved completely while the unexposed (masked) resist remained intact, defining the cantilever geometry. After development, the wafer was rinsed with DI water, dried with a nitrogen gun, and hard-baked at 120 C for 3 min to crosslink the resist, remove residual solvents, and maximise its mechanical and chemical resistance for the subsequent etch steps.

Steps 8–9: Silicon Nitride Dry Etch and Silicon Release Etch

Dry Etching Sequence — Day 8

Step 8 — Si_3N_4 anisotropic dry etch:	Pattern the nitride layer using the photoresist as a mask; creates vertical sidewalls.
Step 9 — Si isotropic dry etch (device release):	Undercut the silicon beneath the nitride cantilever beam; frees the beam to vibrate mechanically.

Anisotropic Si_3N_4 dry etch (Step 8). Using the patterned photoresist as a mask, the nitride layer was etched anisotropically in a plasma etcher, transferring the cantilever outline into the hard Si_3N_4 film with vertical sidewalls.

Isotropic silicon release etch (Step 9). A second plasma etch step targeted the underlying silicon isotropically, undercutting the silicon beneath the Si_3N_4 beam. This undercut releases the cantilever, allowing it to deflect freely — the critical step that converts the patterned film into a functional mechanical element.

Step 10: Photoresist Removal

After the etch sequence, the remaining photoresist was dissolved by immersion in **AD-10 developer solvent**, leaving a clean, free-standing silicon nitride cantilever on the 4-inch silicon wafer.

Part B: MEMS Characterisation — Instruments and Methods

Day 8 also included a tour and demonstration of five advanced characterisation instruments used to measure both the structural dimensions and the dynamic mechanical behaviour of MEMS devices fabricated at NTHU.

Setup 1: Open-Loop and Closed-Loop RF Measurement (VNA + Probe)

Instrument Description

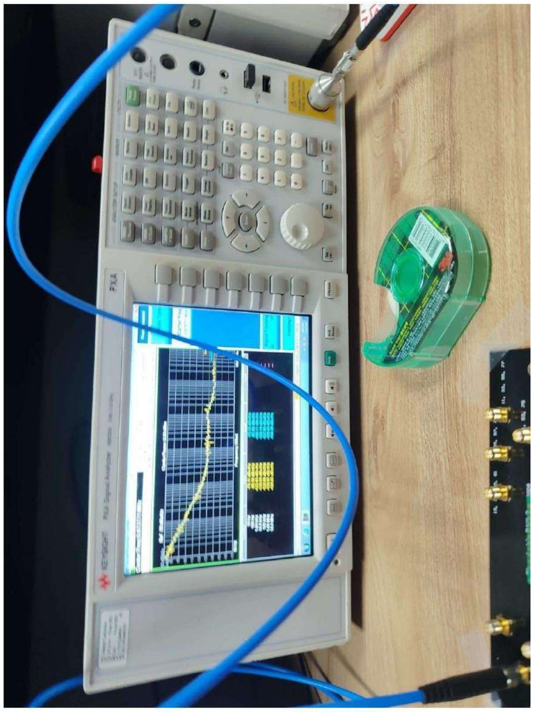
The **Agilent E5071C 4-Port Network Analyser (VNA)** combined with a **Keithley 2612B Dual-Channel Source Meter** and a **Bias Tee** forms the core of the RF measurement bench. The VNA measures S-parameters (scattering parameters) over a wide frequency range, enabling extraction of the resonant frequency, insertion loss, and quality factor (Q-factor) of MEMS resonators. A bias tee allows simultaneous application of a DC bias voltage (from the Keithley source meter) and an RF signal to the device without interference between the two paths.

Open-Loop vs. Closed-Loop Measurement

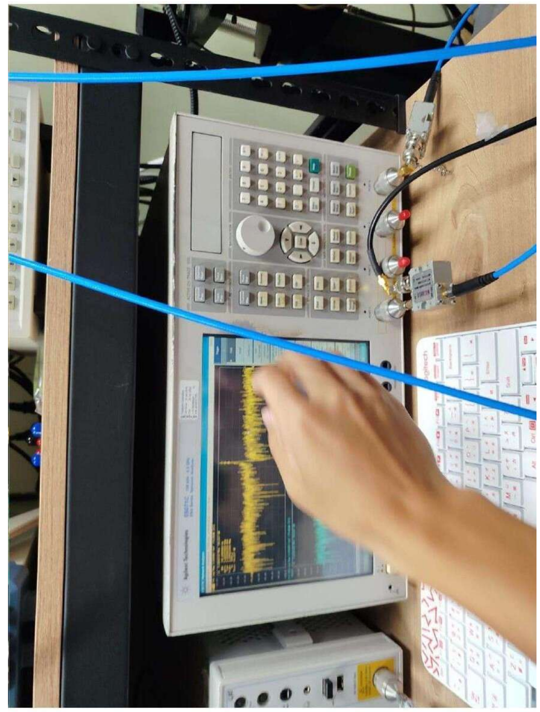
Table 8.2: Comparison of open-loop and closed-loop MEMS resonator measurement.

Parameter	Open-Loop	Closed-Loop
Configuration	VNA sweeps frequency; S21 transmission measured passively.	Device feeds back into sustaining amplifier; self-oscillates.
Output	S-parameter spectrum; extracts f_0 and Q.	Stable oscillation at f_0; phase noise measured.
Use case	Characterise resonant frequency and loss.	Oscillator applications; timing references.

Lab Photographs

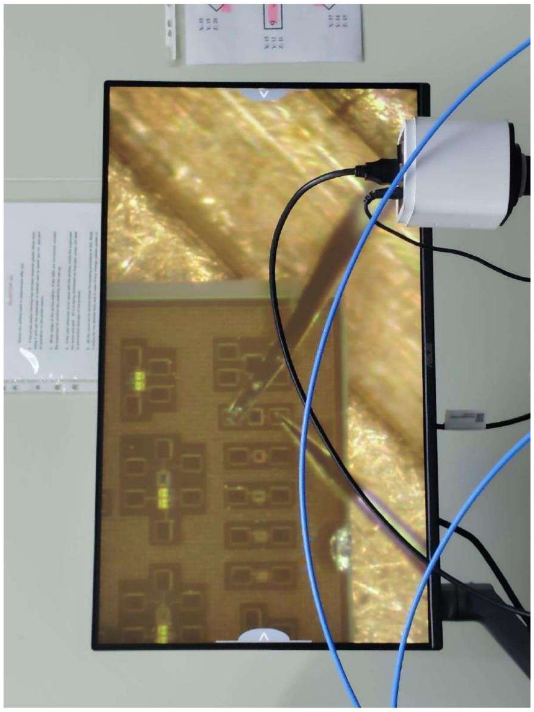


(a) Agilent E5071C VNA display showing S-parameter sweep of a CMOS-MEMS TiN-C resonator. The yellow trace indicates the transmission (S21) response across frequency.

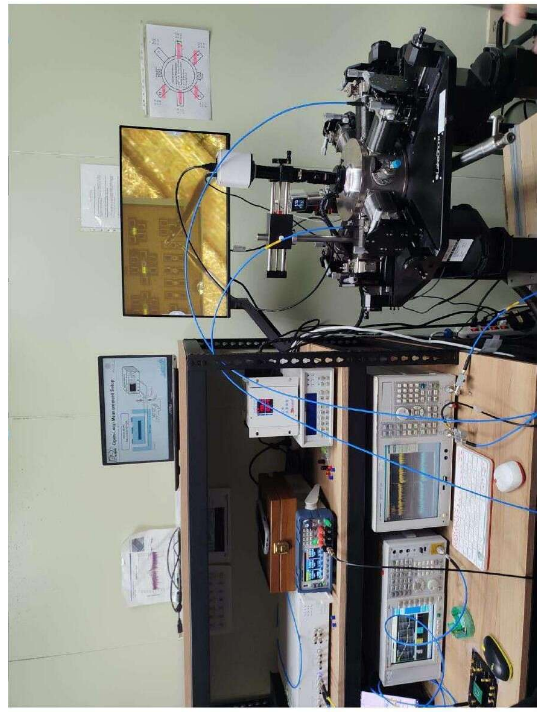


(b) Spectrum analyser display showing the frequency response of the MEMS resonator; the operator adjusts the measurement parameters on the touchscreen.

Figure 8.1: Open-loop RF measurement setup: Agilent E5071C VNA and spectrum analyser used to characterise the frequency response of CMOS-MEMS TiN-C resonators.

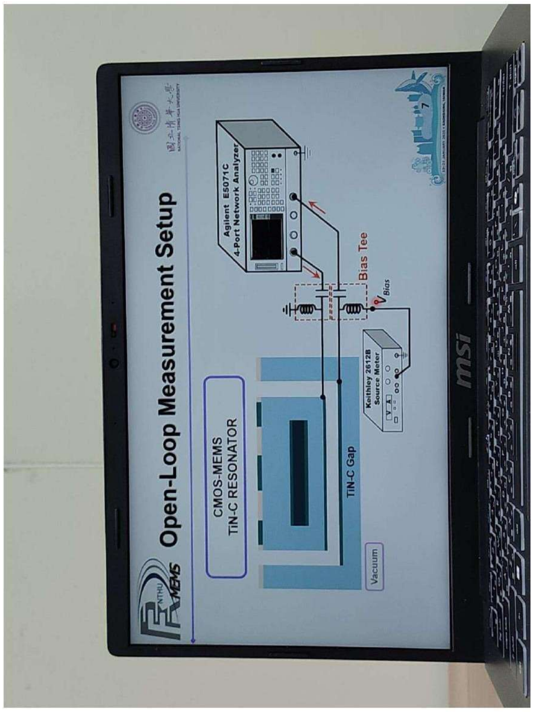


(a) MEMS chip mounted on the probe station stage. Micro-positioner RF probes contact the device pads visible on the gold chip surface. Blue coaxial cables carry the RF signal to and from the VNA.

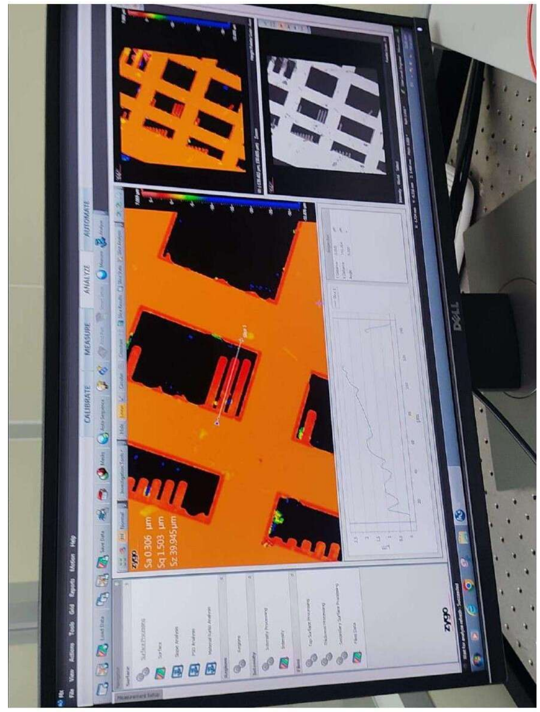


(b) Full measurement bench: RF probe station (top right), VNA, bias tee, Keithley source meter, and spectrum analyser integrated for simultaneous RF and DC biasing.

Figure 8.2: Open-loop measurement setup showing the on-wafer RF probe station and the complete instrument rack used to characterise the CMOS-MEMS resonator.



(a) Laptop screen showing the **Open-Loop Measurement Setup** schematic: Agilent E5071C 4-port VNA connected via bias tee to the CMOS-MEMS TiN-C resonator under vacuum, with Keithley 2612B providing the DC bias voltage V_{Bias}.



(b) Zygo Newview white light interferometer software display showing a false-colour 3D height map of a MEMS surface. The orange/blue contrast reveals step heights corresponding to etched features.

Figure 8.3: Left: schematic diagram of the open-loop CMOS-MEMS measurement configuration. Right: transition to the Zygo WLI instrument showing initial surface height mapping of the MEMS sample.

Setup 2: White Light Interferometer (Zygo NewView)

Instrument Description

The **Zygo NewView White Light Interferometer (WLI)** is a non-contact optical profilometer that uses broadband (white) light to generate high-resolution 3D surface topography maps. It operates on the principle of **vertical scanning interferometry (VSI)**: white light from a Mirau or Michelson interferometric objective interferes with a reference beam; the short coherence length of white light means constructive interference occurs only when the optical path lengths match, i.e., only at the precise surface height. By scanning the objective vertically and recording the fringe peak positions, the system reconstructs the full 3D surface profile with sub-nanometre vertical resolution.

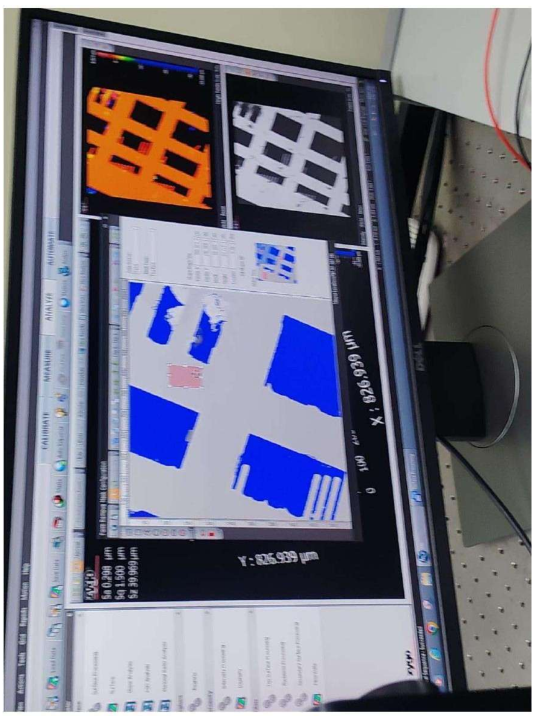
Zygo NewView — Key Specifications

Technique:	Vertical Scanning Interferometry (VSI)
Light source:	Broadband white light (LED)
Vertical resolution:	<1 nm (sub-nanometre)
Lateral resolution:	Objective-dependent (~1 m)
Software:	Zygo MetroPro / Mx
Applications:	Step height, roughness, MEMS topology

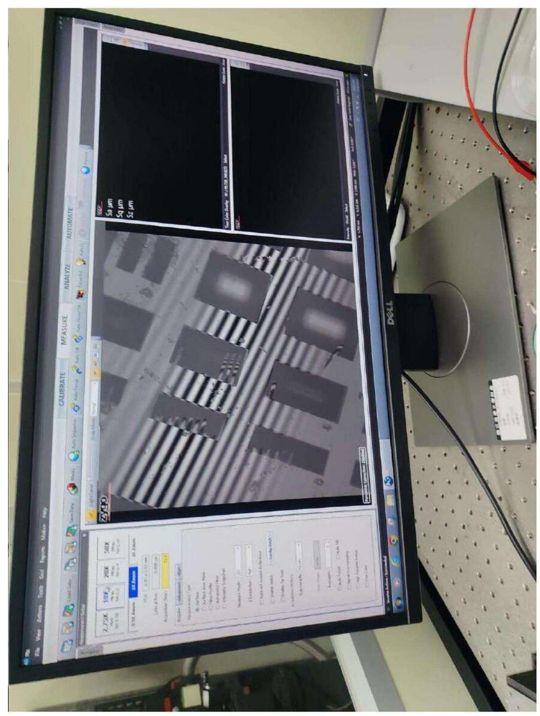
Purpose

The WLI was used to measure the **surface topography and step heights** of the fabricated MEMS cantilever and resonator structures, confirming film thicknesses, etch depths, and any residual stress-induced curvature of the released beams.

Lab Photographs



(a) Zygo MetroPro software displaying a 3D false-colour height map (left: orange/blue surface plot, right: black-and-white intensity map) of a MEMS device. Coordinates read X: 826.939 m, Y: 826.939 m.



(b) Zygo system showing the objective lens over the MEMS chip (top monitor) and the fringe pattern captured during the interferometric scan (bottom). The striped pattern on the chip confirms alignment of the scan region.

Figure 8.4: White Light Interferometer (Zygo NewView) measurements. Left: 3D surface topography map of a MEMS resonator showing step heights and surface roughness. Right: live fringe pattern during the vertical scan.

Setup 3: Laser Doppler Vibrometer (LDV)

Instrument Description

The **Laser Doppler Vibrometer (LDV)** is a non-contact vibration measurement instrument based on the **Doppler effect of light**. A laser beam (reference + probe) is directed at the vibrating MEMS surface. The frequency of the reflected beam is shifted relative to the reference beam by an amount proportional to the surface velocity:

$$\Delta f = \frac{2v}{\lambda} \tag{8.1}$$

where v is the surface velocity and λ is the laser wavelength. By demodulating this frequency shift, the LDV extracts:

Velocity of the moving surface,

Displacement (by integrating velocity over time).

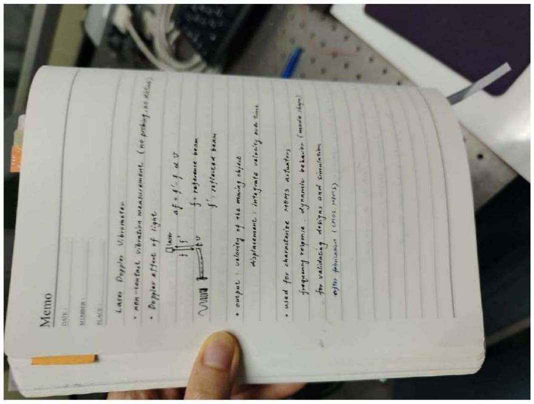
Laser Doppler Vibrometer — Key Facts

Principle:	Doppler shift of a focused laser beam
Probing:	Non-contact (no physical contact with device)
Output:	Velocity and displacement vs. frequency
Applications:	MEMS actuator characterisation, dynamic behaviour, resonant frequency, frequency response, design validation (CMOS-MEMS)

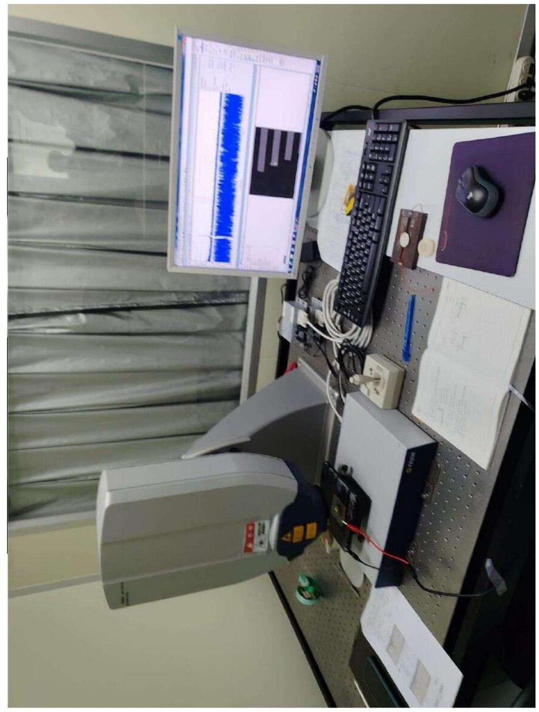
Purpose

The LDV was used to characterise the **dynamic mechanical response** of MEMS actuators and resonators: measuring resonant frequency, mode shapes, and vibration amplitude without any risk of contact damage to the delicate micro-structures. It is especially valuable for validating CMOS-MEMS designs and simulations after fabrication.

Lab Photographs



(a) Lab notebook showing key LDV principles: Doppler shift equation $\Delta f = f' - f = \frac{2v}{\lambda}$, velocity measurement, displacement by integration, and applications in MEMS/CMOS-MEMS characterisation.



(b) Laser Doppler Vibrometer setup: the compact LDV head (bottom, Polytec brand) focuses a laser beam onto the MEMS sample. A monitor above displays the measured frequency spectrum (blue waveform) in real time.

Figure 8.5: Laser Doppler Vibrometer (LDV) demonstration. Left: working principle notes. Right: complete LDV setup with the instrument head focused onto the MEMS chip and live frequency response displayed on the monitor.

Setup 4: Confocal Microscope

Instrument Description

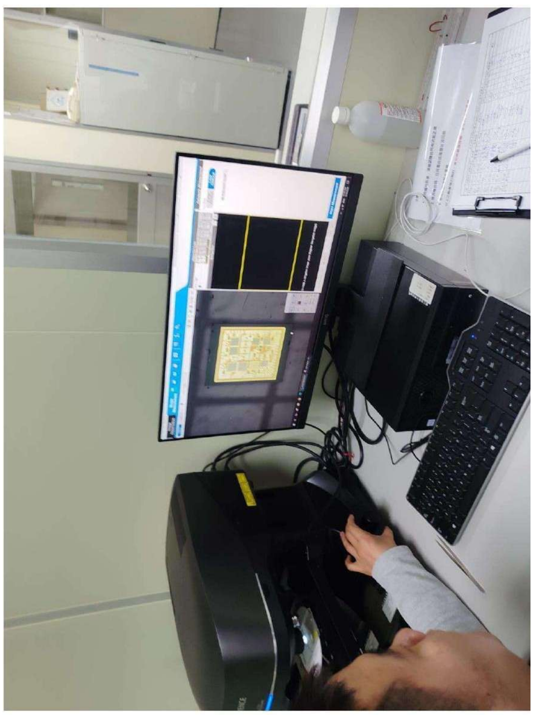
A **confocal microscope** uses a pinhole aperture in the optical path to reject out-of-focus light, collecting only light from a tightly defined focal plane. By scanning the focal plane through the sample depth (z-stack), it constructs a high-resolution 3D image of the surface without physical contact. Unlike a conventional optical microscope, confocal imaging provides true 3D topographic information with excellent lateral and axial resolution.

Confocal Microscope — Key Features	
Technique:	Confocal scanning with pinhole aperture
Imaging:	Non-contact 3D surface profiling
Resolution:	Sub-micrometre lateral; nm-level axial
Software:	Leica LAS X or equivalent
Applications:	MEMS surface topography, step heights, roughness, cantilever inspection

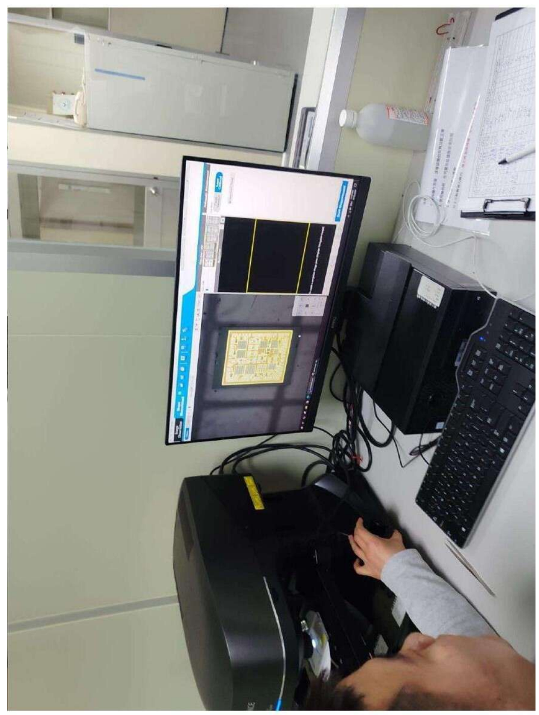
Purpose

The confocal microscope was used to inspect the **surface morphology** of the fabricated MEMS cantilever and resonator, measure step heights and surface roughness, and identify any defects such as undercut non-uniformity or residual photoresist.

Lab Photographs

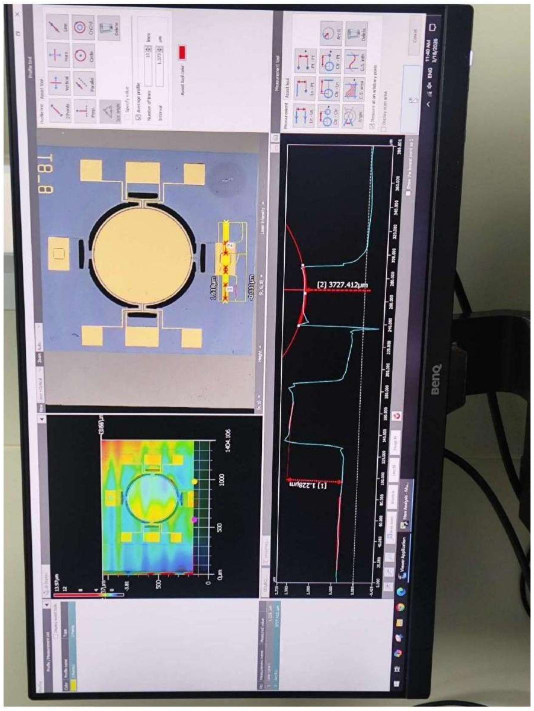


(a) Operator using the confocal microscope (black instrument body, Leica brand) to scan the MEMS chip. The connected monitor displays a live top-down image of the chip surface with patterned metal features visible.

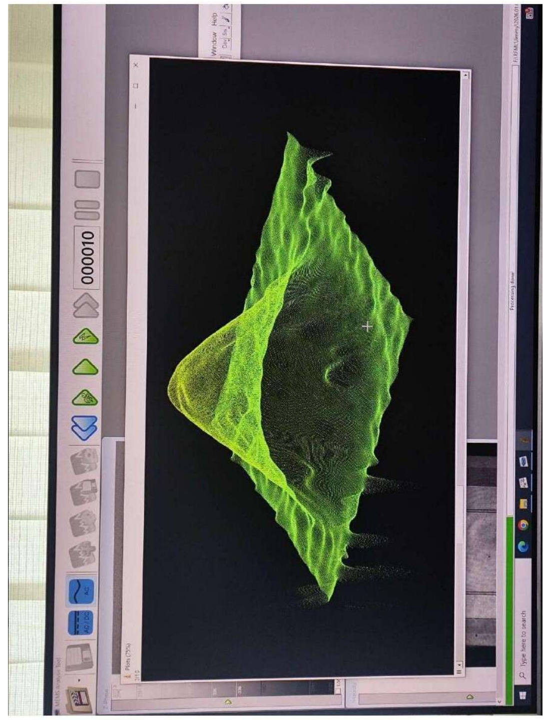


(b) Same confocal microscope setup from a slightly different angle, showing the objective lens positioned over the MEMS sample and the monitor displaying the chip layout for navigation.

Figure 8.6: Confocal microscope setup used to image the surface topography of MEMS devices. The MEMS chip is placed on the motorised stage and scanned in 3D to generate height maps and surface profiles.



(a) Confocal profilometry software display showing a CMOS-MEMS resonator: (top left) 2D optical image with the circular resonator disk visible; (bottom left) false-colour height map (rainbow colourscale); (right) line profile showing step heights of 3727.412 m and 1.228 m.



(b) 3D surface reconstruction of the MEMS resonator rendered by the confocal software (Leica). The green wire-mesh surface plot reveals the 3D geometry of the circular disk and its anchor structures.

Figure 8.7: Confocal microscope results for a CMOS-MEMS disk resonator. Left: 2D optical image, false-colour height map, and step-height line profile. Right: 3D wire-mesh surface reconstruction showing the resonator disk geometry and surrounding anchor structure.

Setup 5: Digital Holographic Microscope (DHM)

Instrument Description

The **Digital Holographic Microscope (DHM)** is an interferometric imaging technique that records both the **amplitude** and **phase** of the light reflected from a sample. Unlike conventional microscopy, DHM captures a hologram (interference pattern between the object beam and a reference beam) in a single camera frame. Numerical reconstruction of the hologram yields a quantitative phase image, which is directly proportional to the **optical path length difference** — and for reflective surfaces, to the **surface height**. This enables:

- Full-field 3D topographic imaging with nanometre-scale height resolution.
- Dynamic measurements (time-resolved phase) to capture vibrating surfaces.
- No sample preparation or physical contact required.

Digital Holographic Microscope — Key Facts

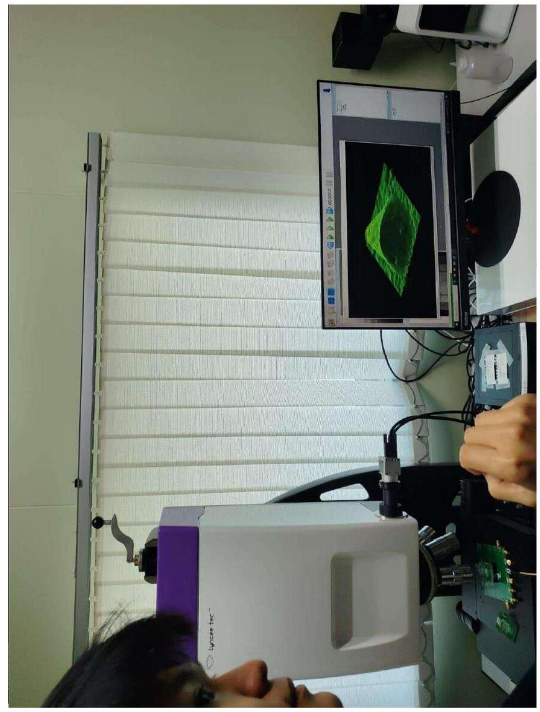
Technique:	Off-axis digital holography
Measurement:	Quantitative phase imaging → surface height map
Height resolution:	Nanometre to sub-nanometre
Acquisition:	Single-shot hologram; no z-scanning needed
Instrument:	Lyncee Tec DHM (Reflection mode)
Applications:	MEMS dynamic and static characterisation, thin film metrology

Purpose

The DHM was used to capture **full-field 3D phase maps** of the MEMS structures, providing quantitative topography information over a large field of view in a single acquisition. It is particularly powerful for observing dynamic deformations and stress-induced bowing of cantilever beams in real time.

Lab Photographs

(a) Digital Holographic Microscope software (Lyncee Tec) displaying the reconstructed 3D phase surface of a MEMS structure rendered as a green wire-mesh. The complex surface topology reflects the actual 3D geometry of the fabricated micro-structure.



(b) The Lyncee Tec DHM instrument (white/purple body) positioned over the MEMS sample on an optical table. A monitor above the instrument displays the reconstructed 3D holographic image in real time.

Figure 8.8: Digital Holographic Microscope (Lyncee Tec DHM) used for full-field 3D characterisation of MEMS devices. The reconstructed green wire-mesh phase map (left) and the complete instrument setup (right) are shown.

Comparison of Characterisation Techniques

Table 8.3: Comparison of the five MEMS characterisation instruments used on Day 8.

Instrument	Quantity Measured	Contact?	Resolution	Best For
VNA + Probe (E5071C)	S-parameters, f_0, Q	RF probe	kHz–GHz	Resonant frequency, loss
White Light Interferometer (Zygo)	3D surface height	No	<1 nm vertical	Step height, roughness
Laser Doppler Vibrometer	Velocity, displacement	No	pm velocity	Dynamic vibration mode
Confocal Microscope	3D surface image	No	nm– m	Surface morphology, defects
Digital Holographic Microscope	Phase / height map	No	nm full-field	Fast 3D imaging, dynamics

Results and Conclusion

Fabrication Outcome

The complete cantilever fabrication sequence was successfully demonstrated on a 4-inch silicon wafer. Starting from a PVD-deposited Si_3N_4 film, the process progressed through photolithography, nitride dry etching, isotropic silicon release etching, and photoresist removal to yield free-standing Si_3N_4 cantilever beams. The key fabrication insight is that the isotropic silicon undercut (Step 9) is the device-defining step: it transforms the patterned film from a surface feature into a mechanically active suspended beam.

Characterisation Outcome

Exposure to the five characterisation instruments provided a comprehensive understanding of how MEMS devices are evaluated:

VNA / RF probing extracts the electrical resonant frequency and Q-factor — the core performance metrics for resonator applications.

WLI and confocal microscopy provide non-contact 3D surface maps to verify etch depths and detect stress-induced curvature.

LDV confirms the mechanical vibration mode and amplitude at drive frequency, verifying that the cantilever can oscillate without fracture.

DHM enables real-time full-field phase imaging — valuable for rapid inspection of dynamic deformations over large areas.

Conclusion

Day 8 provided end-to-end experience of the MEMS development cycle: from designing and fabricating a silicon nitride cantilever through a six-step micro-machining process, to characterising its structural and dynamic properties using five complementary techniques. The combination of optical, interferometric, and laser-based instruments demonstrated that no single tool is sufficient; together they give a complete picture of the device's geometry, surface quality, and mechanical performance — all without damaging the fragile micro-structures.

Report compiled by Group 3 — Indo-Taiwan Advanced Semiconductor Manufacturing Program, NTHU Taiwan.
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